

# NONSOFC GROUPS EXIST

ANONYMOUS

**ABSTRACT.** We prove that nonsoc groups exist: for every  $m \geq 1$ , the group  $\text{EL}_{m+1}(L_{\mathbb{F}_2}(1, 2))$  of elementary matrices over the binary Leavitt algebra is infinite, finitely generated, has Kazhdan’s property (T), and is not sofc. The proof is engineered for verification: the two auxiliary matrices on which the construction turns are explicit products of elementary matrices, the remaining memberships reduce to one Whitehead identity and exhibited commutators of units, and no  $K$ -theory is used.

The engine is a compression–centralizer criterion. If a Kazhdan group is generated by an infinite Kazhdan subgroup  $\Gamma$  together with finitely many elements that conjugate  $\Gamma$  into itself, then in a hypothetical sofc approximation the expander decompositions supplied by Kun’s theorem, for  $\Gamma$  and for the ambient group, can be matched almost bijectively: a median normalization of component sizes is almost monotone along compressor arcs, exact conservation under permutations makes the estimate two-sided, and an  $\ell^1$  co-area inequality pins the normalized sizes at their median. Localizing to one matched component yields a sofc approximation of  $K \times J$  whose  $K$ -edges expand, and the Kun–Thom obstruction forces the commuting subgroup  $J$  to be locally embeddable into finite groups (LEF). The Leavitt relations realize the scheme with  $J$  a two-generator subgroup of a corner copy of Thompson’s group  $V$  that is not LEF for a finite reason: in a finite group the two  $F$ -relators force commutation, and the witness pair does not commute.

Self-similarity spreads the conclusion: for each fixed finite field  $k$ , the unit group of  $L_k(1, 2)$ , every  $\text{GL}_r$ , and every  $\text{EL}_r$  with  $r \geq 2$  are mutually isomorphic Kazhdan nonsoc groups. Every finitely generated nonsoc group is a quotient of an infinite finitely presented nonsoc group, Kazhdan whenever the original is; so not every finitely presented group is sofc, and neither sofcity, local embeddability, nor residual finiteness is closed under quotients. The published proofs of the two decomposition theorems are audited: two local gaps are repaired, and the spectral implication of a published equivalence is refuted by counterexample. An appendix gives an elementary and effective proof—the first we are aware of—that  $K_1(L_k(1, 2)) = 0$  over every field.

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## 1. INTRODUCTION

A countable discrete group is *sofic* if every finite portion of its multiplication table admits arbitrarily accurate and asymptotically faithful models by permutations of finite sets. The notion arose in Gromov’s work on symbolic dynamics [10]; Weiss introduced the term and asked whether every countable group is sofic [21]. The class contains all amenable and all residually finite groups, and it is stable under a long list of standard constructions; see [7, 18]. Whether every group is sofic has remained a central open problem of modern group theory; recent work examines the structure a minimal counterexample would have, conditional on existence [8].

We answer the question in the negative. Two consequences follow at once: not every finitely presented group is sofic (Theorem C), and none of the classes of sofic, LEF, or residually finite groups is closed under quotients (Remark 10.6).

**Main results.** Let  $k$  be a field and let

$$(1) \quad L = L_k(1, 2) = k\langle s_0, s_1, t_0, t_1 \mid t_i s_j = \delta_{ij}, s_0 t_0 + s_1 t_1 = 1 \rangle$$

be the binary Leavitt algebra [15]. For a unital ring  $A$  and  $r \geq 2$ , write

$$x_{ij}(a) = I_r + aE_{ij} \in \mathrm{GL}_r(A), \quad a \in A, \quad i \neq j,$$

and let  $\mathrm{EL}_r(A) \leq \mathrm{GL}_r(A)$  be the subgroup generated by these elementary matrices.

**Theorem A.** *Let  $R = L_{\mathbb{F}_2}(1, 2)$ . For every  $m \geq 1$ , the group*

$$\mathrm{EL}_{m+1}(R)$$

*is infinite, finitely generated, has property (T), and is not sofic. In particular  $G = \mathrm{EL}_4(R)$  is not sofic, and not every countable discrete group is sofic.*

The proof of Theorem A is engineered for audit. The two auxiliary matrices of the construction are exhibited as explicit products of elementary matrices; every remaining elementary membership follows from one explicit Whitehead identity together with explicit commutator certificates—every cylinder transposition is a single commutator of units; the ranks below four are reached by an explicit isomorphism between the elementary groups of any two ranks  $\geq 2$  over  $R$  (Proposition 5.8); and no  $K$ -theory is used.

**Theorem B.** *Let  $k$  be a finite field and let  $L = L_k(1, 2)$ .*

- (i) *The unit group  $L^\times$  is infinite, finitely generated, has property (T), and is not sofic.*

Endpoint/MainResults  
universalLeavittEL4\_not\_isSofic  
universalLeavitt\_profile  
ambient\_profile  
ambient\_full\_profile  
Countability of the witness,  
which the final clause needs,  
is the first conjunct of  
ambient\_full\_profile.  
Endpoint/ManuscriptStatement  
nts  
theoremB\_exact  
theoremB\_Theta  
theoremB\_Theta\_apply  
Endpoint/MainResults  
binaryLeavitt\_finiteField\_profile  
ofile  
binaryLeavittUnits\_not\_isSofic  
Leavitt/UnitsGLProfile  
binaryLeavittUnits\_profile  
binaryLeavittGL\_profile  
theoremB\_exact is the printed  
statement in one type;  
theoremB\_Theta is  $\Theta_C$  for  
every complete leaf set, with  
theoremB\_Theta\_apply the  
displayed formula.

- (ii) For every  $r \geq 1$ , every ordered complete binary leaf set  $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$  induces an explicit group isomorphism

$$(2) \quad \Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \xrightarrow{\cong} L^{\times}, \quad (a_{ij}) \mapsto \sum_{i,j=1}^r \alpha_i a_{ij} \alpha_j^*.$$

For  $r \geq 2$  one has  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ . Hence all the groups  $\mathrm{GL}_r(L)$ ,  $r \geq 1$ , and  $\mathrm{EL}_r(L)$ ,  $r \geq 2$ , are mutually isomorphic, and each is an infinite finitely generated property (T) nonsocic group.

**Theorem C.** Every finitely generated nonsocic group is a quotient of an infinite finitely presented nonsocic group. Every nonsocic group with property (T) is a quotient of an infinite finitely presented nonsocic group with property (T). In particular, there is an infinite finitely presented property (T) group  $H$  that is not sofic;  $H$  may be chosen to surject onto  $L^{\times}$ , and hence to have a quotient isomorphic to every group listed in Theorem B.

**Strategy.** The proof has three independent layers.

The first layer is analytic.

**Theorem D** (Kazhdan compression–centralizer criterion). Let  $G$  be a finitely generated property-(T) group, and let  $\Gamma \leq G$  be an infinite finitely generated property-(T) subgroup. Suppose there is a finite set  $Q \subseteq G$  with

$$G = \langle \Gamma, Q \rangle \quad \text{and} \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix  $q_0 \in Q$ , put  $K = q_0\Gamma q_0^{-1}$ , and let  $J \leq \Gamma$  be finitely generated with

$$[K, J] = 1, \quad K \cap J = \{1\}.$$

If  $G$  is sofic, then  $J$  is LEF.

The point of departure is the tension between the two theorems that govern sofic approximations of Kazhdan groups. Kun’s theorem [13] proves Bowen’s conjecture: every sofic approximation of an infinite property-(T) group is essentially a vertex-disjoint union of uniformly expanding components. The Kun–Thom theorem [14] requires more: if the restriction of a sofic approximation of  $\Lambda \times \Delta$  to a Kazhdan factor  $\Lambda$  is essentially one expander, then  $\Delta$  is locally embeddable into finite groups (LEF). A disjoint union of equal-sized expanders is not enough, because the commuting factor may permute the components—this is exactly what happens in a product approximation.

Compression closes the gap. A hypothetical sofic approximation of  $G$  yields a Kun decomposition for  $\Gamma$  and another for  $G$ . Transporting the  $\Gamma$ -components by a compressor  $q$  gives one-sided inclusions into other  $\Gamma$ -components. Write  $s(y)$  for the size of the  $\Gamma$ -component containing  $y$ ; on each ambient  $G$ -component, normalize by a vertex-weighted median  $m$  of the sizes and set

$$(3) \quad f(y) = \frac{s(y)}{s(y) + m}.$$

Endpoint/ManuscriptState

nts

theoremC\_first

theoremC\_second

theoremC\_exact

theoremC\_covers\_theoremB\_groups

Endpoint/MainResults

exists\_finitelyPresented\_nonsocic\_group

sofic\_group

exists\_infinite\_finitelyPresented\_nonsocic\_ambient\_cover

Covers/KazhdanCover

exists\_kazhdan\_finitelyPresented\_cover\_of\_not\_isSofic

nted\_cover\_of\_not\_isSofic

The final assertion is

theoremC\_covers\_theoremB\_groups: one  $H$ , surjecting

onto  $L^{\times}$  and onto every  $\mathrm{GL}_r$

and  $\mathrm{EL}_r$  of Theorem B.

Endpoint/ManuscriptState

nts

theoremD\_subgroups

Criterion/CriterionAssembly

isLEF\_of\_isSofic

not\_isSofic\_of\_not\_isLEF

theoremD\_subgroups is the

printed subgroup form;

countability of  $G$ , implicit in

the sofic vocabulary of

Section 2, is its one explicit

instance argument, and finite

generation of  $G$  is derived

rather than assumed. The

LEF conclusion is the

textbook notion by

isLEF\_iff\_textbook.

The one-sided inclusions make  $f$  almost monotone along compressor arcs. But a compressor acts by a permutation, so total increase equals total decrease *exactly*. The resulting small  $\ell^1$  edge variation, fed into an elementary co-area inequality on the ambient expanders, pins  $f$  at its median value  $1/2$ . Hence transported and target components have asymptotically equal sizes, and one-sided containment becomes almost-bijective matching. A weighted diagonal selection extracts a single matched component on which every multiplication, faithfulness, invariance, conjugacy, and graph-edit error is negligible. Restricting and completing the permutations yields a sofic approximation of  $K \times J$  whose  $K$ -edges form an essential expander sequence, and the Kun–Thom obstruction forces  $J$  to be LEF. Theorem D is proved in Section 4 as a corollary of a local statement, Theorem 4.1, which isolates the exact graph-theoretic hypotheses and records precisely what property (T) is used for. Remarks 4.6 and 4.7 test the criterion against sofic examples and show that each hypothesis earns its place.

The second layer is algebraic, and it is where the Leavitt relations enter. Fix  $m \geq 2$  and a nontrivial unital ring  $A$  carrying a binary Leavitt family—four elements satisfying the two Leavitt relations; no copy of  $L$  is assumed. A left-comb binary partition with  $m+1$  leaves produces a compressor  $u \in \mathrm{GL}_{m+1}(A)$  that conjugates the block subgroup  $\Gamma \cong \mathrm{EL}_m(A)$  into the corner  $a \mapsto s_0 a t_0$  of itself. A block involution  $z$  centralizes the compressed image  $K$ , so  $v = zu$  is a second compressor with the same image, and  $vu^{-1} = z$  generates every elementary matrix involving the last coordinate; whenever  $u$  and  $z$  lie in  $\mathrm{EL}_{m+1}(A)$ , it follows that  $G = \mathrm{EL}_{m+1}(A) = \langle \Gamma, u, v \rangle$ . The complementary corner  $a \mapsto p_0 + s_1 a t_1$  carries a commuting copy of Thompson’s group  $V$ , and inside it the witness: the two-generator standard copy of Thompson’s  $F$ , which is not LEF for a finite reason—in a finite group the two  $F$ -relators force their two elements to commute, and the witness pair does not commute. The witness is elementary over every ambient ring: its two generators are products of explicitly exhibited commutators of units—every cylinder transposition is a single commutator—and the Whitehead lemma places diagonal products of commutators of units in  $\mathrm{EL}_2$ . The only membership hypotheses of the general theorem therefore concern  $u$  and  $z$ . Over  $\mathbb{F}_2$  they are discharged by explicit four-term elementary factorizations, and an explicit rank equivalence between the elementary groups of any two ranks  $\geq 2$ —block flattening along the self-similarity of the Leavitt relations—then spreads the conclusion from rank four to every rank, which gives Theorem A with no  $K$ -theory. Over every finite field the memberships also follow from  $K_1(L) = 0$  and the GE property of purely infinite simple rings; the  $K_1$  input itself admits an elementary, effective proof, written out in full in Appendix A. Matrix self-similarity  $M_r(L) \cong L$  then spreads nonsoficity and property (T) to the unit group and to all  $\mathrm{GL}_r$ , and a separate two-by-two realization shows that the scheme already closes in rank two.

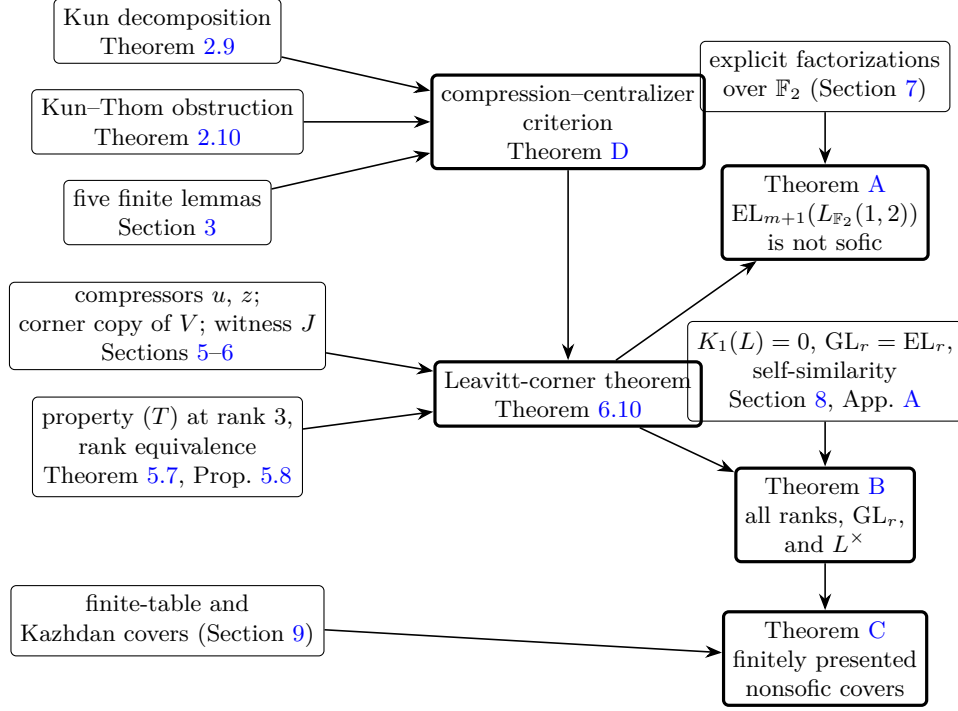


FIGURE 1. The architecture of the proof. The analytic layer (top left) produces the criterion; the algebraic layer (middle left) realizes its hypotheses over any ring carrying a binary Leavitt family; the memberships are discharged twice, explicitly in characteristic two and  $K$ -theoretically over every finite field; and the finite-table layer converts nonsocficity into finitely presented covers.

The third layer is a finite multiplication-table construction. Every finitely generated nonsocfic group  $G$  has a finite table fragment that admits no sufficiently accurate permutation model. Imposing precisely that finite table as relators gives a finitely presented group mapping onto  $G$ ; the map proves infinitude and preserves distinctness of the named elements, while the forbidden finite table proves nonsocficity directly. When  $G$  is Kazhdan, Shalom’s finite-presentation theorem first supplies a finitely presented Kazhdan cover, and imposing the same table inside its kernel produces a finitely presented Kazhdan nonsocfic cover. The covers are proved nonsocfic directly, from the forbidden table itself; quotient closure of socficity—which in fact fails outright once nonsocfic groups exist (Remark 10.6)—plays no role.

Figure 1 charts how the three layers assemble.

**External inputs, and how to verify this paper.** The proof is organized so that its trust surface is explicit and a skeptical reader can audit it efficiently.

Theorems A and D rest on the following external inputs: Kun’s expander decomposition theorem [13], cited in one-way form and with two proof repairs recorded in Remark 2.11; the Kun–Thom centralizer obstruction [14], with the repair recorded there as well; property (T) at rank three [9], stated and used in the finite-field case; and, for context only, the classical facts about Thompson’s group  $V$  [11, 6, 5]—the main line replaces them by the two-relator obstruction of Lemma 5.12 and the explicit commutator certificates of Lemma 6.6. Everything else is self-contained. In the proof of Theorem A, the two auxiliary matrices  $u$  and  $z$  are factored explicitly into elementary matrices, and every remaining elementary membership follows from the explicit Whitehead identity of Lemma 5.5 together with the commutator certificates of Lemma 6.6. The analytic sections use no spectral theory: the only analytic inequality proved and used in the matching argument is the  $\ell^1$  co-area estimate of Lemma 2.7, established in a few lines from the layer-cake formula, while the approximation-theoretic depth is confined to the cited theorems of Kun and Kun–Thom.

Theorem B additionally needs  $K_1(L) = 0$ , the collapse  $GL_r = EL_r$ , and the perfectness of  $L^\times$ , classically consequences of [1, 3, 2]. None remains a dependency: Appendix A proves all three elementarily and effectively— $K_1$  by the two-exit elimination, the collapse at every rank, and perfectness through the three-leaf self-similarity—so the  $K$ -theoretic citations are retained purely as independent confirmation.

Theorem C is self-contained apart from Shalom’s finitely presented Kazhdan covers [19], used only for the Kazhdan refinement.

A reader who wants to check the headline claim should verify, in order: the five finite lemmas of Section 3, each elementary and a few lines long; the proof of Theorem 4.1 in Section 4, which assembles them and invokes Kun’s theorem twice and the Kun–Thom theorem once, in exactly the forms stated in Section 2.5; the matrix identities of Sections 5 and 6, each a finite computation from the two Leavitt relations; and the two explicit factorizations of Section 7. That path proves Theorem A in full. Sections 8 and 9 lie downstream and can be audited separately.

**Organization.** Section 2 fixes the sofic, expander, and LEF vocabulary and states the theorems of Kun and Kun–Thom in the forms used. Section 3 isolates the five elementary lemmas of the matching mechanism. Section 4 proves the local compression–centralizer theorem and deduces Theorem D. Section 5 develops Leavitt leaf calculus, matrix self-similarity, the elementary rank equivalence, the Whitehead lemma, the characteristic-two block factorizations, and the corner realization of  $V$ . Section 6 presents the compression scheme in adjacent ranks over an arbitrary ring carrying a binary Leavitt family and proves the Leavitt-corner theorem. Section 7 discharges all memberships explicitly over  $\mathbb{F}_2$  and proves Theorem A. Section 8 proves the

two-by-two theorem, the equality  $\text{GL}_r(L) = \text{EL}_r(L)$ , and Theorem B. Section 9 proves Theorem C. Section 10 collects concluding remarks. Appendix A proves  $K_1(L_k(1, 2)) = 0$  elementarily and effectively.

## 2. SOFIC APPROXIMATIONS, EXPANDERS, AND LEF GROUPS

**2.1. Sofic approximations.** For  $\sigma, \tau \in \text{Sym}(Y)$ , let

$$d_H(\sigma, \tau) = \frac{1}{|Y|} |\{y \in Y \mid \sigma(y) \neq \tau(y)\}|$$

denote the normalized Hamming distance.

**Definition 2.1.** Let  $G$  be a countable discrete group. A *sofic approximation* of  $G$  is a sequence of maps  $\sigma_n : G \rightarrow \text{Sym}(Y_n)$ , with  $|Y_n| \rightarrow \infty$ , such that

- (a)  $d_H(\sigma_n(g)\sigma_n(h), \sigma_n(gh)) \rightarrow 0$  for all  $g, h \in G$ ;
- (b)  $d_H(\sigma_n(g), \text{id}) \rightarrow 1$  for all  $g \in G \setminus \{1\}$ .

The group  $G$  is *sofic* if it admits a sofic approximation.

The maps  $\sigma_n$  are not assumed to be homomorphisms. Property (b) self-improves: for a sofic group one may always pass to a sofic approximation in which each  $\sigma_n(g)$ ,  $g \neq 1$ , moves *every* point of a set of density  $1 - o(1)$ , and it is standard that (a) and (b) force  $d_H(\sigma_n(1), \text{id}) \rightarrow 0$ . We will use the following exact normalization.

**Lemma 2.2** (Finite normalization). *Let  $\sigma_n : G \rightarrow \text{Sym}(Y_n)$  be a sofic approximation and let  $F \subseteq G$  be a finite symmetric set. Then there is a sofic approximation  $\sigma'_n : G \rightarrow \text{Sym}(Y_n)$  with  $d_H(\sigma_n, \sigma'_n) \rightarrow 0$  pointwise, such that for all  $n$ :*

$$\sigma'_n(1) = \text{id}, \quad \sigma'_n(g^{-1}) = \sigma'_n(g)^{-1} \quad (g \in F).$$

*Proof.* Set  $\sigma'_n(1) = \text{id}$ , and note that  $d_H(\sigma_n(1), \text{id}) \rightarrow 0$ : applying (a) with  $g = h = 1$  gives  $d_H(\sigma_n(1)^2, \sigma_n(1)) \rightarrow 0$ , and composing with the permutation  $\sigma_n(1)^{-1}$  preserves  $d_H$ .

Choose one representative from each pair  $\{g, g^{-1}\} \subseteq F \setminus \{1\}$  with  $g \neq g^{-1}$ ; put  $\sigma'_n(g) = \sigma_n(g)$  and  $\sigma'_n(g^{-1}) = \sigma_n(g)^{-1}$ . Then  $d_H(\sigma_n(g^{-1}), \sigma_n(g)^{-1}) \rightarrow 0$ , because  $\sigma_n(g^{-1})\sigma_n(g)$  is  $d_H$ -close to  $\sigma_n(1)$ , hence to  $\text{id}$ , and  $d_H$  is invariant under composition with the permutation  $\sigma_n(g)^{-1}$  on the right.

If  $g = g^{-1} \in F \setminus \{1\}$  is an involution, let  $A_n = \{y \mid \sigma_n(g)^2 y = y\}$ . Since  $\sigma_n(g)^2$  is  $d_H$ -close to  $\sigma_n(g^2) = \sigma_n(1)$ , hence to  $\text{id}$ , we have  $|Y_n \setminus A_n| = o(|Y_n|)$ . The set  $A_n$  is  $\sigma_n(g)$ -invariant—if  $\sigma_n(g)^2 y = y$ , then applying  $\sigma_n(g)$  gives  $\sigma_n(g)^2(\sigma_n(g)y) = \sigma_n(g)y$ —and  $\sigma_n(g)$  restricts to an involution of  $A_n$ . Define  $\sigma'_n(g)$  to be  $\sigma_n(g)$  on  $A_n$  and the identity on  $Y_n \setminus A_n$ ; this is an involution of  $Y_n$  differing from  $\sigma_n(g)$  on  $o(|Y_n|)$  points.

Outside  $F \cup \{1\}$  put  $\sigma'_n = \sigma_n$ . Each  $\sigma'_n(g)$  differs from  $\sigma_n(g)$  on  $o(|Y_n|)$  points, so properties (a) and (b) persist.  $\square$

The next lemma is the bridge, used many times below, between statements about products of letter permutations and statements about the permutation of the product.

*Sofic/Sofic*  
*SoficApproximation*  
*IsSofic*

*Sofic/Normalization*  
*normalize*  
*normalize\_map\_one*  
*normalize\_map\_inv*  
*normalize\_close*  
*exists\_normalization*  
*perturb*  
*involutionNormalize*  
*halfOrbit*  
*Sofic/Sofic*  
*map\_one\_close*  
*Matching/InverseNormalizati*  
*on*  
*inverseError\_negligible*  
 *$\sigma'$  is normalize, on the same*  
*finite models: perturb keeps a*  
*Hamming-close reassignment*  
*sofic, involutionNormalize*  
*makes each tested involution*  
*exact, and halfOrbit chooses*  
*the representatives. The*  
*sequel consumes the two*  
*error bounds directly.*

**Lemma 2.3** (Word approximation). *Let  $\sigma_n : G \rightarrow \text{Sym}(Y_n)$  be a sofic approximation, let  $g \in G$ , and let  $g = t_1 t_2 \cdots t_\ell$  be any fixed factorization with  $t_1, \dots, t_\ell \in G$ . Then*

$$|\{y \in Y_n \mid \sigma_n(t_1)\sigma_n(t_2) \cdots \sigma_n(t_\ell)y \neq \sigma_n(g)y\}| = o(|Y_n|).$$

*Proof.* Induction on  $\ell$ , the case  $\ell = 1$  being trivial. For  $\ell \geq 2$  put  $h = t_2 \cdots t_\ell$ . Composing with the permutation  $\sigma_n(t_1)$  on the left does not change disagreement sets, so  $\sigma_n(t_1) \cdots \sigma_n(t_\ell)$  agrees with  $\sigma_n(t_1)\sigma_n(h)$  outside the  $o(|Y_n|)$  points supplied by the inductive hypothesis; and  $\sigma_n(t_1)\sigma_n(h)$  agrees with  $\sigma_n(t_1 h) = \sigma_n(g)$  outside  $o(|Y_n|)$  points by Definition 2.1(a). The union of the two sets has size  $o(|Y_n|)$ .  $\square$

Whenever an element of a subgroup is written below as a word in a generating set and a property is verified letter by letter, Lemma 2.3 is what transfers the conclusion from the product of the letter permutations to the permutation of the element itself; it is cited explicitly at each such point.

**2.2. Generator graphs and edit distance.** Let  $T \subseteq G$  be a finite set and  $\sigma : G \rightarrow \text{Sym}(Y)$  a map. The  $T$ -generator graph  $X(Y, \sigma, T)$  is the multigraph on the vertex set  $Y$  with one edge  $\{y, \sigma(t)y\}$  for each pair  $(y, t) \in Y \times T$  with  $\sigma(t)y \neq y$ ; parallel edges are retained, fixed points contribute no edge, and every vertex has degree at most  $2|T|$ . An *arc* of  $t \in T$  is an ordered pair  $(y, \sigma(t)y)$ .

*Conventions.* All graphs in this paper are finite multigraphs without loops, and all edge counts—degrees, boundaries, edit distances, and sums over  $E(X)$ —count edges with multiplicity. For multigraphs  $X, X'$  on the same vertex set  $Y$ , the *edit distance*  $\text{ed}(X, X')$  is the minimal number of edge insertions and deletions, with multiplicity, transforming  $X$  into  $X'$ . A sequence of graphs  $X_n$  on  $Y_n$  is *edit-equivalent* to a sequence  $X'_n$  if  $\text{ed}(X_n, X'_n) = o(|Y_n|)$ .

For a finite multigraph  $X$  and  $\emptyset \neq U \subseteq V(X)$ , write  $\partial U$  for the multiset of edges with exactly one endpoint in  $U$ , and

$$h(X) = \min \left\{ \frac{|\partial U|}{|U|} \mid \emptyset \neq U \subseteq V(X), |U| \leq \frac{1}{2}|V(X)| \right\}$$

for the Cheeger constant; for a one-vertex graph the minimum runs over the empty set, and we set  $h(X) = +\infty$ . A sequence of bounded-degree graphs  $X_n$  with  $|V(X_n)| \rightarrow \infty$  is an *expander sequence* if  $\inf_n h(X_n) > 0$ .

**Remark 2.4** (Robustness of the conventions). The conventions are robust in exactly the two directions in which they interface with the cited external theorems. Loops are immaterial: a loop lies in no boundary  $\partial U$ , so inserting or deleting loops changes no Cheeger constant, no component, and no estimate below; arcs with  $\sigma(t)y = y$  are accordingly discarded once and for all in the definition above. Multiplicities are controlled by a uniform bound: in a  $T$ -generator graph every unordered pair carries at most  $2|T|$  parallel edges, one per arc. For multigraphs of multiplicity at most  $M$ : forgetting multiplicities preserves components and connectivity; boundary sizes change

Sofic/Sofic

word\_close

Matching/BlockWordCrossin

g

wordDisagreement\_negligible

all\_wordCrossing\_negligible

by a factor between 1 and  $M$ , so a componentwise Cheeger bound  $h$  in one convention yields at least  $h/M$  in the other; and edit distances transfer with the same factor, one direction at a time: forgetting multiplicities is 1-Lipschitz for edit distance, since each multigraph insertion or deletion changes the underlying simple graph by at most one edge; and conversely, from a multigraph  $X$  of multiplicity at most  $M$  and a simple-graph edit path of length  $e$  from the simplification of  $X$ , one obtains a multigraph  $X'$  lifting its endpoint—keep the multiplicities of retained edges, delete all copies of deleted edges, insert new edges with multiplicity one—at multigraph edit distance at most  $Me$  from  $X$ . Hence  $o(|Y_n|)$  edit-distance statements transfer in both directions against these chosen lifts. We do not assert any comparison for arbitrary multigraphs—two multigraphs can share an underlying simple graph and differ by linearly many parallel copies; the transfer above uses the multiplicity bound, which holds for every graph in this paper. Consequently every statement below, and each of the two external theorems of Section 2.5, may be read in either convention with constants adjusted; we fix the multiplicity-counting convention throughout.

**Definition 2.5.** Let  $(X_n)$  be a sequence of graphs of uniformly bounded degree on vertex sets  $Y_n$  with  $|Y_n| \rightarrow \infty$ . An *expander decomposition* of  $(X_n)$  consists of graphs  $\tilde{X}_n$  on  $Y_n$ , of uniformly bounded degree, with  $\text{ed}(X_n, \tilde{X}_n) = o(|Y_n|)$ , together with a constant  $h > 0$  such that every connected component of every  $\tilde{X}_n$  has Cheeger constant at least  $h$ .

**Definition 2.6.** A sequence of bounded-degree graphs  $(X_n)$  is an *essential expander sequence* if it is edit-equivalent to an expander sequence.

**2.3. Expansion and an  $\ell^1$  co-area estimate.** The single analytic inequality used in this paper is the following co-area estimate.

**Lemma 2.7** ( $\ell^1$  co-area estimate). *Let  $X$  be a finite connected graph and  $f : V(X) \rightarrow \mathbb{R}$ . Let  $c \in \mathbb{R}$  be such that*

$$|\{x \mid f(x) > c\}| \leq \frac{1}{2}|V(X)| \quad \text{and} \quad |\{x \mid f(x) < c\}| \leq \frac{1}{2}|V(X)|.$$

*Then*

$$\sum_{x \in V(X)} |f(x) - c| \leq \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f(x) - f(y)|.$$

*Proof.* If  $X$  has a single vertex, the median hypotheses force  $f \equiv c$  and both sides vanish (with  $1/h(X) = 0$  by convention); assume  $|V(X)| \geq 2$ . Replacing  $f$  by  $f - c$  we may assume  $c = 0$ . Write  $f = f_+ - f_-$  with  $f_{\pm} \geq 0$ . For  $t > 0$ , the level set  $U_t = \{x \mid f_+(x) > t\}$  satisfies  $|U_t| \leq \frac{1}{2}|V(X)|$ , so  $|\partial U_t| \geq h(X)|U_t|$  whenever  $U_t \neq \emptyset$ . By the layer-cake formula,

$$\sum_x f_+(x) = \int_0^\infty |U_t| dt \leq \frac{1}{h(X)} \int_0^\infty |\partial U_t| dt \leq \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f_+(x) - f_+(y)|,$$

since an edge  $\{x, y\}$  lies in  $\partial U_t$  exactly for  $t$  in an interval of length  $|f_+(x) - f_+(y)|$ . The same bound holds for  $f_-$ , and  $|f_+(x) - f_+(y)| + |f_-(x) - f_-(y)| \leq$

Criterion/Criterion  
 ExpanderDecomposition  
 The Lean structure is  
 stronger than the printed  
 definition: it also records the  
 block structure, an edge-edit  
 witness, and the almost  
 invariance (6), supplied by  
 Theorem 2.9 and consumed  
 in Section 4.  
 Criterion/Criterion  
 IsEssentialExpanderSequence  
 MatchingCertificate  
 isEssentialExpanderSequence  
 Matching/EssentialExpander  
 Repair  
 induced\_expands\_eventually  
 repairedActualGraph\_expands\_  
 eventually  
 The matching certificate is  
 proved to exhibit one, and  
 the repair supplies its  
 expander sequence. The Lean  
 predicate carries both printed  
 degree bounds; the  
 generator-graph side is  
 discharged by generatorGraph\_  
 hasDegreeBound with the value  
 $2|S_K|$ .  
 Matching/FiniteGraph  
 coarea  
 coarea\_mul  
 Matching/DirectedCoarea  
 nonnegative\_coarea  
 The Lean statement holds for  
 every finite multigraph: the  
 proof uses only the Cheeger  
 lower bound.

$|f(x) - f(y)|$  pointwise on edges, because  $f_+$  and  $f_-$  have disjoint supports. Adding the two estimates gives the claim.  $\square$

**2.4. LEF groups.** A group  $J$  is *LEF* (locally embeddable into finite groups) [20] if for every finite subset  $F \subseteq J$  there are a finite group  $P$  and a map  $\varphi : F \rightarrow P$  that is injective and satisfies  $\varphi(g)\varphi(h) = \varphi(gh)$  whenever  $g, h, gh \in F$ . Every LEF group is sofic; finite and residually finite groups are LEF.

**Lemma 2.8.** *A finitely presented LEF group is residually finite.*

*Proof.* Let  $J = \langle T \mid W \rangle$  with  $T, W$  finite and  $T$  symmetric, and let  $g \in J \setminus \{1\}$ ; fix a word  $w_g$  in  $T$  representing  $g$ . Let  $F \subseteq J$  be the finite set consisting of 1, the elements of  $T$ , all partial products  $t_1 \cdots t_i$  of every relator  $t_1 \cdots t_\ell \in W$ , and all partial products of  $w_g$ ; in particular  $g \in F$ . Let  $\varphi : F \rightarrow P$  be a local embedding into a finite group. First,  $\varphi(1)\varphi(1) = \varphi(1)$  gives  $\varphi(1) = 1_P$ . Next, the assignment  $t \mapsto \varphi(t)$  kills every relator: its partial products  $p_i = t_1 \cdots t_i$  lie in  $F$  along with the letters, so  $\varphi(p_{i+1}) = \varphi(p_i)\varphi(t_{i+1})$  by multiplicativity, and induction gives  $\varphi(t_1) \cdots \varphi(t_\ell) = \varphi(p_\ell) = \varphi(1) = 1_P$ . Hence  $t \mapsto \varphi(t)$  extends to a homomorphism  $\psi : J \rightarrow P$ . The same induction along the partial products of  $w_g$ , all of which lie in  $F$ , gives  $\psi(g) = \varphi(g)$ , and  $\varphi(g) \neq \varphi(1) = 1_P$  by injectivity of  $\varphi$ . Thus  $g$  survives in a finite quotient, and  $J$  is residually finite.  $\square$

**2.5. The theorems of Kun and Kun–Thom.** We use two structure theorems for sofic approximations of Kazhdan groups.

**Theorem 2.9** (Kun [13]). *Let  $\Gamma$  be an infinite finitely generated property-(T) group with a finite symmetric generating set  $S$ , and let  $\sigma_n : \Gamma \rightarrow \text{Sym}(Y_n)$  be a sofic approximation. Then the  $S$ -generator graphs  $X(Y_n, \sigma_n, S)$  admit an expander decomposition. Moreover, the modified graphs  $\tilde{X}_n$  may be chosen of degree at most  $d$ , with the Cheeger constants of all components bounded below by a single constant  $h > 0$ ; the constants  $d$  and  $h$  depend only on  $(\Gamma, S)$ .*

Kun’s theorem proves Bowen’s conjecture: sofic approximations of infinite Kazhdan groups are essentially disjoint unions of expanders. Theorem 2.9 is deliberately a *one-way* statement: the four-way equivalence of [13, Theorem 3] contains a spectral implication that fails as written and is not imported anywhere in this paper; Remark 2.11 records this precisely, together with two local repairs to the published proofs of the theorems of this subsection. The companion result of Kun and Thom is an obstruction for centralizers.

**Theorem 2.10** (Kun–Thom [14]). *Let  $\Lambda$  be an infinite finitely generated property-(T) group with finite symmetric generating set  $S_\Lambda$ , and let  $\Delta$  be a finitely generated group. Suppose  $\Lambda \times \Delta$  admits a sofic approximation  $\pi_n : \Lambda \times \Delta \rightarrow \text{Sym}(D_n)$  such that the  $S_\Lambda$ -generator graphs  $X(D_n, \pi_n|_\Lambda, S_\Lambda)$  form an essential expander sequence. Then  $\Delta$  is LEF.*

Sofic/FinitelyPresentedLEF  
finitelyPresented\_isLEF\_resi  
duallyFinite  
For any generator set and  
finitely many relators, with  
an explicit finite  
symmetric-group quotient.  
Kun/KunDecomposition  
exists\_expanderDecomposition  
Kun/KunFixedDecomposition  
expanderDecomposition  
Proved internally, in the  
full-sequence one-way form  
used here. The Lean  
statement normalizes the  
generating set to contain 1  
and, like the printed  
statement, assumes  $\Gamma$   
infinite.  
KunThom/KunThomTheore  
m  
isLEF\_of\_exactProductExpansi  
on  
KunThom/KunThomEssentia  
l  
isLEF\_of\_matchingCertificate  
Proved internally, with the  
matching-core repair; the  
generating set is normalized  
to contain 1, as in  
Theorem 2.9.

Theorem 2.10 states the theorem of [14] in its *essential* form, which is exactly how it is invoked in Section 4. The statement in [14] assumes that the  $\Lambda$ -graphs form a genuine expander sequence; the essential form is nonetheless the published theorem’s: the published proof opens under the weaker hypothesis that the sequence is essentially expanding and is carried out under that hypothesis throughout. The heuristic content: a genuine expander is essentially connected, so the commuting factor  $\Delta$  cannot shuffle components and is forced to act almost by automorphisms of finite structures, which yields LEF approximations.

*Interface.* The proof in [14] is carried out for *regularly labeled* graphs: each generator label has exactly one outgoing directed edge at every vertex. An approximation normalized as in Lemma 2.2 supplies this structure. Every generator acts by a genuine permutation, and an inverse pair of generators acts by an inverse pair of permutations, so the two directed arcs of a pair combine into one undirected labeled edge; identity loops are discarded by the conventions of Section 2.2. Distinct labels rarely collide: for generators  $g, h$  with  $h \neq g^{\pm 1}$ , a collision  $\sigma_n(g)y = \sigma_n(h)y$  makes  $y$  a preimage under the permutation  $\sigma_n(h)$  of a fixed point of  $\sigma_n(h)^{-1}\sigma_n(g)$ , which agrees with  $\sigma_n(h^{-1}g)$  outside a vanishing fraction of points by approximate multiplicativity and has a vanishing fraction of fixed points by asymptotic faithfulness. These properties are verified for the localized approximation at the end of the proof of Theorem 4.1.

**Remark 2.11** (Audit of the two cited proofs). Because the two theorems above carry the entire approximation-theoretic weight of this paper, their published proofs were audited line by line, with three findings; we record them here both to delimit exactly what is imported and to credit the arguments of [13, 14], which survive intact.

First, Theorem 3 of [13] is stated as a four-way equivalence, and its spectral implication (4)  $\Rightarrow$  (1) fails as written: it invokes the Cheeger–Alon–Milman bound, which separates the nonconstant Markov spectrum from  $+1$  but not from  $-1$ . The counterexample is explicit. In  $K_{n,n}$  with sides  $\{l_i\}$ ,  $\{r_i\}$ , replace the two edges  $\{l_0, r_0\}$ ,  $\{l_1, r_1\}$  by  $\{l_0, l_1\}$ ,  $\{r_0, r_1\}$ ; this preserves degrees, creates a triangle (so the graph is genuinely non-bipartite for  $n \geq 3$ ), and keeps  $h \geq 1$  for  $n \geq 6$ : a set  $U$  with  $2|U| \leq 2n$  and  $a, b$  vertices on the two sides has  $K_{n,n}$ -boundary  $n(a+b) - 2ab \geq n|U|/2$ , since  $4ab \leq (a+b)^2 \leq n(a+b)$ , and the switch removes at most two boundary occurrences. Meanwhile the vector equal to  $+1$  on one side and  $-1$  on the other has normalized Markov Rayleigh quotient  $-1 + 4/n^2$ , so the least eigenvalue runs to  $-1$  while the expansion stays uniform—non-bipartiteness alone does not repair the implication. Repeating each switched graph enough times gives a sequence satisfying (2)–(4) while violating (1). This does not affect the present paper, nor Theorem 1 of [13]: the decomposition theorem needs only the forward implications, and Theorem 2.9 above is exactly that one-way statement—the only form used here.

Kun/KunSpectralCounterexample  
switched  
hasCheegerLowerBound\_one  
not\_isBipartite  
rayleigh\_testVector  
no\_uniform\_spectral\_gap  
The switched family with its  
uniform Cheeger bound 1,  
non-bipartiteness, Rayleigh  
quotient  $-1 + 4/n^2$ , and the  
packaged refutation: no  
spectral gap at  $-1$  follows  
from any uniform Cheeger  
bound. The reading into the  
numbered conditions of [13,  
Theorem 3], and the  
repetition of components,  
remain prose.

Second, the proof of the forward direction in [13] normalizes a Markov defect (Lemma 10) without first excluding the degenerate case in which that defect vanishes; the repair separates the zero-defect case, where the displacement estimate holds for homogeneous reasons, before dividing.

Third, the centralizer argument of [14] constructs a correspondence between finite models and later treats it as the graph of a permutation, which it need not yet be; the repair measures the row and column fibers of the correspondence, extracts its injective matching core, and completes that core to a genuine permutation, after which the argument proceeds as published.

Both repairs are local to the proofs; a reader auditing this paper by hand should read [13, 14] with the three points above in mind.

### 3. CONSERVATIVE COMPONENT MATCHING: THE TOOLKIT

This section isolates five elementary lemmas about finite graphs, permutations, and sequences. They are stated once, in the exact quantitative forms used in Section 4; the only genuinely group-theoretic input of the paper appears there.

#### 3.1. Expander refinement.

**Lemma 3.1** (Expander refinement). *Let  $X$  be a finite graph, every connected component of which has Cheeger constant at least  $h > 0$ . Let  $V(X) = A_1 \sqcup \cdots \sqcup A_r$  be a partition, and let  $e$  be the number of edges of  $X$  whose endpoints lie in different parts. Then there is an assignment  $C \mapsto i(C)$  of a part to each component of  $X$ , with  $|C \cap A_{i(C)}|$  maximal, such that*

$$\sum_C |C \setminus A_{i(C)}| \leq \frac{4e}{h}.$$

*Proof.* Every edge of  $X$  lies inside a component, so the crossing edges distribute among the components; let  $e_C$  be the number inside  $C$ , with  $\sum_C e_C = e$ . Fix a component  $C$ , let  $a_1 \geq a_2 \geq \cdots \geq a_k$  be the sizes of its nonempty intersections with the parts, and let  $A_{i(C)}$  realize  $a_1$ . We claim

$$|C| - a_1 \leq \frac{4e_C}{h},$$

which proves the lemma upon summation. If  $C$  meets only one part, then  $C \setminus A_{i(C)} = \emptyset$  and the claim is trivial; assume  $C$  meets at least two parts. If a subset  $W \subseteq C$  is a union of part-intersections with  $|W| \leq \frac{1}{2}|C|$ , then every edge of  $\partial_{X[C]}W$  joins different parts, so  $e_C \geq |\partial_{X[C]}W| \geq h|W|$ .

If  $a_1 \geq \frac{1}{2}|C|$ , apply this to  $W = C \setminus A_{i(C)}$ , a union of part-intersections of size  $|C| - a_1 \leq \frac{1}{2}|C|$ ; this gives  $|C| - a_1 \leq e_C/h$ .

If  $a_1 < \frac{1}{2}|C|$ , accumulate part-intersections in any order until the running total first reaches at least  $\frac{1}{4}|C|$ ; the resulting union  $W$  satisfies  $\frac{1}{4}|C| \leq |W| < \frac{1}{4}|C| + a_1 < \frac{3}{4}|C|$ , so  $\min(|W|, |C \setminus W|) \geq \frac{1}{4}|C|$ , and whichever of  $W, C \setminus W$

Matching/Refinement  
exists\_dominant\_cell  
le\_crossing\_of\_cell  
Matching/ComponentRefinement  
refineComponent  
Stated per component:  
exists\_dominant\_cell is the  
assignment  $C \mapsto i(C)$  with  
its bound on one component,  
and refineComponent applies  
it componentwise, as  
Section 4 consumes it.

has size at most  $\frac{1}{2}|C|$  is again a union of part-intersections. Applying the boundary bound to it gives  $e_C \geq \frac{h}{4}|C| \geq \frac{h}{4}(|C| - a_1)$ .  $\square$

### 3.2. Permutation conservation.

**Lemma 3.2** (Permutation conservation). *Let  $Y$  be a finite set,  $\pi \in \text{Sym}(Y)$ , and  $f : Y \rightarrow \mathbb{R}$ . Then*

$$\sum_{y \in Y} (f(y) - f(\pi y))_+ = \sum_{y \in Y} (f(\pi y) - f(y))_+ = \frac{1}{2} \sum_{y \in Y} |f(\pi y) - f(y)|.$$

*Proof.* Both  $\sum_y f(y)$  and  $\sum_y f(\pi y)$  equal the same number, so  $\sum_y (f(y) - f(\pi y)) = 0$ ; split into positive and negative parts.  $\square$

This exact identity is what converts a one-sided (monotonicity) estimate along the arcs of a permutation into a two-sided  $\ell^1$  estimate.

### 3.3. Median pinning.

**Lemma 3.3** (Median pinning). *Let  $Z$  be a finite graph in which every connected component  $B$  satisfies  $h(Z[B]) \geq h > 0$ . Let  $f : V(Z) \rightarrow \mathbb{R}$ , and suppose that on every component at most half of the vertices satisfy  $f > \frac{1}{2}$  and at most half satisfy  $f < \frac{1}{2}$ . Then*

$$\sum_{x \in V(Z)} \left| f(x) - \frac{1}{2} \right| \leq \frac{1}{h} \sum_{\{x,y\} \in E(Z)} |f(x) - f(y)|.$$

*Proof.* Every edge of  $Z$  lies within a component. Apply Lemma 2.7 with  $c = \frac{1}{2}$  on each component and sum.  $\square$

### 3.4. Weighted selection.

**Lemma 3.4** (Weighted selection). *For each  $n$ , let  $\{D_{n,i}\}_{i \in I_n}$  be disjoint nonempty subsets of a finite set  $Y_n$ ,  $|Y_n| = N_n \rightarrow \infty$ , with  $\sum_{i \in I_n} |D_{n,i}| \geq (1 - o(1))N_n$ . Suppose that for every fixed  $M$ ,*

$$\sum_{i: |D_{n,i}| \leq M} |D_{n,i}| = o(N_n).$$

*Let, for each  $r \in \mathbb{N}$ , nonnegative error weights  $E_{n,i}^{(r)}$  be given with*

$$\sum_{i \in I_n} E_{n,i}^{(r)} = o(N_n) \quad \text{for every fixed } r.$$

*Then there exist indices  $i(n) \in I_n$ , for all  $n$  in a cofinite set, such that  $|D_{n,i(n)}| \rightarrow \infty$  and, for every fixed  $r$ ,*

$$E_{n,i(n)}^{(r)} = o(|D_{n,i(n)}|).$$

Matching/PermutationConser  
vation  
permutation\_conservation\_ful  
1

Matching/Pinning  
median\_pinning  
Matching/ComponentPinning  
normalized\_pinning\_global

Matching/Selection  
exists\_selection  
exists\_selection\_diverging  
diagonalLevel\_diverges  
diagonalLevel\_error  
exists\_selection\_diverging is  
the form carrying the  
divergence  $|D_{n,i(n)}| \rightarrow \infty$ ;  
exists\_selection is the same  
selection without it.

*Proof.* Replacing  $E_{n,i}^{(r)}$  by  $\max_{r' \leq r} E_{n,i}^{(r')}$ —which only increases the weights, preserves the hypothesis for each fixed  $r$  because the maximum of finitely many families, each of total weight  $o(N_n)$ , has total weight at most their sum, and only strengthens the conclusion—we may and do assume that  $E_{n,i}^{(r)}$  is nondecreasing in  $r$ .

For integers  $k \geq 1$ , call an index  $i \in I_n$   $k$ -bad if  $|D_{n,i}| \leq k$  or  $E_{n,i}^{(k)} > \frac{1}{k}|D_{n,i}|$ , and  $k$ -good otherwise; let

$$q_k(n) = \frac{1}{N_n} \sum_{i \text{ } k\text{-bad}} |D_{n,i}|.$$

For each fixed  $k$ ,

$$q_k(n) \leq \frac{1}{N_n} \left( \sum_{i: |D_{n,i}| \leq k} |D_{n,i}| + k \sum_{i \in I_n} E_{n,i}^{(k)} \right) \xrightarrow{n \rightarrow \infty} 0,$$

the second summand bounding the mass of the second kind of badness by Markov's inequality, and both summands being  $o(N_n)$  by hypothesis. Moreover the set of  $k$ -bad indices is nondecreasing in  $k$ :  $|D_{n,i}| \leq k$  implies  $|D_{n,i}| \leq k+1$ , and  $E_{n,i}^{(k)} > \frac{1}{k}|D_{n,i}|$  implies  $E_{n,i}^{(k+1)} \geq E_{n,i}^{(k)} > \frac{1}{k+1}|D_{n,i}|$ , using monotonicity in  $r$ . Hence  $q_k(n)$  is nondecreasing in  $k$  for each fixed  $n$ .

Now make one simultaneous choice. Since  $q_k(n) \rightarrow 0$  as  $n \rightarrow \infty$  for each fixed  $k$ , there are integers  $n_1 < n_2 < \dots$  with  $q_k(n) \leq \frac{1}{k}$  for all  $n \geq n_k$ . For  $n \geq n_1$  put  $k_n = \max \{k \mid n_k \leq n\}$ , a finite maximum since the  $n_k$  strictly increase; then  $k_n \rightarrow \infty$  and, by the definition of  $n_{k_n}$ ,  $q_{k_n}(n) \leq \frac{1}{k_n} \rightarrow 0$ .

For  $n$  large, the  $k_n$ -bad indices carry mass  $q_{k_n}(n)N_n \leq N_n/k_n = o(N_n)$ , while all indices together carry mass at least  $(1 - o(1))N_n$ ; hence some  $k_n$ -good index  $i(n)$  exists. Then  $|D_{n,i(n)}| > k_n \rightarrow \infty$ , and for every fixed  $r$ : as soon as  $k_n \geq r$ , monotonicity in  $r$  gives

$$E_{n,i(n)}^{(r)} \leq E_{n,i(n)}^{(k_n)} \leq \frac{1}{k_n}|D_{n,i(n)}| = o(|D_{n,i(n)}|). \quad \square$$

### 3.5. Localization by completion.

**Lemma 3.5** (Localization by completion). *Let  $H$  be a countable group and  $\sigma_n : H \rightarrow \text{Sym}(Y_n)$  maps. Suppose  $D_n \subseteq Y_n$  are subsets with  $|D_n| \rightarrow \infty$  such that, for every fixed  $g, h \in H$ :*

- (i)  $|\{y \in D_n \mid \sigma_n(g)y \notin D_n\}| = o(|D_n|)$ ;
- (ii)  $|\{y \in D_n \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y\}| = o(|D_n|)$ ;
- (iii)  $|\{y \in D_n \mid \sigma_n(g)y = y\}| = o(|D_n|)$  whenever  $g \neq 1$ .

For  $g \in H$  define  $\pi_n(g) \in \text{Sym}(D_n)$  by letting  $\pi_n(g)$  agree with  $\sigma_n(g)$  on  $\{y \in D_n \mid \sigma_n(g)y \in D_n, \sigma_n(g)y \neq \sigma_n(g)y' \forall y' \neq y \in D_n\}$  and extending the resulting partial injection of  $D_n$  into itself to a permutation of  $D_n$  arbitrarily. Then  $\pi_n : H \rightarrow \text{Sym}(D_n)$  is a sofic approximation of  $H$ . Moreover, if  $g, g^{-1} \in H$  satisfy  $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$ , the completions may be chosen with  $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ , and then

$$|\{y \in D_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D_n|).$$

Matching/Localization  
exists\_completion  
involutiveCompletion  
Criterion/LocalizedApproximation  
toSoficApproximation

*Proof.* Since  $\sigma_n(g)$  is injective on  $Y_n$ , the described set consists precisely of the points of  $D_n$  sent into  $D_n$  by  $\sigma_n(g)$ , and its complement in  $D_n$  has size  $o(|D_n|)$  by (i); the partial map is an injection, and any extension to a permutation of  $D_n$  differs from  $\sigma_n(g)$  on  $o(|D_n|)$  points of  $D_n$ . Then (ii) transfers: for fixed  $g, h$ , the points of  $D_n$  where  $\pi_n(g)\pi_n(h)y \neq \pi_n(gh)y$  are contained in the union of the  $o(|D_n|)$ -sized exceptional sets for  $g, h, gh$  (points where  $\pi \neq \sigma$ , including the  $\sigma_n(h)$ -preimage of the exceptional set of  $g$ ) and the set in (ii); similarly (iii) transfers faithfulness. When  $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$  and  $g \neq g^{-1}$ , the two partial injections attached to  $g$  and to  $g^{-1}$  are mutually inverse, so a completion of one can be transported to the other. When  $g = g^{-1}$ , so that  $\sigma_n(g)^2 = \text{id}$ , the set  $A_{n,g} = \{y \in D_n \mid \sigma_n(g)y \in D_n\}$  is  $\sigma_n(g)$ -invariant—if  $y$  and  $\sigma_n(g)y$  both lie in  $D_n$ , then so do  $\sigma_n(g)y$  and  $\sigma_n(g)^2y = y$ —so  $\sigma_n(g)$  restricts to an involution of  $A_{n,g}$ ; extending by the identity on  $D_n \setminus A_{n,g}$  yields an involutive completion  $\pi_n(g) = \pi_n(g)^{-1}$  that differs from  $\sigma_n(g)$  on  $o(|D_n|)$  points of  $D_n$  by (i).  $\square$

#### 4. THE COMPRESSION-CENTRALIZER CRITERION

This section proves the analytic core of the paper. We first state a local version that isolates the exact graph-theoretic hypotheses; property (T) of the ambient group is a sufficient condition for the second hypothesis, and the local statement records precisely what it is used for.

##### 4.1. Statement.

**Theorem 4.1** (Local compression-centralizer theorem). *Let  $G$  be a finitely generated group and let  $\Gamma \leq G$  be an infinite finitely generated property-(T) subgroup with finite symmetric generating set  $S$ . Let  $Q \subseteq G$  be a finite set with*

$$(4) \quad G = \langle \Gamma, Q \rangle \quad \text{and} \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

*Fix  $q_0 \in Q$ , put  $K = q_0\Gamma q_0^{-1}$ , and let  $J \leq \Gamma$  be finitely generated with*

$$(5) \quad [K, J] = 1, \quad K \cap J = \{1\}.$$

*Put  $A = S \cup Q \cup Q^{-1}$ . Suppose  $G$  admits a sofic approximation  $\sigma_n : G \rightarrow \text{Sym}(Y_n)$  such that*

- (i) *the  $S$ -generator graphs  $X(Y_n, \sigma_n, S)$  admit an expander decomposition, and*
- (ii) *the  $A$ -generator graphs  $X(Y_n, \sigma_n, A)$  admit an expander decomposition.*

*Then  $J$  is LEF.*

**Corollary 4.2** (Kazhdan compression-centralizer criterion; Theorem D). *In the setting of (4) and (5), suppose in addition that  $G$  and  $\Gamma$  have property (T). If  $G$  is sofic, then  $J$  is LEF. Equivalently: if  $J$  is not LEF, then  $G$  is not sofic.*

Criterion/LocalCriterion

LocalCriterionData

Criterion/CriterionAssembly

isLEF\_of\_soficApproximation

Criterion/CriterionAssembly

isLEF\_of\_isSofic

not\_isSofic\_of\_not\_isLEF

*Proof.* Suppose  $G$  is sofic and fix a sofic approximation  $\sigma_n$ . By (4) the finite symmetric set  $A = S \cup Q \cup Q^{-1}$  generates  $G$ , and  $G$  is infinite, containing the infinite subgroup  $\Gamma$ . The restriction  $\sigma_n|_\Gamma$  is a sofic approximation of the infinite Kazhdan group  $\Gamma$ , so Theorem 2.9 yields hypothesis (i); applying Theorem 2.9 to  $G$  itself with the generating set  $A$  yields hypothesis (ii). Theorem 4.1 concludes.  $\square$

**Remark 4.3.** Only the componentwise  $\ell^1$  median inequality of Lemma 3.3 is extracted from hypothesis (ii); that hypothesis may accordingly be weakened to any decomposition of the  $A$ -generator graphs, up to  $o(|Y_n|)$  edge edits, into components satisfying a uniform  $\ell^1$  Poincaré inequality around medians. This separates the approximation-theoretic input, which property (T) supplies through Kun's theorem, from the compression mechanism, which is combinatorial.

The remainder of the section proves Theorem 4.1. Write  $N_n = |Y_n|$ . All  $o(N_n)$  estimates are uniform over the fixed finite data  $(S, Q, q_0)$  and the decomposition constants; the set  $Q$  itself is never symmetrized inside the compression steps, since the compression hypothesis  $q\Gamma q^{-1} \leq \Gamma$  is assumed only for  $q \in Q$ .

**4.2. The two component partitions.** Put  $S_K = q_0 S q_0^{-1}$ , a finite symmetric generating set of  $K$ , and fix a finite symmetric generating set  $S_J$  of the finitely generated group  $J$ . By Lemma 2.2 we may and do assume the exact identities

$$\sigma_n(1) = \text{id}, \quad \sigma_n(g^{-1}) = \sigma_n(g)^{-1} \quad (g \in A \cup S_K \cup S_J),$$

after a pointwise  $o(N_n)$  modification; each relevant generator graph changes by  $o(N_n)$  edges, so hypotheses (i) and (ii) persist.

By hypothesis (i), fix an expander decomposition of the  $S$ -generator graphs: graphs  $\tilde{X}_n$  on  $Y_n$  of degree at most  $d_\Gamma$ , with

$$\text{ed}(X(Y_n, \sigma_n, S), \tilde{X}_n) = o(N_n),$$

such that every component of  $\tilde{X}_n$  has Cheeger constant at least  $h_\Gamma > 0$ . Let  $\mathcal{P}_n$  denote the partition of  $Y_n$  into the components of  $\tilde{X}_n$ , and write  $P_n(y)$  for the block containing  $y$ . We refer to the blocks of  $\mathcal{P}_n$  as  $\Gamma$ -components.

By hypothesis (ii), fix likewise an expander decomposition  $\tilde{Z}_n$  of the  $A$ -generator graphs, with all components of Cheeger constant at least  $h_G > 0$  and uniformly bounded degree; let  $\mathcal{B}_n$  be the partition of  $Y_n$  into the components of  $\tilde{Z}_n$ , with blocks  $B_n(y)$ , the  $G$ -components.

Both partitions are almost invariant in the following sense. For every  $s \in S$ ,

$$(6) \quad |\{y \in Y_n \mid P_n(\sigma_n(s)y) \neq P_n(y)\}| = o(N_n),$$

because apart from the  $o(N_n)$  edge edits, every arc  $(y, \sigma_n(s)y)$  is an edge of  $\tilde{X}_n$  or a loop, and edges of  $\tilde{X}_n$  do not cross  $\mathcal{P}_n$ . For every  $a \in A$ ,

$$(7) \quad |\{y \in Y_n \mid B_n(\sigma_n(a)y) \neq B_n(y)\}| = o(N_n),$$

for the same reason applied to  $\tilde{Z}_n$ .

Both partitions stay fixed from here on:  $\mathcal{P}_n$  and  $\mathcal{B}_n$  are the exact component partitions of  $\tilde{X}_n$  and  $\tilde{Z}_n$ , and every block keeps its connectivity and its Cheeger constant. This is deliberate—the final step of the proof selects a block of  $\mathcal{P}_n$  and uses precisely that it is a genuine component of  $\tilde{X}_n$ . The two partitions are related only pointwise, in two ways: through the almost-invariance estimates (6) and (7), and through the median normalization of Section 4.4, which compares a point with its compressor image exactly when the two lie in the same  $G$ -component; the  $o(N_n)$  points where this fails are carried as explicit exceptional sets through every estimate below. In particular, neither partition is assumed to refine the other.

**4.3. One-sided refinement along a compressor.** Fix  $q \in Q$ . The group identity  $qs = (qsq^{-1})q$  and approximate multiplicativity give, for each  $s \in S$ , the *conjugacy defect* bound

$$(8) \quad \left| \left\{ y \in Y_n \mid \sigma_n(q)\sigma_n(s)y \neq \sigma_n(qsq^{-1})\sigma_n(q)y \right\} \right| = o(N_n).$$

Indeed,  $\sigma_n(q)\sigma_n(s)$  agrees with  $\sigma_n(qs)$  outside  $o(N_n)$  points, and  $\sigma_n(qs) = \sigma_n((qsq^{-1})q)$  agrees with  $\sigma_n(qsq^{-1})\sigma_n(q)$  outside  $o(N_n)$  points, both by Definition 2.1(a); here  $qsq^{-1}$  belongs to the fixed finite set  $qSq^{-1} \subseteq \Gamma$ , by the compression hypothesis (4).

Transport the graphs  $\tilde{X}_n$  by the permutation  $\sigma_n(q)$ : the transported graph  $\sigma_n(q)\tilde{X}_n$  has vertex set  $Y_n$  and one edge  $\{\sigma_n(q)x, \sigma_n(q)y\}$  for each edge  $\{x, y\}$  of  $\tilde{X}_n$ , multiplicities preserved; its components are  $\{\sigma_n(q)C \mid C \in \mathcal{P}_n\}$ , with the same component Cheeger constants  $\geq h_\Gamma$  and degree bound  $d_\Gamma$ . By (8) and the edit bounds, apart from  $o(N_n)$  edge occurrences the edges of the transported graph are of the form  $\{\sigma_n(q)y, \sigma_n(\gamma)\sigma_n(q)y\}$  with  $\gamma \in qSq^{-1} \subseteq \Gamma$ ; for each of the finitely many  $\gamma \in qSq^{-1} \subseteq \Gamma$ , fix a word in  $S$  representing it; by Lemma 2.3,  $\sigma_n(\gamma)$  agrees with the product of the letter permutations outside  $o(N_n)$  points, and by (6) applied to each letter in turn, that product preserves the block  $P_n(\cdot)$  outside  $o(N_n)$  points. Hence all but  $o(N_n)$  of the transported edges lie inside blocks of  $\mathcal{P}_n$ .

Apply Lemma 3.1 to the transported graph with the partition  $\mathcal{P}_n$ . This produces a map

$$\Phi_{q,n} : \mathcal{P}_n \rightarrow \mathcal{P}_n, \quad \Phi_{q,n}(C) = \text{the block meeting } \sigma_n(q)C \text{ in maximal size,}$$

with total leakage

$$(9) \quad \sum_{C \in \mathcal{P}_n} e_{q,n}(C) = o(N_n), \quad \text{where} \quad e_{q,n}(C) := |\sigma_n(q)C \setminus \Phi_{q,n}(C)|.$$

Thus each transported  $\Gamma$ -component is almost contained in a single target  $\Gamma$ -component. The containment is a priori one-sided: a large target could still absorb several small transported components. Ruling this out is the business of the median normalization.

**4.4. Median normalization and conservation.** For  $y \in Y_n$ , let  $s_n(y) = |P_n(y)|$  be the size of its  $\Gamma$ -component. For a  $G$ -component  $B \in \mathcal{B}_n$ , let  $m_n(B) \geq 1$  be a *vertex-weighted median* of  $s_n$  on  $B$ : a value such that

$$|\{y \in B \mid s_n(y) > m_n(B)\}| \leq \frac{1}{2}|B| \text{ and } |\{y \in B \mid s_n(y) < m_n(B)\}| \leq \frac{1}{2}|B|.$$

Define the *median normalization*

$$(10) \quad f_n(y) = \frac{s_n(y)}{s_n(y) + m_n(B_n(y))} \in (0, 1).$$

Since  $t \mapsto t/(t+m)$  is strictly increasing, on every  $G$ -component  $B$  at most half the vertices satisfy  $f_n > \frac{1}{2}$  and at most half satisfy  $f_n < \frac{1}{2}$ ; that is,  $\frac{1}{2}$  is a median of  $f_n$  on every component of  $\tilde{Z}_n$ .

*Variation along  $S$ .* For  $s \in S$ ,

$$(11) \quad \sum_{y \in Y_n} |f_n(\sigma_n(s)y) - f_n(y)| = o(N_n),$$

since the summand vanishes unless  $P_n(\sigma_n(s)y) \neq P_n(y)$  or  $B_n(\sigma_n(s)y) \neq B_n(y)$ , a set of size  $o(N_n)$  by (6) and (7), and  $0 < f_n < 1$ .

*One-sided variation along  $q \in Q$ .* Call  $y \in Y_n$  a *good point* for  $q$  if

$$(12) \quad \sigma_n(q)y \in \Phi_{q,n}(P_n(y)) \quad \text{and} \quad B_n(\sigma_n(q)y) = B_n(y).$$

Let  $y$  be good,  $C = P_n(y)$ ,  $D = \Phi_{q,n}(C)$ , and  $e = e_{q,n}(C)$ ; then

$$s_n(\sigma_n(q)y) = |D| \geq |\sigma_n(q)C \cap D| = |C| - e = s_n(y) - e.$$

Since both  $y$  and  $\sigma_n(q)y$  lie in the same  $G$ -component  $B$ , they are normalized by the same median  $m = m_n(B) \geq 1$ . For  $a \geq (1-\delta)a' \geq 0$  and  $m \geq 1$ ,

$$(13) \quad \frac{a'}{a' + m} - \frac{a}{a + m} \leq \frac{a'}{a' + m} - \frac{(1-\delta)a'}{(1-\delta)a' + m} \leq \delta,$$

where the first inequality is monotonicity of  $t \mapsto t/(t+m)$  and the second is a direct computation: the difference equals  $\delta a' m / ((a' + m)((1-\delta)a' + m)) \leq \delta$ . Applying (13) with  $a' = |C|$ ,  $a = |D| \geq |C| - e$ ,  $\delta = e/|C|$  gives, for every good point  $y$ ,

$$(f_n(y) - f_n(\sigma_n(q)y))_+ \leq \frac{e_{q,n}(P_n(y))}{|P_n(y)|}.$$

Summing over  $Y_n$ —good points of a block  $C$  contribute at most  $e_{q,n}(C)$  in total, points with  $\sigma_n(q)y \notin \Phi_{q,n}(P_n(y))$  number  $e_{q,n}(C)$  per block and contribute at most 1 each, and points whose arc crosses  $\mathcal{B}_n$  contribute at most 1 each—we obtain

$$(14) \quad \begin{aligned} \sum_{y \in Y_n} (f_n(y) - f_n(\sigma_n(q)y))_+ &\leq 2 \sum_{C \in \mathcal{P}_n} e_{q,n}(C) + |\{y \mid B_n(\sigma_n(q)y) \neq B_n(y)\}| \\ &= o(N_n) \end{aligned}$$

by (9) and (7). The compressed image can only grow—but growth is globally impossible:

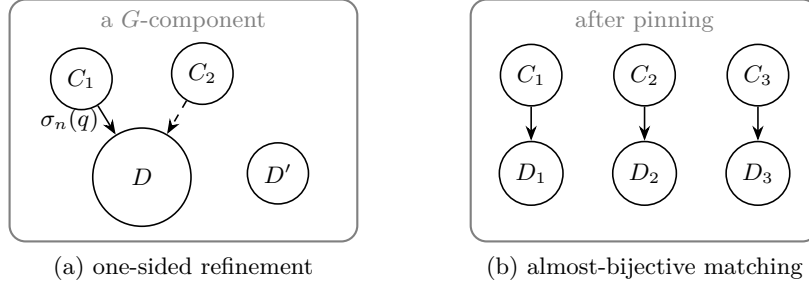


FIGURE 2. Conservative component matching. (a) Transporting a  $\Gamma$ -component by a compressor almost lands it inside a single target component, but a priori a large target could absorb several small sources (dashed). (b) Exact conservation and median pinning force transported and target sizes to agree asymptotically, so the assignment  $\Phi_{q,n}$  becomes injective on almost all mass (Proposition 4.4).

*Conservation.* Since  $\sigma_n(q)$  is a permutation of  $Y_n$ , Lemma 3.2 upgrades (14) to the two-sided estimate

$$(15) \quad \sum_{y \in Y_n} |f_n(\sigma_n(q)y) - f_n(y)| = 2 \sum_{y \in Y_n} (f_n(y) - f_n(\sigma_n(q)y))_+ = o(N_n) \quad (q \in Q),$$

and the change of variables  $y = \sigma_n(q)x$ , legitimate because  $\sigma_n(q^{-1}) = \sigma_n(q)^{-1}$ , extends (15) to all  $q \in Q \cup Q^{-1}$ .

*Pinning.* Every edge of the  $A$ -generator graph  $X(Y_n, \sigma_n, A)$  arises from an arc  $(y, \sigma_n(a)y)$  with  $a \in A = S \cup Q \cup Q^{-1}$ , and each edge arises from at most  $2|A|$  arcs. Hence (11) and (15) give

$$\sum_{\{x,y\} \in E(X(Y_n, \sigma_n, A))} |f_n(x) - f_n(y)| = o(N_n).$$

Since  $0 < f_n < 1$ , editing  $o(N_n)$  edges changes this sum by at most  $o(N_n)$ ; therefore the same bound holds over  $E(\tilde{Z}_n)$ . As  $\frac{1}{2}$  is a median of  $f_n$  on every component of  $\tilde{Z}_n$  and all component Cheeger constants are at least  $h_G$ , Lemma 3.3 yields the concentration estimate

$$(16) \quad \sum_{y \in Y_n} \left| f_n(y) - \frac{1}{2} \right| = o(N_n).$$

In words: after median normalization, almost every  $\Gamma$ -component has size asymptotically equal to the median size on its  $G$ -component, because  $s/(s+m) \rightarrow \frac{1}{2}$  exactly when  $s/m \rightarrow 1$ .

**4.5. Almost-bijective matching.** The mechanism of this subsection is drawn in Figure 2.

Matching/MedianNormalizati  
on  
medianNormalize  
Matching/MatchingPreparati  
on  
acceptable\_symmDiff\_le  
Criterion/Criterion  
symmDiff\_le\_of\_pinned  
matching\_injective  
Stated per component:  
acceptable\_symmDiff\_le is  
(17) for a single acceptable  
component, and  
matching\_injective is the  
injectivity clause.

**Proposition 4.4** (Matching). *Fix  $q \in Q$ . There are subcollections  $\mathcal{G}_{q,n} \subseteq \mathcal{P}_n$  such that:*

- (i)  $\sum_{C \in \mathcal{P}_n \setminus \mathcal{G}_{q,n}} |C| = o(N_n)$ ;
- (ii) for every  $C \in \mathcal{G}_{q,n}$ , with  $D = \Phi_{q,n}(C)$ ,
- (17)  $|\sigma_n(q)C \triangle D| = o(1) \cdot |C|$  uniformly over  $C \in \mathcal{G}_{q,n}$ ;
- (iii) for  $n$  large,  $\Phi_{q,n}$  is injective on  $\mathcal{G}_{q,n}$ ;
- (iv)  $\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1 - o(1)) N_n$ .

*Proof.* Choose  $\eta_n \downarrow 0$  so slowly that the three global errors—in (7) for the element  $q$ , in (9), and in (16)—are all  $o(\eta_n^2 N_n)$ . Let  $W_n \subseteq Y_n$  be the union of the four sets

$$\left\{ y \mid \left| f_n(y) - \frac{1}{2} \right| > \eta_n \right\}, \quad \sigma_n(q)^{-1} \left\{ y \mid \left| f_n(y) - \frac{1}{2} \right| > \eta_n \right\}, \\ \{ y \mid B_n(\sigma_n(q)y) \neq B_n(y) \}, \quad \{ y \mid \sigma_n(q)y \notin \Phi_{q,n}(P_n(y)) \}.$$

Markov's inequality applied to (16) bounds the first set by  $\eta_n^{-1} \cdot o(\eta_n^2 N_n) = o(\eta_n N_n)$ ; the second is a bijective preimage of the first; the last two have sizes  $o(\eta_n^2 N_n)$  by the choice of  $\eta_n$ . Hence  $|W_n| = o(\eta_n N_n)$ .

Discard from  $\mathcal{P}_n$  every block  $C$  with  $e_{q,n}(C) > \eta_n |C|$  or  $|C \cap W_n| > \eta_n |C|$ , and call the remaining collection  $\mathcal{G}_{q,n}$ . The discarded mass is at most  $\eta_n^{-1} \sum_C e_{q,n}(C) + \eta_n^{-1} |W_n| = o(N_n)$ , proving (i).

Let  $C \in \mathcal{G}_{q,n}$  and pick  $y \in C \setminus W_n$ , possible since  $\eta_n < 1$ . Put  $D = \Phi_{q,n}(C)$  and  $B = B_n(y) = B_n(\sigma_n(q)y)$ , and write  $m = m_n(B)$ . Since  $y \notin W_n$ , we have  $\sigma_n(q)y \in D$ , so  $s_n(\sigma_n(q)y) = |D|$ , and

$$\left| \frac{|C|}{|C| + m} - \frac{1}{2} \right| \leq \eta_n, \quad \left| \frac{|D|}{|D| + m} - \frac{1}{2} \right| \leq \eta_n.$$

Solving  $\left| \frac{a}{a+m} - \frac{1}{2} \right| \leq \eta$  gives  $\frac{1-2\eta}{1+2\eta} m \leq a \leq \frac{1+2\eta}{1-2\eta} m$ ; applying this to  $a = |C|$  and to  $a = |D|$  and dividing,

$$(18) \quad |D| = (1 + O(\eta_n)) |C| \quad \text{uniformly over } C \in \mathcal{G}_{q,n}.$$

With  $e = e_{q,n}(C) \leq \eta_n |C|$ ,

$$\begin{aligned} |\sigma_n(q)C \triangle D| &= |\sigma_n(q)C \setminus D| + |D \setminus \sigma_n(q)C| \\ &\leq e + (|D| - |C| + e) \leq 2e + ||D| - |C|| = O(\eta_n) |C|, \end{aligned}$$

proving (ii).

For (iii): if  $C \neq C'$  in  $\mathcal{G}_{q,n}$  had  $\Phi_{q,n}(C) = \Phi_{q,n}(C') = D$ , the disjoint sets  $\sigma_n(q)C \cap D$  and  $\sigma_n(q)C' \cap D$  would, by (17) and (18), each occupy a fraction  $1 - o(1)$  of  $D$ , uniformly—impossible for  $n$  large.

For (iv): by (i), (ii), and (18),

$$\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1 + o(1)) \sum_{C \in \mathcal{G}_{q,n}} |C| = (1 + o(1))(1 - o(1)) N_n = (1 - o(1)) N_n.$$

□

Thus one-sided refinement, conservation, and pinning together force the compressor to permute the  $\Gamma$ -components almost bijectively, with asymptotically equal sizes. One more observation is needed before selecting a single matched pair.

**Lemma 4.5.** *For every fixed  $M$ ,  $\sum \{|C| : C \in \mathcal{P}_n, |C| \leq M\} = o(N_n)$ .*

*Proof.* Let  $g_1, \dots, g_{M+1} \in \Gamma$  be distinct elements; they exist since  $\Gamma$  is infinite. For all but  $o(N_n)$  points  $y$  the points  $\sigma_n(g_1)y, \dots, \sigma_n(g_{M+1})y$  are pairwise distinct: for  $i \neq j$ ,  $\sigma_n(g_i)$  agrees with  $\sigma_n(g_i g_j^{-1})\sigma_n(g_j)$  outside  $o(N_n)$  points by Definition 2.1(a), and  $\sigma_n(g_i g_j^{-1})$  moves  $1 - o(1)$  of all points by Definition 2.1(b). Moreover, fixing a word in  $S$  representing each  $g_i$ , Lemma 2.3 together with (6) applied along its letters shows that for all but  $o(N_n)$  points  $y$  all these points lie in  $P_n(y)$ . A point  $y$  satisfying both conditions has  $|P_n(y)| \geq M + 1$ ; hence the mass of blocks of size at most  $M$  is  $o(N_n)$ .  $\square$

#### 4.6. Selection, localization, and the proof of Theorem 4.1.

*Proof of Theorem 4.1.* Apply Proposition 4.4 with  $q = q_0$ . For  $n$  large this gives a collection of *matched pairs*

$$(19) \quad \{(C, D) : C \in \mathcal{G}_{q_0, n}, D = \Phi_{q_0, n}(C)\},$$

with pairwise distinct sources, pairwise distinct targets, target mass  $(1 - o(1))N_n$ , and  $|\sigma_n(q_0)C \triangle D| = o(1)|C|$  uniformly.

Fix an exhaustion  $L_1 \subseteq L_2 \subseteq \dots$  of the countable group  $KJ \leq \Gamma$  by finite subsets with  $\bigcup_r L_r = KJ$ , each  $L_r$  containing 1, the finite sets  $S_K$  and  $S_J$ , and closed under inverses, and such that  $L_r \cdot L_r \subseteq L_{r+1}$ . To each matched pair  $(C, D)$  and each  $r$  attach the error weight  $E^{(r)}(C, D)$ , the sum of:

- (a)  $|\sigma_n(q_0)C \triangle D|$ ;
- (b) the number of edges of  $\tilde{X}_n$  with an endpoint in  $C$  or  $D$  that are edited relative to the  $S$ -generator graph, plus the number of edges of the transported graph  $\sigma_n(q_0)\tilde{X}_n$  with an endpoint in  $D \cup \sigma_n(q_0)C$  that are edited relative to the transported  $S$ -generator graph;
- (c)  $|\{y \in C \mid \sigma_n(q_0)\sigma_n(s)y \neq \sigma_n(q_0 s q_0^{-1})\sigma_n(q_0)y \text{ for some } s \in S\}|$ ;
- (d)  $|\{y \in D \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y \text{ for some } g, h \in L_r\}|$ ;
- (e)  $|\{y \in D \mid \sigma_n(g)y = y \text{ for some } g \in L_r \setminus \{1\}\}|$ ;
- (f)  $|\{y \in D \mid P_n(\sigma_n(g)y) \neq P_n(y) \text{ for some } g \in L_r\}|$ .

For every fixed  $r$ , the sum of  $E^{(r)}$  over the matched pairs is  $o(N_n)$ : (a) by Proposition 4.4(ii) and (i); (b): the sources are pairwise disjoint, the targets are pairwise disjoint, and a fixed block can occur at most once as a source and at most once as a target; hence each endpoint of an edge lies in at most one source and at most one target, and, an edge having two endpoints, each edited edge is counted for at most four matched pairs—likewise in the transported graph, where the sets  $\sigma_n(q_0)C$  are pairwise disjoint images of the disjoint sources—while the global edit counts are  $o(N_n)$ ; (c) by (8) over

Matching/ComponentDivergence

SoficApproximation.smallBlock

kVertices\_negligible

ExpanderDecomposition.smallBlock

lockVertices\_negligible

the finite set  $S$ , the sources being disjoint; (d) and (e) by softicity of  $\sigma_n$  over the finite set  $L_r \cup L_r^2$ , the targets being disjoint; (f) by Lemma 2.3 together with (6) applied along the letters of a fixed word in  $S$  representing each  $g \in L_r \subseteq \Gamma$ , the targets being disjoint.

The targets  $D$  are pairwise disjoint blocks of  $\mathcal{P}_n$  with total mass  $(1 - o(1))N_n$ , and by Lemma 4.5 the total mass of targets of size at most  $M$  is  $o(N_n)$  for every fixed  $M$ . Lemma 3.4 therefore produces matched pairs  $(C_n, D_n)$  with

$$(20) \quad |D_n| \rightarrow \infty, \quad E^{(r)}(C_n, D_n) = o(|D_n|) \quad \text{for every fixed } r.$$

Put  $U_n = \sigma_n(q_0)C_n$ , so  $|U_n \triangle D_n| = o(|D_n|)$  and  $|U_n| = |C_n| = (1 + o(1))|D_n|$ . Define

$$(21) \quad D'_n = U_n,$$

a set of size  $|C_n| \rightarrow \infty$  that agrees with  $D_n$  up to  $o(|D_n|)$  points. Every error weight in (20) transfers from the pair  $(C_n, D_n)$  to  $D'_n$  at a cost of  $o(|D'_n|)$ , by separate accounting: the point-counted errors (d), (e), (f) are counted over  $D_n$ , and replacing  $D_n$  by  $D'_n$  changes each count by at most  $|D_n \triangle D'_n| = o(|D'_n|)$ ; the error (c) is counted over  $C_n$ , which is unchanged; the edge-counted errors (b) concern graphs of degree at most  $\max(d_\Gamma, 2|S|)$ , so replacing  $D_n$  by  $D'_n$  in the incidence condition changes those counts by at most that degree bound times  $|D_n \triangle D'_n| = o(|D'_n|)$ ; and the quantity (a) does not change.

*Localization.* Consider the countable group  $H = KJ \leq \Gamma$ . By (5),  $K$  and  $J$  commute and intersect trivially, so

$$H = KJ \cong K \times J.$$

For every fixed  $g \in H$ , the transferred error (f) shows that  $\sigma_n(g)$  maps all but  $o(|D'_n|)$  points of  $D'_n$  into  $D_n$ . Injectivity upgrades  $D_n$  to  $D'_n$ : since  $\sigma_n(g)$  is a permutation of  $Y_n$ , at most  $|D_n \setminus D'_n| = o(|D'_n|)$  points of  $D'_n$  can land in  $D_n \setminus D'_n$ , so  $\sigma_n(g)$  maps all but  $o(|D'_n|)$  points of  $D'_n$  into  $D'_n$  itself. The transferred errors (d) and (e) give multiplicativity and faithfulness on  $D'_n$  for every fixed pair  $g, h \in H$ , by choosing  $r$  with  $g, h \in L_r$ . Lemma 3.5, with completions on the symmetric set  $S_K \cup S_J$  chosen compatibly with the exact inverse identities of Section 4.2, yields a softic approximation

$$\pi_n : K \times J \longrightarrow \text{Sym}(D'_n)$$

with  $|\{y \in D'_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D'_n|)$  for every fixed  $g$ , and  $\pi_n(g^{-1}) = \pi_n(g)^{-1}$  for  $g \in S_K \cup S_J$ .

*The  $K$ -edges expand.* Restrict  $\tilde{X}_n$  to  $C_n$  and transport by  $\sigma_n(q_0)$ : the resulting graph  $\Xi_n$  on  $D'_n = \sigma_n(q_0)C_n$  is connected with Cheeger constant at least  $h_\Gamma$  and degree at most  $d_\Gamma$ , because  $C_n$  is a component of  $\tilde{X}_n$ . We claim that  $\Xi_n$  and the  $S_K$ -generator graph  $X(D'_n, \pi_n, S_K)$  differ by  $o(|D'_n|)$  edges. Indeed, an edge of either graph that fails to be an edge of the other must involve one of the following, each affecting  $o(|D'_n|)$  edge occurrences by (20):

- (1) an edge of  $\tilde{X}_n[C_n]$  edited relative to the  $S$ -generator graph on  $C_n$ , or conversely (error (b));
- (2) an arc  $(y, \sigma_n(s)y)$ ,  $y \in C_n$ , whose  $\sigma_n(q_0)$ -transport disagrees with the  $\sigma_n(q_0sq_0^{-1})$ -arc at  $\sigma_n(q_0)y$  (error (c));
- (3) a point of  $D'_n$  where  $\pi_n(g) \neq \sigma_n(g)$  for some  $g \in S_K$  (localization defect);
- (4) an arc of some  $g \in S_K$  leaving  $D'_n$  (transferred error (f) and the symmetric difference (a)).

Hence the  $S_K$ -generator graphs of  $\pi_n|_K$  form an essential expander sequence in the sense of Definition 2.6.

*Conclusion.* The group  $K = q_0\Gamma q_0^{-1} \cong \Gamma$  is an infinite finitely generated property-(T) group with symmetric generating set  $S_K$ , the group  $J$  is finitely generated, and  $\pi_n$  is a sofic approximation of  $K \times J$  whose  $S_K$ -generator graphs form an essential expander sequence. The regular-labeling requirements recorded after Theorem 2.10 hold for  $\pi_n$  as well: each  $\pi_n(g)$  is a genuine permutation of  $D'_n$ , with  $\pi_n(g^{-1}) = \pi_n(g)^{-1}$  for  $g \in S_K \cup S_J$ ; and for  $g, h \in S_K \cup S_J$  with  $h \neq g^{\pm 1}$ , the collision set  $\{y \in D'_n \mid \pi_n(g)y = \pi_n(h)y\}$  has size  $o(|D'_n|)$ , by the transferred errors (d) and (e) applied to the pair  $(h^{-1}, g)$  and the element  $h^{-1}g \in L_2 \setminus \{1\}$ , together with the  $o(|D'_n|)$  agreement between  $\pi_n$  and  $\sigma_n$  on  $S_K \cup S_J \cup \{h^{-1}g\}$ . Theorem 2.10 applies and shows that  $J$  is LEF.  $\square$

#### 4.7. Two remarks on the mechanism.

**Remark 4.6** (The product obstruction). The compression hypothesis cannot be deleted from Theorem D. For a direct product  $\Lambda \times \Delta$  of an infinite Kazhdan group with a residually finite Kazhdan group, product sofic approximations exist whenever  $\Lambda$  is sofic, and in them the  $\Delta$ -factor genuinely permutes the  $\Lambda$ -components: the  $\Lambda$ -component partition has all blocks of equal size, and no single component is almost invariant under the second factor. The role of the compressors  $q \in Q$ , through (4), is to generate the whole ambient group while conjugating  $\Gamma$  into itself, so that the ambient expansion of hypothesis (ii) couples the component structure of  $\Gamma$  to itself across all of  $G$  and forbids exactly this shuffling.

**Remark 4.7** (The hypothesis  $J \leq \Gamma$  is essential). The subgroup  $J$  is required to lie in  $\Gamma$ , and this is used twice: through (6) in error (f), which keeps the  $J$ -arcs inside the matched component, and through the localization step, which restricts  $\sigma_n$  on elements of  $KJ \leq \Gamma$ . The hypothesis cannot be dropped in favor of  $J \leq G$ . Indeed, groups satisfying all the remaining hypotheses with a non-LEF subgroup  $J \leq G$  commuting with  $K$  exist in abundance among sofic groups—already  $J = \text{BS}(2, 3) = \langle a, t \mid ta^2t^{-1} = a^3 \rangle$  is a finitely presented group that is known to be sofic and is non-Hopfian [4], therefore not residually finite, therefore not LEF by Lemma 2.8. The criterion is calibrated exactly so that the commuting subgroup rides inside

the compressed expander, and this is what the Leavitt corner construction of Sections 6–7 arranges algebraically.

## 5. LEAVITT ALGEBRAS, ELEMENTARY GROUPS, AND THOMPSON’S GROUP $V$

Throughout,  $k$  is a field and  $L = L_k(1, 2)$  is the binary Leavitt algebra (1), with generators  $s_0, s_1, t_0, t_1$  and relations  $t_i s_j = \delta_{ij}$ ,  $s_0 t_0 + s_1 t_1 = 1$ . Everything in this section is a consequence of the two relations alone.

**Definition 5.1.** A *binary Leavitt family* in a unital ring  $A$  is a quadruple  $(s_0, s_1, t_0, t_1)$  of elements of  $A$  satisfying

$$t_i s_j = \delta_{ij} \quad (i, j \in \{0, 1\}) \quad \text{and} \quad s_0 t_0 + s_1 t_1 = 1.$$

The generators of  $L$  form a binary Leavitt family in  $L$ , and a unital homomorphic image of  $L$  carries one; but a family is strictly less than a copy of  $L$ —no injectivity is assumed—and the statements below that mention only a family are correspondingly stronger than their specializations to  $L$ . Fix a unital ring  $A$  carrying a binary Leavitt family, and write

$$p_0 = s_0 t_0, \quad p_1 = s_1 t_1,$$

a pair of orthogonal idempotents summing to 1:  $p_i p_j = \delta_{ij} p_i$ , since  $p_0 p_1 = s_0(t_0 s_1)t_1 = 0$  and  $p_i^2 = s_i(t_i s_i)t_i = p_i$ . For a binary word  $w = w_1 \cdots w_\ell \in \{0, 1\}^*$ , put  $s_w = s_{w_1} \cdots s_{w_\ell}$  and  $s_w^* = t_{w_\ell} \cdots t_{w_1}$ , so that  $s_w^* s_{w'} = \delta_{w, w'}$  whenever  $|w| = |w'|$ , and more generally  $s_w^* s_{w'}$  vanishes unless one of  $w, w'$  is a prefix of the other.

Two remarks on size. The algebra  $L$  itself is infinite-dimensional over  $k$ : the elements  $s_0^i s_1$ ,  $i \geq 0$ , are killed on the left by mutually orthogonal projections and are linearly independent. And a family forces infinitude with no reference to  $L$ :

**Lemma 5.2.** A nontrivial unital ring  $A$  carrying a binary Leavitt family is infinite.

*Proof.* The idempotents  $e_i = s_0^i t_0^i$ ,  $i \geq 0$ , are pairwise distinct. Indeed, suppose  $e_i = e_j$  with  $i < j$ . Multiplying on the left by  $t_0^i$  and on the right by  $s_0^i$ , and using  $t_0^i s_0^i = 1$ , gives  $1 = s_0^d t_0^d$  with  $d = j - i \geq 1$ . Then  $t_1 = t_1 \cdot 1 = (t_1 s_0) s_0^{d-1} t_0^d = 0$ , whence  $1 = t_1 s_1 = 0$ , contradicting nontriviality.  $\square$

**5.1. Leaf sets and matrix self-similarity.** A finite set of binary words  $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$  (we use the same symbols for the words and for the corresponding elements  $s_{\alpha_i} \in A$ ) is a *complete leaf set* if the words are the leaves of a finite full rooted binary tree; equivalently, if they are pairwise prefix-incomparable and the cylinders they define partition the boundary  $\{0, 1\}^{\mathbb{N}}$ .

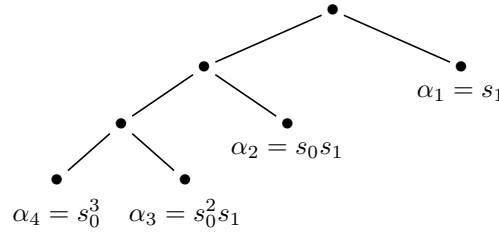


FIGURE 3. The left comb  $\mathcal{C}_4$ : every left child is subdivided. Reading left edges as  $s_0$  and right edges as  $s_1$ , each leaf spells its word along the path from the root; the ordering lists the leaves by depth, with the leftmost branch last.

**Lemma 5.3** (Leaf identities). *Let  $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$  be a complete leaf set. Then in every unital ring  $A$  carrying a binary Leavitt family:*

$$\alpha_i^* \alpha_j = \delta_{ij} \quad \text{and} \quad \sum_{i=1}^r \alpha_i \alpha_i^* = 1.$$

*Proof.* Prefix-incomparability and the relations  $t_i s_j = \delta_{ij}$  give the first identity. The second holds by induction on the tree: for the trivial tree it reads  $1 = 1$ ; replacing a leaf  $\alpha$  by its two children  $\alpha 0, \alpha 1$  changes the sum by

$$s_{\alpha 0} s_{\alpha 0}^* + s_{\alpha 1} s_{\alpha 1}^* - s_{\alpha} s_{\alpha}^* = s_{\alpha} (s_0 t_0 + s_1 t_1 - 1) s_{\alpha}^* = 0. \quad \square$$

**Proposition 5.4** (Matrix self-similarity). *Let  $A$  be a unital ring carrying a binary Leavitt family and let  $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$  be an ordered complete leaf set. Then*

$$\Theta_{\mathcal{C}} : M_r(A) \longrightarrow A, \quad (a_{ij}) \longmapsto \sum_{i,j=1}^r \alpha_i a_{ij} \alpha_j^*,$$

*is an isomorphism of unital rings—of unital  $k$ -algebras when  $A$  is a  $k$ -algebra—with inverse  $x \mapsto (\alpha_i^* x \alpha_j)_{ij}$ . It restricts to group isomorphisms  $\mathrm{GL}_r(A) \cong A^{\times}$  for all  $r \geq 1$ . In particular  $M_r(L) \cong L$  for every  $r \geq 1$ .*

*Proof.* The map  $\Theta_{\mathcal{C}}$  is additive—and  $k$ -linear when  $A$  is a  $k$ -algebra—sends the identity matrix to  $\sum_i \alpha_i \alpha_i^* = 1$  by Lemma 5.3, and is multiplicative:

$$\Theta_{\mathcal{C}}(a) \Theta_{\mathcal{C}}(b) = \sum_{i,j,l,m} \alpha_i a_{ij} \alpha_j^* \alpha_l b_{lm} \alpha_m^* = \sum_{i,m} \alpha_i \left( \sum_j a_{ij} b_{jm} \right) \alpha_m^* = \Theta_{\mathcal{C}}(ab),$$

using  $\alpha_j^* \alpha_l = \delta_{jl}$ . The displayed inverse is two-sided by the same identities:  $\alpha_i^* \Theta_{\mathcal{C}}(a) \alpha_j = a_{ij}$ , and  $\sum_{i,j} \alpha_i (\alpha_i^* x \alpha_j) \alpha_j^* = 1 \cdot x \cdot 1 = x$ . An algebra isomorphism restricts to an isomorphism of unit groups, and  $\mathrm{GL}_r(L) = M_r(L)^{\times}$ .  $\square$

For every  $r \geq 2$  there is a canonical choice, the *left comb*

$$(22) \quad \mathcal{C}_r = (s_1, s_0 s_1, s_0^2 s_1, \dots, s_0^{r-2} s_1, s_0^{r-1}),$$

the leaf set of the tree in which every left child is subdivided; the ordering lists the leaves by depth with the leftmost branch last (Figure 3).

Leavitt/MatrixSelfSimilarity  
ringEquivMatrix  
Leavitt/LeavittSelfSimilarity  
binaryMatrixRingEquiv  
Leavitt/PrefixCode  
prefixRingEquiv  
prefixRingEquiv\_apply  
prefixUnitsEquiv  
Leavitt/SelfSimilarityAlgebra  
matrixAlgEquiv  
prefixAlgEquiv  
Every positive rank, with an  
explicit inverse;  
prefixRingEquiv is  $\Theta_{\mathcal{C}}$  for an  
arbitrary leaf set,  
prefixUnitsEquiv its  
restriction to unit groups,  
and prefixAlgEquiv the  
 $k$ -algebra refinement.

**5.2. Elementary groups over rings.** Let  $A$  be a unital ring. Recall  $x_{ij}(a) = I_r + aE_{ij}$  for  $i \neq j$ , the relations  $x_{ij}(a)x_{ij}(b) = x_{ij}(a+b)$ , and the Steinberg commutator relation

$$[x_{il}(a), x_{lj}(b)] = x_{ij}(ab) \quad (i, j, l \text{ distinct}),$$

where  $[g, h] = ghg^{-1}h^{-1}$ .

**Lemma 5.5** (Whitehead commutator lemma). *Let  $A$  be a unital ring and  $a, b \in A^\times$ . Put*

$$w(a) = x_{12}(a) x_{21}(-a^{-1}) x_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}.$$

Then

$$\text{diag}(a, a^{-1}) = w(a) w(-1) \in \text{EL}_2(A),$$

and

$$\text{diag}(aba^{-1}b^{-1}, 1) = \text{diag}(a, a^{-1}) \text{diag}(b, b^{-1}) \text{diag}((ba)^{-1}, ba) \in \text{EL}_2(A).$$

Consequently, for every  $m \geq 2$ , every product of commutators of units  $c \in [A^\times, A^\times]$ , and every diagonal position, the matrix with  $c$  at that position and 1 elsewhere on the diagonal lies in  $\text{EL}_m(A)$ .

*Proof.* First,  $x_{12}(a)x_{21}(-a^{-1}) = \begin{pmatrix} 1-aa^{-1} & a \\ -a^{-1} & 1 \end{pmatrix} = \begin{pmatrix} 0 & a \\ -a^{-1} & 1 \end{pmatrix}$ , and multiplying by  $x_{12}(a)$  on the right adds the first column, multiplied on the right by  $a$ , to the second, giving  $w(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}$ . Then

$$w(a)w(-1) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The displayed product of three diagonal matrices has first entry  $ab(ba)^{-1} = aba^{-1}b^{-1}$  and second entry  $a^{-1}b^{-1}(ba) = 1$ . For the last statement, embed  $\text{EL}_2(A) \hookrightarrow \text{EL}_m(A)$  in any chosen pair of coordinates and multiply the resulting matrices for the commutator factors of  $c$ .  $\square$

**Lemma 5.6.** *Let  $A$  be a finitely generated unital ring and  $r \geq 3$ . Then  $\text{EL}_r(A)$  is finitely generated: if  $A$  is generated as a unital ring by the finite set  $T$ , then  $\text{EL}_r(A)$  is generated by the finitely many matrices  $x_{ij}(1)$  and  $x_{ij}(t)$ ,  $t \in T$ ,  $i \neq j$ .*

*Proof.* Let  $E$  be the subgroup generated by the listed matrices, and let  $A_0 \subseteq A$  be the set of  $a$  with  $x_{ij}(a) \in E$  for all  $i \neq j$ . By the addition relation,  $A_0$  is an additive subgroup containing 1 and  $T$ ; by the Steinberg relation with a third index  $l \notin \{i, j\}$  (here  $r \geq 3$  is used),  $a, b \in A_0$  implies  $ab \in A_0$ , since  $x_{ij}(ab) = [x_{il}(a), x_{lj}(b)]$  and  $x_{il}(a), x_{lj}(b) \in E$ . Hence  $A_0$  is a unital subring containing  $T$ , so  $A_0 = A$  and  $E = \text{EL}_r(A)$ .  $\square$

**Theorem 5.7** (Property (T) at rank three; Ershov–Jaikin–Zapirain [9]). *Let  $A$  be a finite-type algebra over a finite field. Then  $\text{EL}_3(A)$  has Kazhdan’s property (T).*

Leavitt/Whitehead  
whitehead\_commutator  
whitehead\_diagonal  
Leavitt/MatrixDiagonalization  
exists\_elementary\_whitehead  
Leavitt/DiagonalElementary  
firstDiagonalUnit\_mem\_of\_mem\_commutator  
The two displayed identities are the Whitehead pair; the closing clause—a product of commutators of units at a diagonal position lies in  $\text{EL}_m$ —is firstDiagonalUnit\_mem\_of\_mem\_commutator, stated at the rank the argument uses.  
Leavitt/IntegralGeneration  
elementaryGroup\_eq\_closure\_of\_adjoint\_int  
elementaryGroup\_finitelyGenerated\_int  
Leavitt/ElementaryGroup  
elementaryGroup\_finitelyGenerated  
Any finitely generated unital ring, with the printed generating set exhibited: root matrices at the ring generators and at 1. The characteristic-two form is the one the binary development consumes.  
PropertyT/FreeElementaryPropertyT  
controlSet\_isKazhdanPair  
PropertyT/FiniteFieldElementaryPropertyT  
finiteFieldElementaryThree\_hasKazhdanPropertyT  
Proved internally, with an explicit Kazhdan pair.

This is the case of the Ershov–Jaikin–Zapirain theorem that this paper consumes: [9] proves property (T) for  $\text{EL}_r$  of *every* finitely generated associative unital ring and every  $r \geq 3$ —their statement is for the universal rings  $\mathbb{Z}\langle x_1, \dots, x_d \rangle$ , and the general case follows since  $\text{EL}_r$  of a quotient ring is a quotient group of  $\text{EL}_r$  of the ring and property (T) passes to quotients. We deliberately state only the finite-field, rank-three case: over rings carrying a binary Leavitt family it is all that is ever needed, because the rank equivalence below spreads property (T) from rank three to every rank  $\geq 2$ . Where a statement over an arbitrary finitely generated ring is wanted, the full theorem of [9] may be substituted; we flag each such use.

Over a ring carrying a binary Leavitt family, the rank of an elementary group is immaterial: matrix self-similarity collapses all ranks at the level of the *elementary* groups themselves, with no  $K$ -theory and no GE input. This is the engine that will spread every property of one rank to all ranks.

**Proposition 5.8** (Elementary rank equivalence). *Let  $A$  be a unital ring carrying a binary Leavitt family and let  $n, m \geq 2$ . Then there is an explicit group isomorphism*

$$\text{EL}_n(A) \cong \text{EL}_m(A),$$

*the composite*

$$\text{EL}_n(A) \cong \text{EL}_n(M_m(A)) = \text{EL}_{nm}(A) = \text{EL}_m(M_n(A)) \cong \text{EL}_m(A),$$

where the outer isomorphisms apply  $\Theta_C^{-1}$  and  $\Theta_{C'}$  of Proposition 5.4 to each matrix coefficient, and the middle equalities are block flattening. In particular finite generation, infinitude, property (T), and (non)soficity hold at one rank  $\geq 2$  if and only if they hold at every rank  $\geq 2$ .

*Proof. Coefficient transport.* A unital ring isomorphism  $\varphi : B \rightarrow B'$  carries  $x_{ij}(b)$  to  $x_{ij}(\varphi b)$ , hence induces an isomorphism  $\text{EL}_p(B) \cong \text{EL}_p(B')$  for every  $p$ . Apply this with  $\varphi = \Theta_{C_m}^{-1} : A \rightarrow M_m(A)$ , which exists by Proposition 5.4.

*Block flattening.* Identify  $M_p(M_q(A)) = M_{pq}(A)$  by the index pairing  $(i, c)$ ,  $1 \leq i \leq p$ ,  $1 \leq c \leq q$ . We claim  $\text{EL}_p(M_q(A)) = \text{EL}_{pq}(A)$  as subgroups of  $\text{GL}_{pq}(A)$ , for every  $p \geq 2$  and  $q \geq 1$ . One inclusion is direct: a block transvection  $I + aE_{ij}$ ,  $i \neq j$ ,  $a \in M_q(A)$ , equals the product  $\prod_{c,d} x_{(i,c),(j,d)}(a_{cd})$  of  $q^2$  commuting entry transvections. For the other inclusion, an entry transvection  $x_{(i,c),(j,d)}(r)$  with  $i \neq j$  is a block transvection with the single-entry coefficient  $rE_{cd}$ ; and one with  $i = j$  (so  $c \neq d$ ) is produced by the Steinberg relation through any third block index  $l \neq i$ , which exists because  $p \geq 2$ :

$$x_{(i,c),(i,d)}(r) = [x_{(i,c),(l,c)}(r), x_{(l,c),(i,d)}(1)],$$

both factors being cross-block transvections, hence in  $\text{EL}_p(M_q(A))$  by the previous sentence. Applying block flattening with  $(p, q) = (n, m)$  and—after reindexing along the bijection  $(i, c) \mapsto (c, i)$ , which conjugates by a permutation matrix and matches entry transvections to entry transvections—with  $(p, q) = (m, n)$ , gives the middle equalities. The final claim holds because all four properties are isomorphism-invariant.  $\square$

Leavitt/LeavittRankEquivalence  
 nce  
 rankSuccEquiv  
 rankSuccToProduct  
 rankSucc\_propertyT\_of\_rankSucc  
 cc

**5.3. Characteristic-two block factorizations.** The following two identities, valid in characteristic two, are the engine of the explicit spine.

**Lemma 5.9.** *Let  $A$  be a unital ring of characteristic 2 carrying a binary Leavitt family. Then, in  $M_2(A)$ :*

- (a)  $x_{12}(s_1) x_{21}(t_1) x_{12}(s_1) = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix};$
- (b)  $x_{21}(1 + t_0) x_{12}(1) x_{21}(1 + s_0) x_{12}(t_0) = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}.$

*Proof.* (a) Since  $1 + p_1 = p_0$  in characteristic 2,

$$\begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 + s_1 t_1 & s_1 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix},$$

and then

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & p_0 s_1 + s_1 \\ t_1 & t_1 s_1 + 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix},$$

using  $p_0 s_1 = s_0(t_0 s_1) = 0$  and  $t_1 s_1 + 1 = 1 + 1 = 0$ .

(b) Compute the product of the first two factors:

$$\begin{pmatrix} 1 & 0 \\ 1 + t_0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix}$$

(using  $1 + t_0 + 1 = t_0$  in characteristic 2). Multiplying by the third factor,

$$\begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 + s_0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 1 + t_0 + t_0 + t_0 s_0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix},$$

where the  $(1, 1)$ -entry is  $1 + 1 + s_0 = s_0$  and the  $(2, 1)$ -entry is  $(1 + t_0) + t_0(1 + s_0) = 1 + t_0 + t_0 + t_0 s_0 = 1 + 1 = 0$ , using  $t_0 s_0 = 1$ . Finally,

$$\begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & t_0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & s_0 t_0 + 1 \\ 0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix},$$

since  $s_0 t_0 + 1 = p_0 + 1 = p_1 = s_1 t_1$  in characteristic 2.  $\square$

**5.4. Thompson's group  $V$  in a Leavitt corner.** Thompson's group  $V$  is the group of homeomorphisms of the boundary  $\{0, 1\}^{\mathbb{N}}$  given by *tree tables*: a pair of ordered complete leaf sets  $(\beta_1, \dots, \beta_\ell)$ ,  $(\alpha_1, \dots, \alpha_\ell)$  with the same number  $\ell$  of leaves, acting by the prefix substitutions  $\beta_i x \mapsto \alpha_i x$ ; each such map carries the cylinder of  $\beta_i$  bijectively onto the cylinder of  $\alpha_i$  and is a homeomorphism of the Cantor set. Two tables define the same element exactly when they have a common refinement, refinement being simultaneous subdivision of a source leaf and its target leaf.

**Proposition 5.10** (Tree-table embedding). *Let  $A$  be a nontrivial unital ring carrying a binary Leavitt family. For  $g \in V$  given by a table  $((\beta_i), (\alpha_i))$ , put*

$$(23) \quad U_g = \sum_{i=1}^{\ell} \alpha_i \beta_i^* \in A.$$

Leavitt/Leavitt  
characteristicTwo\_involution  
characteristicTwo\_compressor  
compressor\_factorization

The factorization is proved in every characteristic.

Leavitt/FamilyVEmbedding  
vEmbeddingOfFamily  
vEmbeddingOfFamily\_injective  
vEmbeddingOfFamily\_tableEquiv

tableSum\_eq\_of\_mapsCylinder  
tableSum\_mul

wordS\_injective

Leavitt/ThompsonVEmbedding

ng  
vEmbedding  
vEmbedding\_injective  
isComplete\_of\_covers

Leavitt/ThompsonV  
tableEquiv

Over an arbitrary nontrivial ring carrying a binary family, with vEmbeddingOfFamily\_tableEquiv the formula (23), well definedness

tableSum\_eq\_of\_mapsCylinder, and faithfulness the word calculus wordS\_injective.

The ThompsonVEmbedding declarations are the earlier version over the presented algebra  $L_k(1, 2)$ .

Then  $U_g$  is well defined (independent of the table),  $U_g U_h = U_{gh}$ ,  $U_1 = 1$ , each  $U_g$  is a unit with  $U_g^{-1} = U_{g^{-1}}$ , and  $g \mapsto U_g$  is an injective homomorphism  $V \hookrightarrow A^\times$ .

*Proof.* Refining a table replaces a summand  $\alpha\beta^*$  by  $\alpha 0(\beta 0)^* + \alpha 1(\beta 1)^* = \alpha(s_0 t_0 + s_1 t_1)\beta^* = \alpha\beta^*$ ; since any two tables of the same element admit a common refinement,  $U_g$  is well defined. For multiplicativity, choose tables of  $g$  and  $h$  such that the source leaf set of  $g$  equals the target leaf set of  $h$  (refine both accordingly); then

$$U_g U_h = \sum_{i,j} \alpha_i \beta_i^* \gamma_j \delta_j^* = \sum_i \alpha_i \delta_i^* = U_{gh},$$

using  $\beta_i^* \gamma_j = \delta_{ij}$  for the common leaf set, and the composed table represents  $gh$ . Also  $U_1 = \sum_i \alpha_i \alpha_i^* = 1$  by Lemma 5.3, so each  $U_g$  is a unit with inverse  $U_{g^{-1}}$ . For injectivity, suppose  $U_g = 1$  with a table  $((\beta_i), (\alpha_i))$ . Multiplying  $\sum_i \alpha_i \beta_i^* = 1$  on the right by  $\beta_j$  gives  $s_{\alpha_j} = s_{\beta_j}$  in  $A$  for each  $j$ —the products  $\beta_i^* \beta_j = \delta_{ij}$  collapse the sum. Equality of these elements forces equality of the binary words. If neither word is a prefix of the other, multiplying  $s_{\alpha_j} = s_{\beta_j}$  on the left by  $s_{\alpha_j}^*$  gives  $1 = s_{\alpha_j}^* s_{\beta_j} = 0$ , contradicting the nontriviality of  $A$ . If, say,  $\beta_j = \alpha_j \gamma$  with  $\gamma$  nonempty, the same multiplication gives  $1 = s_\gamma$ ; choosing  $i \in \{0, 1\}$  different from the first letter of  $\gamma$  then gives  $t_i = t_i s_\gamma = 0$ , whence  $1 = t_i s_i = 0$ , again a contradiction. The case  $\alpha_j = \beta_j \gamma$  is symmetric. Hence  $\alpha_j = \beta_j$  as words for every  $j$ , and  $g$  is the identity substitution. This is the standard realization of the Higman–Thompson groups inside Leavitt algebras and their Cuntz-algebra completions; compare [16, 17].  $\square$

**Lemma 5.11** (Corner embedding). *Let  $A$  be a nontrivial unital ring carrying a binary Leavitt family. For  $g \in V$  put*

$$\rho(g) = p_0 + s_1 U_g t_1 \in A.$$

*Then  $\rho : V \rightarrow A^\times$  is an injective group homomorphism, and for all  $g \in V$ :*

$$t_1 \rho(g) s_1 = U_g, \quad \rho(g) s_0 = s_0, \quad t_0 \rho(g) = t_0.$$

*Proof.* All computations take place in the subring generated by the Leavitt family, using  $t_1 s_0 = 0$ ,  $t_0 s_1 = 0$ ,  $t_1 s_1 = t_0 s_0 = 1$ ,  $p_0 s_0 = s_0$ ,  $t_0 p_0 = t_0$ ,  $p_0 s_1 = 0$ ,  $t_1 p_0 = 0$ . Multiplicativity:

$$\begin{aligned} \rho(g) \rho(h) &= (p_0 + s_1 U_g t_1)(p_0 + s_1 U_h t_1) \\ &= p_0^2 + p_0 s_1 U_h t_1 + s_1 U_g t_1 p_0 + s_1 U_g t_1 s_1 U_h t_1 \\ &= p_0 + s_1 U_g U_h t_1 = \rho(gh), \end{aligned}$$

by Proposition 5.10. Also  $\rho(1) = p_0 + p_1 = 1$ , so  $\rho(g) \in A^\times$  with inverse  $\rho(g^{-1})$ . The three displayed identities are immediate:  $t_1 \rho(g) s_1 = t_1 p_0 s_1 + t_1 s_1 U_g t_1 s_1 = U_g$ ;  $\rho(g) s_0 = p_0 s_0 + s_1 U_g t_1 s_0 = s_0$ ;  $t_0 \rho(g) = t_0 p_0 + t_0 s_1 U_g t_1 = t_0$ . Injectivity:  $\rho(g) = 1$  forces  $U_g = t_1 \rho(g) s_1 = t_1 s_1 = 1$ , hence  $g = 1$  by Proposition 5.10.  $\square$

Leavitt/LeavittCorner

cornerHom

cornerHom\_s0

t0\_cornerHom

t1\_cornerHom\_s1

Thus  $\rho$  realizes  $V$  inside the corner  $p_0 + s_1(\cdot)t_1$ , acting as the identity on the complementary corner  $s_0(\cdot)t_0$ : this complementarity is the algebraic seed of the centralizer relation  $[K, J] = 1$ .

The non-LEF input of the whole paper is the following finite obstruction. Its two relators are the standard defining relators of Thompson's group  $F$ , used here purely as relators.

**Lemma 5.12** (Two-relator obstruction). *Let  $a, b$  be elements of a group satisfying*

$$(24) \quad [ab^{-1}, a^{-1}ba] = 1 \quad \text{and} \quad [ab^{-1}, a^{-2}ba^2] = 1.$$

*If  $a$  and  $b$  do not commute, the group is not LEF.*

*Proof.* We first show that in a *finite* group the relations (24) force  $[a, b] = 1$ . Put  $c_n = a^{-n}ba^n$ , so  $c_0 = b$  and  $a^{-1}c_na = c_{n+1}$ . A relation  $[ab^{-1}, c_n] = 1$  rearranges:  $ab^{-1}c_nba^{-1} = c_n$  is exactly

$$b^{-1}c_nb = a^{-1}c_na = c_{n+1}.$$

Thus the two relations (24) are the cases  $n = 1, 2$  of the family  $(C_n)$ :  $b^{-1}c_nb = c_{n+1}$ . Conjugating  $(C_1)$  by  $a^{-n}$  gives, since  $a^{-n}ba^n = c_n$ ,

$$(A_n) : \quad c_n^{-1}c_{n+1}c_n = c_{n+2} \quad (n \geq 1).$$

Now  $(C_n)$  holds for every  $n \geq 1$ , by strong induction with base  $(C_1), (C_2)$ : using  $(A_{n-1})$ , then  $(C_{n-1})$  and  $(C_n)$ , then  $(A_n)$ ,

$$b^{-1}c_{n+1}b = b^{-1}(c_{n-1}^{-1}c_nc_{n-1})b = c_n^{-1}c_{n+1}c_n = c_{n+2}.$$

In a finite group  $a^N = 1$  for some  $N \geq 1$ , so  $c_N = c_0 = b$ , and  $(C_N)$  reads  $b^{-1}bb = c_{N+1} = c_1$ ; that is,  $b = a^{-1}ba$ , so  $[a, b] = 1$ .

Now suppose the ambient group were LEF. Take  $F$  to consist of  $1, [a, b]$ , and every product of at most twenty letters drawn from  $\{a^{\pm 1}, b^{\pm 1}\}$ —a finite set containing  $a, b$ , and all partial products needed to evaluate both relators and  $[a, b]$  letter by letter. A local embedding  $\varphi : F \rightarrow P$  into a finite group is multiplicative on these partial products, so  $\varphi(a), \varphi(b)$  satisfy (24) in  $P$ , whence  $[\varphi(a), \varphi(b)] = 1_P$  by the previous paragraph; but  $\varphi([a, b]) = [\varphi(a), \varphi(b)] = 1_P = \varphi(1)$  and injectivity force  $[a, b] = 1$ , a contradiction.  $\square$

**Proposition 5.13.** *Thompson's group  $V$  is not LEF.*

*Proof.* Let  $x_0, x_1 \in V$  be the tree tables

$$\begin{aligned} x_0 : \{ & 00 \mapsto 0, 01 \mapsto 10, 1 \mapsto 11 \}, \\ x_1 : \{ & 0 \mapsto 0, 100 \mapsto 10, 101 \mapsto 110, 11 \mapsto 111 \} \end{aligned}$$

( $x_1$  acts as  $x_0$  inside the cylinder 1 and as the identity outside it; Figure 4 draws the pair of trees of  $x_0$ ). Both relators (24) hold for  $(a, b) = (x_0, x_1)$ : each side is a composition of finitely many prefix substitutions, and refining all tables to a common leaf depth and comparing leaf images is a finite

Sofic/ThompsonFObstruction  
not\_isLEF\_of\_two\_relations  
finite\_commute\_of\_two\_relations  
conjugacy\_relation\_all

Leavitt/ThompsonVWitness  
thompsonV\_not\_isLEF  
Leavitt/ThompsonVWitness  
not\_isLEF\_cornerWitnessSubgroup  
Criterion/Scheme  
not\_isLEF\_cornerSubgroup  
Proved for V itself, with no  
simplicity or presentation  
input.

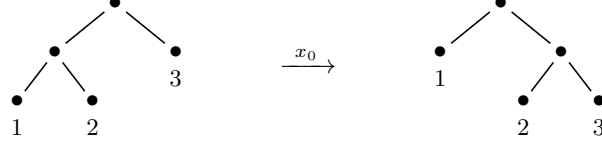


FIGURE 4. The tree table of  $x_0$ . Matching leaf labels give the prefix substitution: the source leaves 00, 01, 1 map to the target leaves 0, 10, 11, so  $x_0 = \{00 \mapsto 0, 01 \mapsto 10, 1 \mapsto 11\}$ . The element  $x_1$  applies the same table inside the cylinder 1 and fixes its complement.

computation. The pair does not commute: on the point  $011^\infty$  of the Cantor set,

$$x_0x_1(011^\infty) = x_0(011^\infty) = 101^\infty, \quad x_1x_0(011^\infty) = x_1(101^\infty) = 1101^\infty.$$

Lemma 5.12 applies.  $\square$

**Remark 5.14** (The classical facts about  $V$ , and the classical route). The group  $V$  is infinite, simple, and finitely presented [11, 6], and 2-generated [5]; it is perfect, since  $[V, V]$  is a nontrivial normal subgroup of a simple group. These classical facts give a second proof of Proposition 5.13: a finitely presented LEF group is residually finite (Lemma 2.8), while an infinite simple group is not. Nothing in the main line of this paper consumes them—the proofs of Theorems A–D use only Lemma 5.12 and the explicit pair above—but they are the conceptual frame: the witness subgroup built in Section 6 is the image of the standard copy of Thompson’s  $F$  inside the corner copy of  $V$ .

## 6. THE COMPRESSION SCHEME IN ADJACENT RANKS

This section presents the algebraic heart of the paper in its natural generality. The matrix identities below are verifiable by finite computation from the two Leavitt relations, over an arbitrary ambient ring. Beyond them, the section uses only the facts about Thompson’s group  $V$  established in Section 5 and the two membership hypotheses isolated in Theorem 6.10, the latter discharged explicitly in Section 7 and  $K$ -theoretically in Section 8.

Throughout,  $A$  is a nontrivial associative unital ring carrying a binary Leavitt family  $(s_0, s_1, t_0, t_1)$  in the sense of Definition 5.1—in particular  $A$  may be, but need not contain a copy of, the Leavitt algebra  $L = L_k(1, 2)$ —and we retain the notation  $p_0 = s_0t_0$ ,  $p_1 = s_1t_1$ . Fix  $m \geq 2$  and put

$$n = m + 1, \quad G = \text{EL}_n(A), \quad \Gamma = \{\hat{\gamma} = \text{diag}(\gamma, 1) \mid \gamma \in \text{EL}_m(A)\} \leq G.$$

Indeed  $\Gamma \leq G$ : an elementary matrix in the first  $m$  coordinates is an elementary matrix of rank  $n$ . The group  $\Gamma \cong \text{EL}_m(A)$  is infinite, since  $a \mapsto x_{12}(a)$  is injective and  $A$  is infinite by Lemma 5.2.

We use the left comb (22) with  $n$  leaves:

$$(25) \quad \alpha_i = s_0^{i-1} s_1 \quad (1 \leq i \leq m), \quad \alpha_n = s_0^m,$$

a complete leaf set, so that  $\alpha_i^* \alpha_j = \delta_{ij}$  and  $\sum_{i=1}^n \alpha_i \alpha_i^* = 1$  by Lemma 5.3.

**6.1. The comb compressor.** Define the column  $b \in A^{m \times 1}$ , the row  $c \in A^{1 \times m}$ , and the matrices  $u, u' \in M_n(A)$ :

$$(26) \quad b = \begin{pmatrix} s_1 \alpha_1^* \\ \vdots \\ s_1 \alpha_m^* \end{pmatrix}, \quad c = (\alpha_1 t_1 \quad \cdots \quad \alpha_m t_1),$$

$$u = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix}, \quad u' = \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix}.$$

**Lemma 6.1.** *The following identities hold in  $A$ :*

$$bc = p_1 I_m, \quad cb = 1 - \alpha_n \alpha_n^*, \quad t_0 b = 0, \quad cs_0 = 0, \quad b \alpha_n = 0, \quad \alpha_n^* c = 0.$$

Consequently  $uu' = u'u = I_n$ ; thus  $u \in \text{GL}_n(A)$  with  $u^{-1} = u'$ .

*Proof.* Entrywise:  $(bc)_{ij} = s_1 \alpha_i^* \alpha_j t_1 = \delta_{ij} s_1 t_1 = \delta_{ij} p_1$ . Next,  $cb = \sum_{i=1}^m \alpha_i t_1 s_1 \alpha_i^* = \sum_{i=1}^m \alpha_i \alpha_i^* = 1 - \alpha_n \alpha_n^*$  by Lemma 5.3. The entries of  $t_0 b$  are  $t_0 s_1 \alpha_i^* = 0$ ; those of  $cs_0$  are  $\alpha_i t_1 s_0 = 0$ ; those of  $b \alpha_n$  are  $s_1 \alpha_i^* \alpha_n = 0$  and of  $\alpha_n^* c$  are  $\alpha_n^* \alpha_i t_1 = 0$ , by orthogonality of distinct leaves. Then

$$uu' = \begin{pmatrix} s_0 t_0 I_m + bc & b \alpha_n \\ \alpha_n^* c & \alpha_n^* \alpha_n \end{pmatrix} = \begin{pmatrix} (p_0 + p_1) I_m & 0 \\ 0 & 1 \end{pmatrix} = I_n,$$

and likewise

$$u'u = \begin{pmatrix} t_0 s_0 I_m & t_0 b \\ cs_0 & cb + \alpha_n \alpha_n^* \end{pmatrix} = I_n.$$

**Proposition 6.2** (Compression). *For every  $\gamma \in \text{EL}_m(A)$ ,*

$$(27) \quad u \hat{\gamma} u^{-1} = p_1 \widehat{I_m + s_0 \gamma t_0},$$

and on elementary generators  $u \hat{x}_{ij}(a) u^{-1} = \hat{x}_{ij}(s_0 a t_0)$  for  $i \neq j \leq m$ . Consequently

$$(28) \quad K := u \Gamma u^{-1} = \langle \hat{x}_{ij}(s_0 a t_0) : a \in A, i \neq j \leq m \rangle \leq \Gamma,$$

and conjugation by  $u$  is an isomorphism  $\Gamma \xrightarrow{\cong} K$ ; in particular  $K \cong \text{EL}_m(A)$ .

*Proof.* Block multiplication using Lemma 6.1:

$$u \hat{\gamma} u^{-1} = \begin{pmatrix} s_0 \gamma & b \\ 0 & \alpha_n^* \end{pmatrix} \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix} = \begin{pmatrix} s_0 \gamma t_0 + bc & b \alpha_n \\ \alpha_n^* c & \alpha_n^* \alpha_n \end{pmatrix}$$

$$= \begin{pmatrix} s_0 \gamma t_0 + p_1 I_m & 0 \\ 0 & 1 \end{pmatrix}.$$

For  $\gamma = x_{ij}(a) = I_m + a E_{ij}$ , the block is  $p_1 I_m + s_0 t_0 I_m + s_0 a t_0 E_{ij} = I_m + s_0 a t_0 E_{ij} = x_{ij}(s_0 a t_0)$ . Since  $\Gamma$  is generated by the  $\hat{x}_{ij}(a)$  and conjugation is a group isomorphism onto its image,  $K$  is generated by the displayed elements, which lie in  $\Gamma$ .  $\square$

Leavitt/LeavittMatrixCompression  
matrixCompressionHom  
matrixCompressionUnit  
matrixCompression\_recover  
Leavitt/RankFourCompressor  
s  
compressor  
compressor\_val  
compressor is a product of  
elementary transvections,  
invertible by construction,  
and compressor\_val identifies  
it with the matrix (26).  
Leavitt/RankFourCompressor  
s  
compressor\_conjugation  
compressionEnd\_injective

The compressed copy  $K$  sits inside  $\Gamma$  with all matrix entries confined to the corner  $s_0(\cdot)t_0 = p_0Ap_0$  of  $A$ ; the complementary corner  $p_1Ap_1$  is untouched, and this is the room in which the centralizing subgroup  $J$  will live.

**6.2. The involution  $z$  and generation.** Let  $z \in M_n(A)$  agree with the identity matrix outside the coordinates 1 and  $n$ , where it is

$$z|_{\{1,n\}} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}; \quad \text{that is, } z = I_n + (p_0 - 1)E_{11} + s_1E_{1n} + t_1E_{n1} - E_{nn}.$$

**Lemma 6.3.**  $z^2 = I_n$ ; in particular  $z \in \text{GL}_n(A)$ . Moreover  $z$  centralizes  $K$ .

*Proof.* On the coordinates  $\{1, n\}$ ,

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}^2 = \begin{pmatrix} p_0^2 + s_1t_1 & p_0s_1 \\ t_1p_0 & t_1s_1 \end{pmatrix} = \begin{pmatrix} p_0 + p_1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

using  $p_0s_1 = 0$ ,  $t_1p_0 = 0$ ,  $t_1s_1 = 1$ ; the remaining coordinates are fixed. For the centralizing statement, put  $r = s_0at_0$  and note

$$p_0r = r = rp_0, \quad t_1r = 0, \quad rs_1 = 0.$$

Let  $i \neq j \leq m$ . If  $1 \notin \{i, j\}$ , then  $z$  acts as the identity on the coordinates  $i, j$  and  $z\hat{x}_{ij}(r)z = \hat{x}_{ij}(r)$ . If  $i = 1$ : the matrix  $rE_{1j}$  satisfies  $z(rE_{1j})z = (p_0rE_{1j} + t_1rE_{nj})z = rE_{1j}z = rE_{1j}$ , since the  $j$ -th row of  $z$  is  $e_j^\top$  for  $2 \leq j \leq m$ . If  $j = 1$ :  $z(rE_{i1})z = rE_{i1}z = rp_0E_{i1} + rs_1E_{in} = rE_{i1}$ . In each case  $z\hat{x}_{ij}(r)z^{-1} = \hat{x}_{ij}(r)$ , and these elements generate  $K$  by (28).  $\square$

Define the second compressor

$$(30) \quad v = zu.$$

Then, by Proposition 6.2 and Lemma 6.3,

$$(31) \quad v\Gamma v^{-1} = z(u\Gamma u^{-1})z^{-1} = zKz^{-1} = K = u\Gamma u^{-1} :$$

the two compressors have one common image. The quotient  $vu^{-1} = z$  is precisely what generates the missing last row and column:

**Proposition 6.4** (Generation). *The subgroup of  $\text{GL}_n(A)$  generated by  $\Gamma$  and  $z$  contains  $\text{EL}_n(A)$ . Consequently, if  $u$  and  $z$  lie in  $\text{EL}_n(A)$ , then*

$$\text{EL}_n(A) = \langle \Gamma, z \rangle = \langle \Gamma, u, v \rangle.$$

*Proof.* For  $2 \leq j \leq m$  and  $a \in A$ , we claim

$$z\hat{x}_{1j}(s_1a)z = x_{nj}(a), \quad z\hat{x}_{j1}(at_1)z = x_{jn}(a).$$

For the first:  $z(s_1aE_{1j})z = (p_0s_1aE_{1j} + t_1s_1aE_{nj})z = aE_{nj}z = aE_{nj}$ , using  $p_0s_1 = 0$ ,  $t_1s_1 = 1$ , and that the  $j$ -th row of  $z$  is  $e_j^\top$ . For the second:  $z(at_1E_{j1})z = at_1E_{j1}z = at_1p_0E_{j1} + at_1s_1E_{jn} = aE_{jn}$ , using  $t_1p_0 = 0$ ,  $t_1s_1 = 1$ , and that the  $j$ -th column of  $z$  is  $e_j$ . Thus  $\langle \Gamma, z \rangle$  contains  $x_{nj}(a)$  and  $x_{jn}(a)$  for all  $2 \leq j \leq m$  and  $a \in A$ , as well as all  $\hat{x}_{ij}(a) = x_{ij}(a)$  with

Leavitt/Leavitt  
z\_sq  
Leavitt/RankFourCompressor  
s  
involution\_sq  
Leavitt/DiagonalCornerCompression  
matrixCompression\_commutates\_f  
firstDiagonalCorner

Leavitt/RankFourCompressor  
s  
coreEmbedding\_compressorSet\_  
generate

$i \neq j \leq m$ . The remaining elementary matrices are supplied by the Steinberg relation, using  $m \geq 2$  so that the coordinate  $2 \notin \{1, n\}$  exists:

$$x_{1n}(a) = [x_{12}(a), x_{2n}(1)], \quad x_{n1}(a) = [x_{n2}(a), x_{21}(1)].$$

Hence  $\langle \Gamma, z \rangle \supseteq \text{EL}_n(A)$ . If moreover  $u, z \in \text{EL}_n(A)$ , then  $\langle \Gamma, z \rangle \subseteq \text{EL}_n(A)$ , giving the first equality; and since  $z = vu^{-1}$ , also  $\langle \Gamma, u, v \rangle \supseteq \langle \Gamma, z \rangle = \text{EL}_n(A) \supseteq \langle \Gamma, u, v \rangle$ .  $\square$

**6.3. The commuting corner witness.** For  $g \in V$ , recall  $\rho(g) = p_0 + s_1 U_g t_1 \in A^\times$  from Lemma 5.11, and define

$$j(g) = \text{diag}(\rho(g), 1, \dots, 1) \in \text{GL}_n(A), \quad J = \langle j(x_0), j(x_1) \rangle,$$

where  $x_0, x_1 \in V$  are the two tree tables of Proposition 5.13. Thus  $J$  is a two-generator subgroup of the corner copy  $j(V)$  of  $V$ ; it is the image of the standard copy of Thompson's  $F$ . Two lemmas place its generators inside  $\Gamma$  by explicit commutator certificates.

For prefix-incomparable words  $v, w$ , write

$$T(v, w) = s_v t_w + s_w t_v + (1 - p_v - p_w) \in A^\times,$$

the *cylinder transposition*: it is the table unit  $U_g$  of the element of  $V$  exchanging the cylinders  $v$  and  $w$  and fixing everything else, and  $T(v, w)^2 = 1$ .

**Lemma 6.5** (Transpositions are single commutators). *Let  $v$  and  $w$  be prefix-incomparable. Then*

$$T(v, w) = [T(v0, v1) T(w0, w1), T(v0, w0)]$$

*is a single commutator of units of  $A$ .*

*Proof.* All five transpositions are table units, and multiplication of table units composes the underlying prefix maps (Proposition 5.10), so the identity may be verified at the level of maps of  $\{0, 1\}^{\mathbb{N}}$ . Splitting the two cylinders into their children,  $T(v, w)$  acts as the double transposition  $(v0\ w0)(v1\ w1)$  of the four pairwise-incomparable child cylinders, fixing everything else; and in the symmetric group of these four cylinders, with  $x = (v0\ v1)(w0\ w1)$  and  $y = (v0\ w0)$ ,

$$xyx^{-1} = (v1\ w1), \quad \text{so} \quad [x, y] = (v1\ w1)(v0\ w0) = T(v, w)\text{'s action.}$$

All the maps involved fix the complement of the four child cylinders, so the identity holds outright.  $\square$

**Lemma 6.6** (Commutator certificates for the witness pair). *The table units of  $x_0$  and  $x_1$  factor as*

$$U_{x_0} = T(0, 10) T(0, 11) T(0, 1), \quad U_{x_1} = \kappa_{[1]}(U_{x_0}),$$

where  $\kappa_{[1]}(u) = p_0 + s_1 u t_1$  is the corner insertion at the cylinder 1—multiplicative and unital by the computation of Lemma 5.11. Consequently  $U_{x_0}$  and  $U_{x_1}$  are products of three commutators of units of  $A$  each, and so are  $\rho(x_0)$  and  $\rho(x_1)$ .

Leavitt/ThompsonWitness  
cylinderSwap\_is\_commutator  
cylinderSwap\_mem\_commutator

Leavitt/ThompsonWitness  
rootRotation  
cornerWitnessSubgroup  
cornerWitnessSubgroup\_le\_commutator

*Proof.* For the factorization, compose the three transpositions as prefix maps, rightmost first: a point  $00v$  goes to  $10v$ , stays at  $10v$ , then goes to  $0v$ ; a point  $01v$  goes to  $11v$ , then to  $0v$ , then to  $10v$ ; a point  $1v$  goes to  $0v$ , then to  $11v$ , and stays. The composite is exactly the table  $\{00 \mapsto 0, 01 \mapsto 10, 1 \mapsto 11\} = x_0$ , and the displayed unit identity follows through Proposition 5.10. The identity  $U_{x_1} = \kappa_{[1]}(U_{x_0})$  is the definition of  $x_1$  (act as  $x_0$  inside the cylinder 1, and note  $1 - p_1 = p_0$ ). Each transposition is a single commutator by Lemma 6.5, so  $U_{x_0}$  is a product of three commutators, and  $\kappa_{[1]}$ , being unital and multiplicative, carries each commutator certificate to a commutator certificate. Finally  $\rho(g) = p_0 + s_1 U_g t_1 = \kappa_{[1]}(U_g)$  itself, so the same transport gives the claim for  $\rho(x_0)$  and  $\rho(x_1)$ .  $\square$

**Proposition 6.7.**  $J \leq \Gamma$ : both  $j(x_0)$  and  $j(x_1)$  lie in  $\text{EL}_m(A)$ , embedded in the first  $m$  coordinates. The subgroup  $J$  is generated by two elements, and it is not LEF.

*Proof.* By Lemma 6.6,  $\rho(x_0)$  and  $\rho(x_1)$  are products of commutators of units of  $A$ ; the Whitehead commutator lemma (Lemma 5.5, with  $m \geq 2$ ) places  $\text{diag}(\rho(x_i), 1, \dots, 1)$  in  $\text{EL}_m(A)$ . Hence  $J \leq \Gamma$ . The composite  $g \mapsto j(g)$  is an injective homomorphism (Lemma 5.11), so  $j(x_0), j(x_1)$  satisfy the two relators (24) and do not commute, exactly as  $x_0, x_1$  do in the proof of Proposition 5.13; Lemma 5.12 shows  $J$  is not LEF.  $\square$

**Remark 6.8** (The full corner copy of  $V$ ). Granting the classical perfectness of  $V$  (Remark 5.14), the same Whitehead argument places *every*  $j(g)$ ,  $g \in V$ , in  $\text{EL}_m(A)$ , so the full corner copy  $j(V) \cong V$  is a subgroup of  $\Gamma$ , over every ring carrying a binary Leavitt family. The main theorems use only the two-generator witness  $J$ , whose certificates are explicit.

**Proposition 6.9.**  $[K, J] = 1$  and  $K \cap J = \{1\}$ . Consequently  $KJ \cong K \times J$ .

*Proof. Centralizing.* Conjugating an elementary matrix by a diagonal matrix multiplies its coefficient:  $j(g) \hat{x}_{il}(r) j(g)^{-1} = \hat{x}_{il}(\rho(g)^{[i=1]} r \rho(g^{-1})^{[l=1]})$ , where the exponent  $[i = 1]$  is 1 if  $i = 1$  and 0 otherwise. For  $r = s_0 a t_0$ , Lemma 5.11 gives  $\rho(g)r = \rho(g)s_0 a t_0 = s_0 a t_0 = r$  and  $r\rho(g^{-1}) = s_0 a t_0 \rho(g^{-1}) = s_0 a t_0 = r$ . Hence  $j(g)$  commutes with every generator  $\hat{x}_{il}(s_0 a t_0)$  of  $K$  from (28).

*Trivial intersection.* Suppose  $j(g) \in K$ . By (27), every element of  $K$  has the form  $p_1 \widehat{I_m} + s_0 \gamma t_0$  with  $\gamma \in \text{EL}_m(A)$ ; comparing  $(1, 1)$ -entries,  $\rho(g) = p_1 + s_0 \gamma_{11} t_0$ . Sandwiching between  $t_1$  and  $s_1$ : on one side,  $t_1(\rho(g) - 1)s_1 = t_1(s_1 U_g t_1 - p_1)s_1 = U_g - 1$ , using  $t_1 p_0 = 0$ ; on the other,  $t_1(p_1 + s_0 \gamma_{11} t_0 - 1)s_1 = t_1(s_0 \gamma_{11} t_0 - p_0)s_1 = 0$ , using  $t_1 s_0 = 0$  and  $t_1 p_0 = 0$ . Hence  $U_g = 1$ , so  $g = 1$  by Proposition 5.10, and  $j(g) = 1$ .

The two computations quantify over every  $g \in V$ , so they prove the claims for the full corner copy  $j(V)$ ; the witness  $J = \langle j(x_0), j(x_1) \rangle \leq j(V)$  inherits both. Since  $K$  and  $J$  commute and intersect trivially, the multiplication map  $K \times J \rightarrow KJ$  is an isomorphism.  $\square$

Leavitt/UniversalRankFour  
witnessEmbedding  
witnessEmbedding\_injective  
Leavitt/DiagonalElementary  
firstDiagonalUnit\_mem\_of\_mem  
\_commutator  
Leavitt/ThompsonWitness  
not\_isLEF\_cornerWitnessSubgr  
oup

Criterion/Scheme  
commute\_compressed\_corner  
compressed\_inf\_corner

#### 6.4. The Leavitt-corner theorem.

**Theorem 6.10** (Leavitt-corner theorem). *Let  $A$  be a nontrivial associative unital ring carrying a binary Leavitt family, let  $m \geq 2$ , and put  $n = m + 1$ . Let  $u, z \in \mathrm{GL}_n(A)$  be the matrices (26) and (29). Assume:*

- (i)  $\mathrm{EL}_m(A)$  and  $\mathrm{EL}_n(A)$  have property (T);
- (ii)  $u, z \in \mathrm{EL}_n(A)$ ; this holds, for instance, whenever  $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$ .

*Then  $\mathrm{EL}_n(A)$  is not sofic.*

*Proof.* Put  $G = \mathrm{EL}_n(A)$  and let  $\Gamma, u, z, v = zu, K, J$  be as above. By hypothesis (i),  $G$  and  $\Gamma \cong \mathrm{EL}_m(A)$  are property-(T) groups, hence finitely generated, and  $\Gamma$  is infinite. By hypothesis (ii) and Proposition 6.4,  $G = \langle \Gamma, u, v \rangle = \langle \Gamma, Q \rangle$  with  $Q = \{u, v\} \subseteq G$ . By Proposition 6.2 and (31),

$$u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma,$$

so both elements of  $Q$  conjugate  $\Gamma$  into itself. Fix  $q_0 = u$ , so  $K = q_0\Gamma q_0^{-1}$ . By Propositions 6.7 and 6.9, the subgroup  $J \leq \Gamma$  is generated by two elements, satisfies  $[K, J] = 1$  and  $K \cap J = \{1\}$ , and is not LEF. If  $G$  were sofic, Corollary 4.2 (Theorem D) would force  $J$  to be LEF, a contradiction.  $\square$

**Corollary 6.11.** *Let  $A$  be a nontrivial finite-type algebra over a finite field carrying a binary Leavitt family, and let  $m \geq 2, n = m + 1$ . If  $u, z \in \mathrm{EL}_n(A)$ —for instance, if  $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$ —then  $\mathrm{EL}_n(A)$  is an infinite, finitely generated, property-(T) group that is not sofic.*

*Proof.* Theorem 5.7 gives property (T) for  $\mathrm{EL}_3(A)$ , and the elementary rank equivalence (Proposition 5.8) transports it to  $\mathrm{EL}_m(A)$  and  $\mathrm{EL}_n(A)$  for every  $m \geq 2$ , supplying hypothesis (i) of Theorem 6.10; hence  $\mathrm{EL}_n(A)$  is not sofic and has property (T). It is finitely generated by Lemma 5.6 (here  $n \geq 3$ , and a finite-type algebra over a finite field is a finitely generated ring), and infinite, since  $a \mapsto x_{12}(a)$  embeds the ring  $A$ , which is infinite by Lemma 5.2. Over an arbitrary finitely generated unital ring and  $m \geq 3$ , the same conclusion holds verbatim, granting the full theorem of [9] in place of Theorem 5.7.  $\square$

The corollary’s reach is wider than its statement: once one rank is nonsolic, the self-similarity transport of Section 5 spreads the conclusion over the same algebra to every elementary group of rank at least two, to the unit group  $A^\times$ , and to every positive-rank  $\mathrm{GL}_r(A)$ ; the formal development records this full reach, with the countability of  $A$  derived from finite type over the finite field rather than assumed.

**Remark 6.12** (The trust surface of the scheme). The matrix identities of this section are verifiable by finite computation from the two Leavitt relations; beyond them, the section inputs only the Whitehead lemma, the two-relator obstruction and the explicit commutator certificates of Section 5, and hypotheses (i) and (ii) of Theorem 6.10. The witness is elementary for every ambient ring: Proposition 6.7 uses only Lemmas 6.5–6.6 and the Whitehead lemma, and no perfectness theorem; with the classical perfectness

Leavitt/GeneralCornerTheore  
m  
corner\_not\_isSofic  
compressionSetup  
coreEmbedding\_compressorSet\_  
generate  
witnessEmbedding  
corner\_not\_isSofic\_of\_charTw  
o  
Leavitt/GeneralScheme  
uUnit  
zUnit  
uMatrix\_mul\_uPrimeMatrix  
Criterion/CompressionSetup  
CompressionSetup  
Formalized at every adjacent  
pair  $(m, m + 1)$  with  $m \geq 2$ ,  
at exactly the printed  
hypotheses: nontrivial ring,  
binary family, (T) at both  
ranks, and the membership  
hypothesis (ii); finite  
generation and countability  
of the groups are derived  
from (T). The witness is the  
Whitehead diagonal in the  
plane  $(0, 1)$ , and corner\_not\_  
isSofic\_of\_charTwo  
discharges (ii) in  
characteristic two through  
the elementary words of  
Leavitt/GeneralRankWords.

Endpoint/ManuscriptState  
 ments  
 cor\_fgring\_exact  
 cor\_fgring\_countableFree  
 Criterion/CriterionAssembly  
 not\_isSofic\_of\_not\_isLEF  
 Leavitt/FamilyRankFour  
 compressionSetup  
 ambient\_not\_isSofic  
 elementary\_not\_isSofic  
 units\_not\_isSofic  
 gl\_not\_isSofic  
 FamilyRankFour assembles the  
 corollary over any nontrivial  
 finite-type algebra with a  
 binary family, discharging  
 the memberships rather than  
 assuming them; countability  
 is derived from finite type  
 over the finite field  
 (countable\_of\_finiteType),  
 not assumed. The ranks  
 reached are every  $r \geq 2$ ,  
 together with  $A^\times$  and every  
 $GL_r$ .

of  $V$ , the full corner copy of  $V$  is elementary as well (Remark 6.8). The entire membership burden therefore falls on hypothesis (ii), which concerns exactly two matrices,  $u$  and  $z$ : Section 7 discharges it by explicit elementary factorizations in characteristic two, and Section 8 discharges it over every finite field via  $K_1(L) = 0$  and the GE property, with the  $K_1$  input available in elementary form (Remark 8.4).

## 7. THE EXPLICIT REALIZATION: NO $K$ -THEORY

In this section  $k = \mathbb{F}_2$  and  $R = L_{\mathbb{F}_2}(1, 2)$ ; the matrix identities and the argument extend without change to any finite field of characteristic two. Fix  $m \geq 3$ , put  $n = m + 1$ , and let  $G = \text{EL}_n(R)$ ,  $\Gamma$ ,  $u$ ,  $z$ ,  $v$ ,  $K$ ,  $J$  be as in Section 6 with  $A = R$ . We now exhibit the two matrices  $u$  and  $z$  as explicit products of elementary matrices, discharging hypothesis (ii) of Theorem 6.10 with no  $K$ -theoretic input.

**Proposition 7.1** (Explicit elementary factorization of the compressor). *For  $1 \leq i \leq m$ , put*

$$U_i = x_{ni}(1 + t_0) x_{in}(1) x_{ni}(1 + s_0) x_{in}(t_0) \in \text{EL}_n(R).$$

*Then*

$$u = U_m U_{m-1} \cdots U_1.$$

*In particular  $u \in \text{EL}_n(R)$ , as a product of  $4m$  elementary matrices.*

*Proof.* By Lemma 5.9(b), applied in the pair of coordinates  $(i, n)$ , the matrix  $U_i$  agrees with the identity outside the coordinates  $i$  and  $n$  and equals  $\begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}$  on them; explicitly,

$$U_i = I_n + (s_0 + 1)E_{ii} + s_1 t_1 E_{in} + (t_0 + 1)E_{nn},$$

signs being immaterial in characteristic two. Put  $P_k = U_k U_{k-1} \cdots U_1$ . We claim, by induction on  $k$ , that

$$P_k = \sum_{i \leq k} (s_0 E_{ii} + s_1 t_1 t_0^{i-1} E_{in}) + \sum_{k < i \leq m} E_{ii} + t_0^k E_{nn}.$$

For  $k = 1$  this is the displayed form of  $U_1$ . For the inductive step, compute the rows of  $P_{k+1} = U_{k+1} P_k$ . For  $i \notin \{k+1, n\}$ , the  $i$ -th row of  $U_{k+1}$  is  $e_i^\top$ , so the  $i$ -th row of  $P_{k+1}$  equals that of  $P_k$ . The  $(k+1)$ -st row of  $U_{k+1}$  is  $s_0 e_{k+1}^\top + s_1 t_1 e_n^\top$ ; since the  $(k+1)$ -st row of  $P_k$  is  $e_{k+1}^\top$  and the  $n$ -th row of  $P_k$  is  $t_0^k e_n^\top$ , the  $(k+1)$ -st row of  $P_{k+1}$  is  $s_0 e_{k+1}^\top + s_1 t_1 t_0^k e_n^\top$ . The  $n$ -th row of  $U_{k+1}$  is  $t_0 e_n^\top$ , so the  $n$ -th row of  $P_{k+1}$  is  $t_0^{k+1} e_n^\top$ . This is the claimed form for  $k+1$ .

Finally, for the left comb (25),  $s_1 t_1 t_0^{i-1} = s_1 (s_0^{i-1} s_1)^* = s_1 \alpha_i^*$  for  $1 \leq i \leq m$ , and  $t_0^m = \alpha_n^*$ . Hence

$$P_m = \sum_{i=1}^m (s_0 E_{ii} + s_1 \alpha_i^* E_{in}) + \alpha_n^* E_{nn} = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix} = u,$$

by (26). □

**Lemma 7.2** (Explicit elementary factorization of the involution).

$$z = x_{1n}(s_1) x_{n1}(t_1) x_{1n}(s_1) \in \text{EL}_n(R).$$

*Proof.* Lemma 5.9(a), applied in the pair of coordinates  $(1, n)$ , produces exactly the matrix  $z$  of (29).  $\square$

**Theorem 7.3** (The explicit spine). *For every  $m \geq 3$ , the group  $\text{EL}_{m+1}(R)$  over  $R = L_{\mathbb{F}_2}(1, 2)$  is infinite, finitely generated, has property (T), and is not sofic.*

*Proof.* The ring  $R = L_{\mathbb{F}_2}(1, 2)$  is finitely generated (by  $s_0, s_1, t_0, t_1$  over the prime field). Proposition 7.1 and Lemma 7.2 verify  $u, z \in \text{EL}_{m+1}(R)$ , and Corollary 6.11 applies.  $\square$

*Proof of Theorem A.* Theorem 7.3 gives every  $m \geq 3$ ; the case  $m = 3$  is the group  $G = \text{EL}_4(R)$ . For  $m \in \{1, 2\}$ , the elementary rank equivalence of Proposition 5.8 is an explicit isomorphism  $\text{EL}_{m+1}(R) \cong \text{EL}_4(R)$ , and all four properties transport along it; in particular  $\text{EL}_2(R)$  and  $\text{EL}_3(R)$  have property (T) although Theorem 5.7 says nothing at rank two. The two auxiliary matrices are exhibited as explicit products of elementary matrices:  $u$  by Proposition 7.1 and  $z$  (hence  $v = zu$ ) by Lemma 7.2. The remaining memberships, of the two witness matrices  $j(x_0)$  and  $j(x_1)$ , follow from the explicit commutator certificates of Lemmas 6.5 and 6.6 together with the Whitehead identity of Lemma 5.5, which converts each certificate into an explicit elementary word. No  $K$ -theory is used.  $\square$

**Remark 7.4** (Scope of the explicit spine). The factorizations of this section are identities in characteristic two and hold over any field of characteristic two; finiteness of the field is used only through finite generation of the ring in Theorem 5.7. Thus  $\text{EL}_{m+1}(L_k(1, 2))$  is infinite, finitely generated, Kazhdan, and nonsofic, with all memberships explicit, for every finite field  $k$  of characteristic two and every  $m \geq 3$ —and then for every  $m \geq 1$  by the elementary rank equivalence (Proposition 5.8). For rank  $n = 3$  (that is,  $m = 2$ ), hypothesis (i) of Theorem 6.10 for  $\text{EL}_2$  is not supplied by Theorem 5.7; it is exactly here that the rank equivalence substitutes. Section 8 reaches the remaining groups of Theorem B—the unit group and every  $\text{GL}_r$ , over every finite field—through self-similarity.

## 8. RANK TWO AND THE FULL PANORAMA

The compression–centralizer mechanism closes already in rank two, over a different and smaller group  $\Gamma$ : the full diagonal unit group, with property (T) transported through self-similarity rather than supplied by Theorem 5.7. Since all the groups in question turn out to be isomorphic, the value of the rank-two realization is not a new group but a map of the mechanism’s boundary—it shows how little room the criterion actually needs. Combined with matrix self-similarity and the vanishing of  $K_1$  (Proposition 8.2, with the elementary proof of Appendix A), this yields Theorem B.

Leavitt/GeneralRankWords  
 piece  
 piece\_val  
 comb  
 comb\_mem  
 comb\_val  
 combMatrix  
 partialComb\_val  
 Leavitt/RankFourCompressor  
 s  
 compressor  
 compressorMatrix  
 compressorPiece  
 compressorPiece\_val  
 compressor\_val  
 Every rank: piece and comb  
 are the displayed products for  
 arbitrary  $m$ ,  $\text{comb\_val}$  is the  
 printed row induction  
 ( $\text{partialComb\_val}$ ), all via a  
 coordinate-plane embedding  
 of the two-by-two identities.  
 The rank-four instance, valid  
 in every characteristic, is  
 kept for the concrete spine.

Leavitt/GeneralRankWords

involutionWord  
involutionWord\_mem  
involutionWord\_val  
involutionWord\_sq

Leavitt/RankFourCompressor  
s

involution  
involutionMatrix  
involution\_sq

Leavitt/RawSwapCompressor  
s

rawInvolutionWord\_val  
Every rank, with the  
displayed value and  $z^2 = 1$ ;  
the rank-four instance is  
valid in every characteristic.  
RawSwapCompressors  
additionally proves that the  
uncorrected

three-transvection word  
already centralizes,  
conjugates, and generates, so  
the order-two relation is a  
convenience rather than an  
input.

Endpoint/MainResults  
universalLeavittEL4\_not\_isSo  
fic

universalLeavitt\_profile  
Leavitt/RankTwoCompressio  
n

rankTwo\_not\_isSofic  
rankTwoInvolution\_conj\_upper  
Nil  
rankTwoInvolution\_conj\_lower  
Nil

### 8.1. The two-by-two realization.

**Theorem 8.1** (Two-by-two Leavitt-corner theorem). *Let  $A$  be a nontrivial associative unital ring carrying a binary Leavitt family. Suppose that  $A^\times$  and  $\text{EL}_2(A)$  have property (T) and that  $\text{GL}_2(A) = \text{EL}_2(A)$ . Then  $\text{EL}_2(A)$  is not sofic.*

*Proof.* Put  $G = \text{EL}_2(A) = \text{GL}_2(A)$  and

$$\Gamma = \{\text{diag}(a, 1) \mid a \in A^\times\} \leq G.$$

Then  $\Gamma \cong A^\times$  is a property-(T) group, hence finitely generated, and it is infinite because  $\rho : V \hookrightarrow A^\times$  is injective (Lemma 5.11).

*The compressor.* Put

$$u = \begin{pmatrix} s_0 & p_1 \\ 0 & t_0 \end{pmatrix}, \quad u' = \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix}.$$

Using  $p_1 s_0 = s_1(t_1 s_0) = 0$ ,  $t_0 p_1 = (t_0 s_1)t_1 = 0$ ,  $t_0 s_0 = 1$ , and  $p_0 + p_1 = 1$ :

$$uu' = \begin{pmatrix} s_0 t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = I_2, \quad u'u = \begin{pmatrix} t_0 s_0 & t_0 p_1 \\ p_1 s_0 & p_1 + s_0 t_0 \end{pmatrix} = I_2,$$

so  $u \in \text{GL}_2(A) = G$  with  $u^{-1} = u'$ . For  $a \in A$ , put

$$\kappa(a) = p_1 + s_0 a t_0.$$

Then  $\kappa(1) = 1$  and  $\kappa(a)\kappa(a') = p_1 + s_0 a a' t_0 = \kappa(a a')$  by the same identities, so  $\kappa$  is unital and multiplicative and restricts to a group homomorphism  $A^\times \rightarrow A^\times$ , injective because  $t_0 \kappa(a) s_0 = t_0 p_1 s_0 + t_0 s_0 a t_0 s_0 = a$ ; and

$$\begin{aligned} u \text{diag}(a, 1) u^{-1} &= \begin{pmatrix} s_0 a & p_1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix} \\ &= \begin{pmatrix} s_0 a t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = \text{diag}(\kappa(a), 1). \end{aligned}$$

Hence

$$K := u \Gamma u^{-1} = \{\text{diag}(\kappa(a), 1) \mid a \in A^\times\} \leq \Gamma.$$

*The involution.* Put

$$z = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}, \quad \text{so} \quad z^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = I_2,$$

using  $p_0 s_1 = 0$ ,  $t_1 p_0 = 0$ ,  $t_1 s_1 = 1$ ; thus  $z \in \text{GL}_2(A) = G$  and  $z^{-1} = z$ . For any  $c \in A$ ,

$$(32) \quad z \text{diag}(c, 1) z = \begin{pmatrix} p_0 c p_0 + p_1 & p_0 c s_1 \\ t_1 c p_0 & t_1 c s_1 \end{pmatrix}.$$

The element  $\kappa(a)$  satisfies  $p_0 \kappa(a) = \kappa(a) p_0 = s_0 a t_0$ ,  $t_1 \kappa(a) = t_1$ , and  $\kappa(a) s_1 = s_1$ ; substituting into (32) gives  $z \text{diag}(\kappa(a), 1) z = \text{diag}(s_0 a t_0 + p_1, 1) = \text{diag}(\kappa(a), 1)$ . Hence  $z$  centralizes  $K$ , and  $v := zu \in G$  satisfies  $v \Gamma v^{-1} = z K z^{-1} = K$ .

*Generation.* For any  $a \in A$ , the elements  $1 + p_0 a t_1$  and  $1 + s_1 a p_0$  are units of  $A$ , since  $(p_0 a t_1)^2 = 0$  and  $(s_1 a p_0)^2 = 0$  (using  $t_1 p_0 = 0$  and  $p_0 s_1 = 0$ );

so the diagonal matrices  $\text{diag}(1 + p_0at_1, 1)$  and  $\text{diag}(1 + s_1ap_0, 1)$  lie in  $\Gamma$ . Substituting  $c = 1 + p_0at_1$  and  $c = 1 + s_1ap_0$  into (32) and using  $t_1p_0 = 0$ ,  $p_0s_1 = 0$ ,  $t_1s_1 = 1$ ,  $p_0^2 = p_0$ :

$$(33) \quad z \text{diag}(1 + p_0at_1, 1) z = \begin{pmatrix} 1 & p_0a \\ 0 & 1 \end{pmatrix} = x_{12}(p_0a),$$

$$(34) \quad z \text{diag}(1 + s_1ap_0, 1) z = \begin{pmatrix} 1 & 0 \\ ap_0 & 1 \end{pmatrix} = x_{21}(ap_0).$$

Next, the *corner swap*

$$d = s_1t_0 + s_0t_1 \in A^\times, \quad d^2 = s_1t_1 + s_0t_0 = 1,$$

gives  $D = \text{diag}(d, 1) \in \Gamma$  with  $D^{-1} = D$ . Conjugation by  $D$  multiplies an upper coefficient on the left by  $d$  and a lower coefficient on the right by  $d$ ; since  $dp_0s_0t_1 = d s_0t_1 = p_1$  and  $s_1t_0p_0d = s_1t_0d = p_1$ ,

$$D x_{12}(p_0(s_0t_1a)) D = x_{12}(p_1a), \quad D x_{21}((as_1t_0)p_0) D = x_{21}(ap_1),$$

with the inner matrices supplied by (33) and (34). Finally

$$x_{12}(a) = x_{12}(p_0a) x_{12}(p_1a), \quad x_{21}(a) = x_{21}(ap_0) x_{21}(ap_1),$$

so  $\langle \Gamma, z \rangle$  contains every elementary matrix; hence  $\langle \Gamma, z \rangle = G$ , and since  $z = vu^{-1}$ ,

$$G = \langle \Gamma, u, v \rangle, \quad u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma.$$

*The commuting corner.* Put  $J = \langle \text{diag}(\rho(x_0), 1), \text{diag}(\rho(x_1), 1) \rangle \leq \Gamma$ ; membership is automatic here, since  $\rho(g) \in A^\times$ . By Lemma 5.11 the pair inherits the relators (24) and the noncommutation of  $x_0, x_1$ , so  $J$  is a two-generator non-LEF subgroup by Lemma 5.12. The subgroups  $K$  and  $J$  commute—indeed  $K$  commutes with the whole diagonal corner copy:

$$\begin{aligned} \kappa(a)\rho(g) &= (p_1 + s_0at_0)(p_0 + s_1U_g t_1) = s_1U_g t_1 + s_0at_0 \\ &= (p_0 + s_1U_g t_1)(p_1 + s_0at_0) = \rho(g)\kappa(a), \end{aligned}$$

using  $p_1s_1 = s_1$ ,  $t_1p_1 = t_1$ ,  $t_0p_0 = t_0$ ,  $p_0s_0 = s_0$ ,  $t_0s_1 = 0$ ,  $t_1s_0 = 0$ . And  $K \cap J$  is trivial: if  $\kappa(a) = \rho(g)$ , then sandwiching between  $t_0$  and  $s_0$  gives  $a = t_0\kappa(a)s_0 = t_0\rho(g)s_0 = 1$ , so  $\rho(g) = \kappa(1) = 1$  and  $g = 1$  by injectivity of  $\rho$ .

*Conclusion.* The data  $G, \Gamma, Q = \{u, v\}$ ,  $q_0 = u$ ,  $K, J$  satisfy all hypotheses of Corollary 4.2:  $G$  and  $\Gamma$  are finitely generated Kazhdan groups,  $\Gamma$  is infinite,  $G = \langle \Gamma, Q \rangle$ , each  $q \in Q$  conjugates  $\Gamma$  into itself with  $q_0\Gamma q_0^{-1} = K$ , and  $J \leq \Gamma$  is finitely generated with  $[K, J] = 1$  and  $K \cap J = \{1\}$ . If  $G$  were sofic,  $J$  would be LEF, contradicting Lemma 5.12.  $\square$

**8.2.  $K_1$  vanishes and  $\text{GL} = \text{EL}$ .** Every ring-theoretic fact this subsection needs is proved in Appendix A; for context we recall the theorem of Ara–Goodearl–Pardo [3], of which the facts below are the  $L_k(1, 2)$ -instances: a purely infinite simple unital ring  $R$  is a GE-ring (every matrix in  $\text{GL}_r(R)$  is a product of elementary matrices and one  $\text{diag}(a, 1, \dots, 1)$ ), and

$R^\times/[R^\times, R^\times] \cong K_1(R)$ . Neither statement is invoked in any proof of this paper.

**Proposition 8.2.** *Let  $k$  be any field and  $L = L_k(1, 2)$ . Then:*

- (i)  $K_1(L) = 0$ ;
- (ii) *the unit group  $L^\times$  is perfect*;
- (iii)  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$  for every  $r \geq 2$ .

*Proof.* Parts (i) and (iii) are proved, elementarily and in full, in Appendix A: Theorem A.2 establishes  $K_1(L) = 0$  in Whitehead form—every unit  $u$  has  $\mathrm{diag}(u, 1) \in \mathrm{EL}_2(L)$ —and Corollary A.23 deduces  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$  for every  $r \geq 2$ , by rank-two elimination and flattening transport. The only input of the appendix beyond the Leavitt relations is strong division (Lemma A.3), which holds because  $L$  is purely infinite simple: the graph criteria of [1] are immediate for the rose  $R_2$  with one vertex and two loops—the single vertex admits no nontrivial hereditary saturated subset, every cycle has an exit, and the vertex lies on a cycle.

Part (ii) is Corollary A.23 as well:  $\Theta_{\mathcal{C}_3}$  identifies  $L^\times$  with  $\mathrm{GL}_3(L) = \mathrm{EL}_3(L)$ , and every elementary generator is a commutator by the Steinberg relation, so  $L^\times$  is its own commutator subgroup. (Classically, (ii) also follows from (i) and the Ara–Goodearl–Pardo identification of  $K_1$  with the abelianized unit group.)

For an independent confirmation of (i), the  $K$ -theory of Leavitt path algebras [2] gives the exact sequence

$$K_1(k) \xrightarrow{1-N^T} K_1(k) \longrightarrow K_1(L) \longrightarrow K_0(k) \xrightarrow{1-N^T} K_0(k),$$

where  $N$  is the adjacency matrix—here the  $1 \times 1$  matrix  $(2)$ , so  $1 - N^T = -1$ , an isomorphism on both  $K_0(k) \cong \mathbb{Z}$  and  $K_1(k) \cong k^\times$ , whence  $K_1(L) = 0$ .  $\square$

**Remark 8.3.** For an independent recent account of generation phenomena for unit groups and matrix groups over Leavitt algebras, consistent with Proposition 8.2 and with the self-similarity of Proposition 5.4, see [12].

**Remark 8.4** (An elementary route to  $K_1(L) = 0$ ). The classical computation of  $K_1(L)$  goes through the localization sequence of [2]. Appendix A replaces this input by a complete elementary proof—a Gaussian elimination over the binary tree, in the spirit of a Wiener–Hopf factorization whose “symbol” is a matrix pencil with two shift variables. In outline: a width reduction narrows every unit, modulo the Whitehead class group  $H$  of Definition A.1, to one whose value lies in the degree window  $[-1, 1]$ ; at a sufficiently deep leaf set such a narrow unit becomes a matrix pencil with scalar coefficient matrices  $(A_0, A_1, C, B_0, B_1)$  over the five-dimensional space  $kt_0 \oplus kt_1 \oplus k \oplus ks_0 \oplus ks_1$ , indexed by a pair of complete leaf sets; and a single loop with two exits eliminates the pencil, splitting one column per pass (the index shift of the Wiener–Hopf analogy) until either a reshape by two tree tables lands the value in a one-sided degree window, or the stacked matrix  $[B_0; B_1]$  acquires a scalar left inverse and the *inverse* unit lands there instead; one-sided windows

KOne/RefineLoopDischarge  
K1\_trivial  
stableUnits\_eq\_top\_holds  
KOne/AllRanksElementary  
elementaryGroup\_eq\_top  
glAll\_eq\_elementary  
binaryLeavittUnits\_perfect

KOne/RefineLoopDischarge  
K1\_trivial  
narrowReduction\_holds  
scalarReduction\_holds

die by a block-move induction whose base is the keystone Lemma A.16. Every step is carried out in Appendix A, so the  $K_1$ -theoretic input of [2] is eliminated from the trust surface of Theorem B; the citation is retained as an independent confirmation.

**Remark 8.5** (Effective Wiener–Hopf factorization over the binary tree).

The proof in Appendix A is an algorithm, and unwinding its run records a factorization of independent interest: *every unit of  $L = L_k(1, 2)$  is an explicit product of cylinder unipotents  $1 + s_v x t_w$ , code-scalar units (matrices over  $k$  transported along a complete leaf set), tree-table units (elements of Thompson’s group  $V$ ), and one unit of nonpositive degree window conjugated by two tree tables*—the moves that the width reduction, the loop, and the two exits actually perform, in the order they perform them. The run is effective, with bounds read off the printed proof: for  $u$  supported on monomials of depth at most  $d$ , the pencil sits over the full code of depth  $m + 1 \leq d + 1$ , so every code involved has at most  $2^{d+2}$  words; each loop pass is one Gaussian elimination over  $k$  at that size; and the loop counter of Theorem A.22 starts at  $2^{m+1}$  columns and exits by  $2^{m+2}$ , so at most  $2^{d+1}$  passes occur. The factorization, and with it an explicit factorization of any  $\text{diag}(u, 1, \dots, 1)$  into elementary matrices, is therefore computed in time polynomial in  $2^d$  and in the number of monomials of  $u$ , in the model that counts arithmetic operations in  $k$  and takes the strong-division witnesses of Lemma A.3, for the explicit elements the moves invoke them on, as given primitives. In the classical Wiener–Hopf picture the loop is index normalization: each refinement is the index shift that trades a constant diagonal for a shift atom, and the two exits are the two triangular factorizations. We are not aware of a prior effective factorization—or of any elementary proof of  $K_1(L_k(1, 2)) = 0$ —in the literature; the localization-sequence computation of [2] is neither elementary nor effective.

### 8.3. Finite fields: all ranks and the unit group.

**Theorem 8.6.** *Let  $k$  be a finite field and  $L = L_k(1, 2)$ . Then  $\text{EL}_2(L) = \text{GL}_2(L)$  is infinite, finitely generated, has property (T), and is not sofic.*

*Proof.* The ring  $L$  is finitely generated, by  $s_0, s_1, t_0, t_1$  and the finitely many scalars. By Theorem 5.7, Proposition 5.8, and Lemma 5.6,  $\text{EL}_4(L)$  is a finitely generated Kazhdan group; by Proposition 8.2(iii),  $\text{EL}_4(L) = \text{GL}_4(L)$ ; and by Proposition 5.4 with the left comb  $\mathcal{C}_4$ ,  $\text{GL}_4(L) \cong L^\times$ . Hence  $L^\times$  is a finitely generated Kazhdan group. Similarly,  $\text{GL}_2(L) = \text{EL}_2(L)$  by Proposition 8.2(iii), and  $\text{GL}_2(L) \cong L^\times$  by Proposition 5.4 with  $\mathcal{C}_2 = (s_1, s_0)$ ; so  $\text{EL}_2(L)$  is Kazhdan as well. Theorem 8.1 with  $A = L$  now shows that  $\text{EL}_2(L)$  is not sofic. It is infinite, containing  $\text{diag}(\rho(V), 1) \cong V$ , and finitely generated, being Kazhdan.  $\square$

**Theorem 8.7** (Panorama). *Let  $k$  be a finite field and  $L = L_k(1, 2)$ . Then the groups*

$$L^\times = \text{GL}_1(L), \quad \text{GL}_r(L) \ (r \geq 1), \quad \text{EL}_r(L) \ (r \geq 2)$$

Leavitt/RankTwoCompression  
n  
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KOne/AllRanksElementary  
binaryLeavittRankTwo\_not\_isSofic  
Leavitt/UnitsGLProfile  
binaryLeavittUnits\_profile  
binaryLeavittGL\_profile  
Instantiated over  $L_k(1, 2)$   
for every finite field, by the  
rank-two compression itself.  
Endpoint/ManuscriptStatement  
nts  
theoremB\_exact  
Endpoint/MainResults  
binaryLeavitt\_finiteField\_profile  
binaryLeavittUnits\_not\_isSofic  
binaryLeavittGL\_not\_isSofic  
universalLeavittEL3\_not\_isSofic  
Leavitt/PrefixCode  
prefixRingEquiv  
prefixRingEquiv\_apply  
prefixUnitsEquiv  
KOne/AllRanksElementary  
elementaryGroup\_eq\_top  
glAll\_eq\_elementary  
The printed statement in one  
type is theoremB\_exact, shared  
with Theorem B. The profiles  
and isomorphisms travel by  
self-similarity; the equalities  
 $\text{GL}_r = \text{EL}_r$  are the  
elementary Whitehead chain  
of Appendix A, internal and  
free of external  $K$ -theory.

are mutually isomorphic—explicitly,  $\Theta_C : \mathrm{GL}_r(L) \rightarrow L^\times$  is an isomorphism for every ordered complete leaf set  $C$  of size  $r$ , and  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$  for  $r \geq 2$ —and each is an infinite, finitely generated, property-(T) group that is not sofic.

*Proof.* The isomorphisms are Proposition 5.4, and the equalities  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$  are Proposition 8.2(iii). Each of the listed groups is therefore isomorphic to  $\mathrm{EL}_2(L)$ , which is infinite, finitely generated, Kazhdan, and nonsofic by Theorem 8.6; all four properties are isomorphism-invariant.  $\square$

*Proof of Theorem B.* Part (i) and part (ii) are exactly Theorem 8.7, with the explicit isomorphism (2) given by Proposition 5.4.  $\square$

**Remark 8.8** (Two routes to rank four). For a finite field of characteristic two and rank  $m + 1 \geq 4$ , the membership hypotheses are discharged twice over: explicitly, with no  $K$ -theory, in Section 7; and via Proposition 8.2 here. Theorem B in its full strength—all ranks down to two, all finite fields, and the unit group itself—uses the second route. The step that makes the panorama characteristic-free is Proposition 6.7: the witness is elementary because its memberships go through single-commutator certificates and the Whitehead lemma, identities valid over every ring, rather than through transvection factorizations of transpositions, which require characteristic two.

## 9. FINITELY PRESENTED NONSOFIC COVERS

Nonsficity of a finitely generated group is witnessed by a finite fragment of its multiplication table; imposing that fragment as a presentation produces finitely presented nonsofic covers. The covers are proved nonsofic directly, from the forbidden table itself; quotient closure of soficity—which in fact fails, by Remark 10.6—plays no role.

**Definition 9.1.** Let  $G$  be a group,  $F \subseteq G$  finite with  $1 \in F$ , and  $\varepsilon > 0$ . An  $(F, \varepsilon)$ -model of  $G$  is a map  $\varphi : F \cup F \cdot F \rightarrow \mathrm{Sym}(Y)$ , for some finite set  $Y$ , such that

$$(35) \quad d_H(\varphi(g)\varphi(h), \varphi(gh)) \leq \varepsilon \quad (g, h \in F),$$

$$(36) \quad d_H(\varphi(g), \varphi(h)) \geq 1 - \varepsilon \quad (g, h \in F, g \neq h).$$

**Lemma 9.2** (Finite-table characterization). *A countable group  $G$  is sofic if and only if it admits an  $(F, \varepsilon)$ -model for every finite  $F \ni 1$  and every  $\varepsilon > 0$ .*

*Proof.* If  $\sigma_n$  is a sofic approximation, then for fixed  $(F, \varepsilon)$  the restriction of  $\sigma_n$  to  $F \cup F \cdot F$  is an  $(F, \varepsilon)$ -model for all large  $n$ : (35) is Definition 2.1(a), and (36) follows from (a) and (b) since  $d_H(\sigma_n(g), \sigma_n(h)) \geq d_H(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(h)) - d_H(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(g)) = d_H(\sigma_n(gh^{-1}), \mathrm{id}) - o(1)$ .

Conversely, exhaust  $G$  by finite symmetric sets  $F_1 \subseteq F_2 \subseteq \dots$  containing 1, and let  $\varphi_n$  be an  $(F_n, 1/n)$ -model on  $Y_n$ ; replacing  $Y_n$  by many disjoint copies of itself we may assume  $|Y_n| \rightarrow \infty$ . Define  $\sigma_n(g) = \varphi_n(g)$  for  $g \in F_n$  and  $\sigma_n(g) = \mathrm{id}$  otherwise. Fix  $g, h \in G$ ; for  $n$  large,  $g, h, gh \in F_n$ , and (35)

Covers/TableCover

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gives Definition 2.1(a). For faithfulness, first note that  $p_n = \varphi_n(1)$  satisfies  $d_H(p_n^2, p_n) \leq 1/n$  by (35) with  $g = h = 1$ , and  $d_H(p_n^2, p_n) = d_H(p_n, \text{id})$  by right invariance of  $d_H$ ; so  $d_H(\varphi_n(1), \text{id}) \leq 1/n$ . Then for  $g \neq 1$  and  $n$  large,

$$d_H(\sigma_n(g), \text{id}) \geq d_H(\varphi_n(g), \varphi_n(1)) - d_H(\varphi_n(1), \text{id}) \geq 1 - \frac{2}{n},$$

by (36). Hence  $\sigma_n$  is a sofic approximation.  $\square$

**Theorem 9.3** (Finite multiplication-table covers). *Let  $G$  be a finitely generated group that is not sofic. Then there is an infinite finitely presented group  $H$  that is not sofic, together with a surjection  $H \twoheadrightarrow G$ .*

Covers/TableCover  
tableGroup\_no\_model  
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truction

*Proof.* Finite groups are sofic, so  $G$  is infinite. By Lemma 9.2 there are a finite set  $F \subseteq G$  with  $1 \in F$  and an  $\varepsilon_0 > 0$  such that  $G$  admits no  $(F, \varepsilon_0)$ -model; enlarging  $F$ , we may assume  $F$  is symmetric and generates  $G$ . Define the finitely presented group

$$(37) \quad H = H_F = \langle x_g \ (g \in F \cup F \cdot F) \mid x_1 = 1, \ x_g x_h = x_{gh} \ (g, h \in F) \rangle.$$

The assignments  $x_g \mapsto g$  satisfy the relations in  $G$ , and  $F$  generates  $G$ , so there is a surjection

$$(38) \quad \pi : H \twoheadrightarrow G, \quad \pi(x_g) = g.$$

In particular  $H$  is infinite, and  $\pi(x_g) = g \neq h = \pi(x_h)$  shows  $x_g \neq x_h$  in  $H$  for distinct  $g, h \in F \cup F \cdot F$ .

Suppose  $H$  were sofic, with sofic approximation  $\sigma_n : H \rightarrow \text{Sym}(Y_n)$ . Define  $\varphi_n : F \cup F \cdot F \rightarrow \text{Sym}(Y_n)$  by  $\varphi_n(g) = \sigma_n(x_g)$ . For  $g, h \in F$ , the relation  $x_g x_h = x_{gh}$  holds in  $H$ , so

$$d_H(\varphi_n(g)\varphi_n(h), \varphi_n(gh)) = d_H(\sigma_n(x_g)\sigma_n(x_h), \sigma_n(x_g x_h)) \rightarrow 0.$$

For distinct  $g, h \in F$ , the element  $x_g x_h^{-1} \neq 1$  of  $H$  gives, as in the proof of Lemma 9.2,

$$d_H(\varphi_n(g), \varphi_n(h)) \geq d_H(\sigma_n(x_g x_h^{-1}), \text{id}) - o(1) \rightarrow 1.$$

Hence for  $n$  large,  $\varphi_n$  is an  $(F, \varepsilon_0)$ -model of  $G$ —note that (35) and (36) refer only to the multiplication table of  $F$  inside  $G$ , which is faithfully copied by the generators of  $H$ . This contradicts the choice of  $(F, \varepsilon_0)$ . Therefore  $H$  is not sofic.  $\square$

**Theorem 9.4** (Kazhdan covers). *Let  $G$  be a nonsofic group with property (T). Then there is an infinite finitely presented property-(T) group  $H$  that is not sofic, together with a surjection  $H \twoheadrightarrow G$ .*

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tation  
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nted\_cover\_of\_not\_isSofic  
Shalom's theorem is proved  
internally, not cited.

*Proof.* Property-(T) groups are finitely generated, and  $G$  is infinite as before. By Shalom's theorem [19],  $G$  is a quotient of a finitely presented property-(T) group: fix a surjection  $\theta : P \twoheadrightarrow G$  with  $P$  finitely presented and Kazhdan. Choose  $(F, \varepsilon_0)$  as in the proof of Theorem 9.3, and for each  $g \in F \cup F \cdot F$  choose a lift  $\tilde{g} \in P$  with  $\theta(\tilde{g}) = g$ , taking  $\tilde{1} = 1$ . Define

$$(39) \quad H = P / \langle\langle \tilde{g} \tilde{h} \tilde{gh}^{-1} : g, h \in F \rangle\rangle.$$

Then  $H$  is finitely presented (finitely many new relators over the finitely presented  $P$ ) and has property (T) (a quotient of  $P$ ). Each relator lies in  $\ker \theta$ , since  $\theta(\tilde{g}\tilde{h}\tilde{g}\tilde{h}^{-1}) = gh(gh)^{-1} = 1$ ; hence  $\theta$  descends to a surjection  $H \twoheadrightarrow G$ , and  $H$  is infinite. Writing  $y_g \in H$  for the image of  $\tilde{g}$ , we have  $y_1 = 1$ ,  $y_g y_h = y_{gh}$  for  $g, h \in F$ , and  $y_g \neq y_h$  for distinct  $g, h \in F$  (their images in  $G$  differ). If  $H$  were sofic, then  $\varphi_n(g) = \sigma_n(y_g)$  would be an  $(F, \varepsilon_0)$ -model of  $G$  for large  $n$ , exactly as in the proof of Theorem 9.3, a contradiction.  $\square$

*Proof of Theorem C.* The first two assertions are Theorems 9.3 and 9.4. For the final assertion, apply Theorem 9.4 to the infinite finitely generated property-(T) nonsocfic group  $L^\times$  of Theorem B, over any fixed finite field; the resulting  $H$  surjects onto  $L^\times$ , and composing with the inverses of the isomorphisms  $\Theta_C$  of Theorem B(ii) exhibits a quotient of  $H$  isomorphic to each group listed there.  $\square$

**Remark 9.5** (Effectivity). The covers are effective in the logical sense: nonsocficity of  $G$  is equivalent, by Lemma 9.2, to the existence of a finite witness  $(F, \varepsilon_0)$ , and any witness yields  $H$  by (37) or (39). The proof establishes the existence of a witness without computing one, and no claim is made that  $L^\times$  itself is finitely presented.

## 10. CONCLUDING REMARKS

**Remark 10.1** (The trust surface of Theorem A). Theorem A is independent of  $\mathrm{GL}_r = \mathrm{EL}_r$  and of  $K_1(L) = 0$ : the memberships of  $u$  and  $z$  are explicit factorizations, the remaining memberships follow from the Whitehead identity and the explicit commutator certificates of the witness pair, and the only analytic inequality proved in the paper is the co-area estimate of Lemma 2.7—the approximation-theoretic depth resides in the cited theorems of Kun and Kun–Thom. The argument concerns permutation models only; hyperlinearity of these groups, and the Connes embedding problem, remain open.

**Remark 10.2** (Two compressors, one image). The pair  $Q = \{u, v\}$  resolves a genuine tension: generation of  $G$  requires elements that move  $\Gamma$ , while compression requires elements that map  $\Gamma$  into itself. The present realization divides the two roles. The compressor  $u$  seats  $K = u\Gamma u^{-1}$ ; the involution  $z = vu^{-1}$  generates the missing last row and column of  $\mathrm{EL}_n$  (Proposition 6.4); and the two compressors share the single image  $K$ , so the criterion sees one coherent compression while  $z$  carries the generation. The criterion itself is indifferent to the size of  $Q$ ; the algebra here simply supplies two natural compressors rather than one.

**Remark 10.3** (The median is canonical up to scale). The normalization  $f = s/(s + m)$  of Section 4 pins at  $1/2$ , and the value  $1/2$  is not a trace or a spectral constant: it is the fixed point of the exchange of source and target sizes, and it is a vertex-weighted median by construction. Median pinning is exactly the statement that a permutation compressing one expander

decomposition into another must asymptotically preserve the size profile—the conservative core of the argument, with exact conservation (Lemma 3.2) converting a one-sided drift estimate into a two-sided one at no cost.

**Remark 10.4** (What property  $(T)$  is used for). Property  $(T)$  enters three times, in three different roles: (a) through Theorem 2.9 applied to  $\Gamma$ , producing the source decomposition that is compressed; (b) through Theorem 2.9 applied to  $G$ , producing the ambient decomposition carrying the co-area inequality—and by Remark 4.3 this use is replaceable by any componentwise  $\ell^1$  Poincaré decomposition; (c) through the Kazhdan factor hypothesis of Theorem 2.10. The local Theorem 4.1 quantifies over the first two uses and leaves only the third to the black box.

**Remark 10.5** (Scope). The Leavitt-corner theorem (Theorem 6.10) applies to an arbitrary nontrivial unital ring carrying a binary Leavitt family, whenever the two adjacent elementary groups are Kazhdan and the two matrices  $u, z$  are elementary; for finite-type algebras over a finite field the Kazhdan hypothesis is automatic (Corollary 6.11), and over an arbitrary finitely generated ring with  $m \geq 3$  it is likewise automatic granting the full theorem of [9]. Finite fields are where all hypotheses are verified simultaneously (Theorem 8.7); characteristic two is where they are verified explicitly (Theorem 7.3); and the construction can already be carried out in rank two (Theorem 8.1).

**Remark 10.6** (Soficity is not closed under quotients). Theorem A settles a companion closure question in the negative. Every group is a quotient of a free group, and free groups are residually finite—hence LEF, hence sofic; presenting the nonsofic group of Theorem A as such a quotient exhibits a residually finite group with a nonsofic quotient. Consequently none of the three classes—residually finite, LEF, sofic—is closed under quotients. This is also why the covers of Section 9 are proved nonsofic directly, from the forbidden table: a quotient-closure argument would have sufficed there, and none is valid.

## APPENDIX A. AN ELEMENTARY AND EFFECTIVE PROOF OF

$$K_1(L_k(1, 2)) = 0$$

This appendix proves the vanishing of  $K_1$  announced in Remark 8.4, in the Whitehead form in which Section 8 uses it, by elementary means: no localization sequence, no spectra, and no input beyond the two Leavitt relations and one classical fact about  $L$  recorded as Lemma A.3 below. The proof is an algorithm—a Gaussian elimination over the binary tree—and the complexity accounting of Remark 8.5 reads off from it. The appendix is written to be read and checked entirely on paper.

Throughout,  $k$  is a field and  $L = L_k(1, 2)$ . Recall from Section 5 the word notation  $s_w, s_w^*$  (we write  $t_w = s_w^*$  for the word  $w$ , so  $t_w s_{w'} = \delta_{w, w'}$  for  $|w| = |w'|$ ), the cylinder idempotents  $p_w = s_w t_w$ , and complete leaf sets.

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exists\_residuallyFinite\_grou  
p\_with\_non\_residuallyFinite\_  
quotient  
Sofic/FreeGroupResiduallyFi  
nite  
freeGroup\_residuallyFinite  
isSofic\_freeGroup  
Residual finiteness of free  
groups is proved from scratch,  
by the ball action on reduced  
words; the three nonclosure  
statements share one  
presentation.*

A *complete code of depth*  $\leq r$  (resp.  $\geq r$ ) is a complete leaf set all of whose words have length  $\leq r$  (resp.  $\geq r$ ).

### A.1. The Whitehead class group.

**Definition A.1.** Let

$$H = \{u \in L^\times \mid \text{diag}(u, 1) \in \text{EL}_2(L)\}.$$

Since  $\text{diag}(uv, 1) = \text{diag}(u, 1) \text{diag}(v, 1)$ ,  $H$  is a subgroup of  $L^\times$ . The main theorem of this appendix is:

**Theorem A.2.**  $H = L^\times$ . *Equivalently, for every  $u \in L^\times$  and every  $r \geq 2$ , the matrix  $\text{diag}(u, 1, \dots, 1)$  is a product of elementary matrices in  $\text{EL}_r(L)$ .*

(The two forms are equivalent because  $\text{EL}_2 \hookrightarrow \text{EL}_r$  in the first two coordinates.) The single input beyond the Leavitt relations is the following division property, a standard consequence of pure infiniteness and simplicity of  $L$ —external to this appendix, which cites it from [1], and internal to the formal development, which proves it from the relations.

**Lemma A.3** (Strong division). *For every nonzero  $x \in L$  there are  $p, q \in L$  with  $pxq = 1$ .*

The first consequence is a Gaussian elimination for invertible two-by-two matrices; it is the rank-two case of the GE property of purely infinite simple rings [3] in elementary form, and rank two is all the sequel needs—every higher rank is reached by flattening (Corollary A.23).

**Lemma A.4** (Elementary diagonalization). *Let  $A$  be a nontrivial unital ring in which every nonzero element  $x$  admits  $p, q$  with  $pxq = 1$ , and let  $X \in \text{GL}_2(A)$ . Then there are  $E, F \in \text{EL}_2(A)$  and  $u \in A^\times$  with*

$$E X F = \text{diag}(u, 1).$$

*Proof.* Write  $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Assume first  $a \neq 0$ , and pick  $p, q$  with  $paq = 1$ . Three elementary operations plant a literal 1 in the corner: the column operation  $C_2 += C_1 \cdot \rho$  with  $\rho = q(1 - pb)$  replaces  $b$  by  $b' = a\rho + b$ , which satisfies  $pb' = paq(1 - pb) + pb = 1$ ; the row operation  $R_2 += wR_1$  with  $w = (1 - d')p$ , where  $d' = c\rho + d$ , replaces  $d'$  by  $wb' + d' = (1 - d')pb' + d' = 1$ ; and with a 1 at position  $(2, 2)$ , the operations  $R_1 += (-b')R_2$  and  $C_1 += C_2 \cdot (-c')$  (where  $c'$  is the current  $(2, 1)$  entry) clear the off-diagonal entries. The result is  $\text{diag}(v, 1)$  with  $v \in A^\times$ , since the product of invertible matrices is invertible and a diagonal matrix is invertible exactly when its diagonal entries are. If  $a = 0$ , then some entry of  $X$  is nonzero (as  $X$  is invertible and  $A \neq 0$ ), and one elementary row or column addition moves a nonzero element into position  $(1, 1)$ ; this reduces to the previous case.  $\square$

**Lemma A.5.**  $\text{EL}_2(L)$  is normal in  $\text{GL}_2(L)$ , and  $H$  is normal in  $L^\times$ .

Leavitt/DiagonalClassGroup  
stableUnits  
ScalarReduction

KOne/RefineLoopDischarge  
K1\_trivial  
narrowReduction\_holds  
scalarReduction\_holds  
stableUnits\_eq\_top\_holds  
KOne/AllRanksElementary  
glAll\_eq\_elementary  
K1\_trivial is the rank-two  
form; the all-rank form of  
the second sentence is carried  
by glAll\_eq\_elementary.  
Leavitt/LeavittSimplicity  
exists\_mul\_mul\_eq\_one

Leavitt/MatrixDiagonalization  
n  
exists\_elementary\_mul\_diag  
binaryLeavitt\_exists\_elementary\_mul\_diag

Leavitt/DiagonalClassGroup  
conj\_mem\_elementaryGroup\_of\_division  
stableUnits\_normal

*Proof.* Conjugation by an elementary matrix preserves  $\text{EL}_2$ ; conjugation by  $\text{diag}(u, v)$  sends  $x_{12}(a)$  to  $x_{12}(uav^{-1})$  and  $x_{21}(a)$  to  $x_{21}(vau^{-1})$ , hence also preserves  $\text{EL}_2$ ; and by Lemma A.4 every element of  $\text{GL}_2(L)$  is a product of elementary matrices and one matrix  $\text{diag}(u, 1)$ . For the second claim, if  $u \in H$  and  $g \in L^\times$ , then  $\text{diag}(gug^{-1}, 1) = \text{diag}(g, 1) \text{diag}(u, 1) \text{diag}(g, 1)^{-1} \in \text{EL}_2(L)$  by the first claim.  $\square$

By Lemma A.5 the quotient  $L^\times/H$  is a group; we write  $[u]$  for the class of  $u$ , so that Theorem A.2 says  $[u] = 1$  for every unit. Three families of moves generate equalities in  $L^\times/H$ .

**Lemma A.6** (Unipotent moves). *Let  $a, b \in L$  with  $ba = 0$ . Then  $1 + ab \in H$ . In particular, for prefix-incomparable words  $v, w$  and any  $x \in L$ , the cylinder unipotent  $1 + s_v x t_w$  lies in  $H$ .*

*Proof.* From  $ba = 0$ ,  $(ab)^2 = a(ba)b = 0$ , so  $1 + ab$  is a unit with inverse  $1 - ab$ , and the four-term identity

$$\text{diag}(1 + ab, 1) = x_{12}(a) x_{21}(b) x_{12}(-a) x_{21}(-b)$$

holds by direct block computation:  $x_{12}(a)x_{21}(b) = \begin{pmatrix} 1+ab & a \\ b & 1 \end{pmatrix}$ , multiplying by  $x_{12}(-a)$  clears the  $(1, 2)$  entry using  $-(1 + ab)a + a = -a(ba) = 0$ , and  $x_{21}(-b)$  clears the  $(2, 1)$  entry. For the particular case take  $a = s_v x$ ,  $b = t_w$ : then  $ba = t_w s_v x = 0$  since  $v, w$  are incomparable.  $\square$

**Lemma A.7** (The flip). *Let  $x, y \in L$  and suppose  $1 + xy \in L^\times$ . Then  $1 + yx \in L^\times$ , with inverse  $1 - y(1 + xy)^{-1}x$ , and*

$$[1 + xy] = [1 + yx] \quad \text{in } L^\times/H.$$

*Proof.* The stated inverse is checked directly:  $(1 + yx)(1 - y(1 + xy)^{-1}x) = 1 + yx - y(1 + xy)(1 + xy)^{-1}x = 1$ , and likewise on the other side. The matrix  $\text{diag}(1 + xy, (1 + yx)^{-1})$  factors into four elementary matrices,

$$x_{12}(x) x_{21}(y) x_{12}(-(1 + xy)^{-1}x) x_{21}(-(1 + yx)y),$$

again by direct computation. Hence

$\text{diag}((1 + xy)(1 + yx)^{-1}, 1) = \text{diag}(1 + xy, (1 + yx)^{-1}) \cdot \text{diag}((1 + yx)^{-1}, 1 + yx)$ , and the second factor is a Whitehead diagonal (Lemma 5.5); both factors lie in  $\text{EL}_2(L)$ , so  $(1 + xy)(1 + yx)^{-1} \in H$ .  $\square$

**Lemma A.8** (Corner insertion). *For a binary word  $w$  and  $u \in L^\times$ , the corner insertion  $\kappa_w(u) = s_w u t_w + (1 - p_w)$  is a unit, and  $[\kappa_w(u)] = [u]$ . Consequently every central unit lies in  $H$ .*

*Proof.* The matrix  $X_w = \begin{pmatrix} s_w & 1-p_w \\ 0 & t_w \end{pmatrix}$  is invertible, with inverse  $\begin{pmatrix} t_w & 0 \\ 1-p_w & s_w \end{pmatrix}$  (using  $t_w s_w = 1$  and  $(1 - p_w)s_w = 0$ ), and it intertwines:

$$X_w \text{diag}(u, 1) X_w^{-1} = \text{diag}(\kappa_w(u), 1).$$

KOne/StableUnitsGenerators  
mem\_stableUnits\_of\_val\_unipo  
tent

KOne/IncomparableUnipoten  
ts

incomparableUnit\_mem

Leavitt/WhiteheadFlip  
flipUnit  
diagPair\_flipUnit\_inv\_mem  
mul\_flipUnit\_inv\_mem\_stableU  
nits

Leavitt/LeavittDiagonalClass  
kappaUnit\_mul\_inv\_mem\_stable  
Units  
central\_mem\_stableUnits  
stableUnits\_eq\_top

Thus  $\text{diag}(\kappa_w(u)u^{-1}, 1) = [X_w, \text{diag}(u, 1)]$ , a commutator of invertible matrices, one of which is elementary-diagonalizable; by Lemmas A.4 and 5.5 such a commutator lies in  $\text{EL}_2(L)$ , so  $\kappa_w(u)u^{-1} \in H$ .

For a central unit  $c$ , centrality gives  $s_0ct_0 + s_1ct_1 = c(s_0t_0 + s_1t_1) = c$ ; on the other hand a direct computation using  $t_1s_0 = 0$  shows

$$\kappa_0(c)\kappa_1(c) = s_0ct_0 + s_1ct_1.$$

Hence  $[c] = [\kappa_0(c)][\kappa_1(c)] = [c]^2$  in  $L^\times/H$ , and  $[c] = 1$ .  $\square$

**A.2. Code moves.** Complete leaf sets play the role of block structures. Two further families of units lie in  $H$ .

**Lemma A.9** (Code scalars). *Let  $\mathcal{D} = (w_1, \dots, w_\ell)$  be a complete leaf set and  $G \in \text{GL}_\ell(k)$  an invertible scalar matrix. Then*

$$\Sigma_{\mathcal{D}}(G) = \sum_{i,j} s_{w_i} G_{ij} t_{w_j}$$

*is a unit of  $L$  lying in  $H$ .*

*Proof.* The assignment  $G \mapsto \Sigma_{\mathcal{D}}(G)$  is unital and multiplicative by the leaf identities (Lemma 5.3), so  $\Sigma_{\mathcal{D}}(G)$  is a unit with inverse  $\Sigma_{\mathcal{D}}(G^{-1})$ . Over the field  $k$ ,  $G$  is a product of transvections and one diagonal matrix (Gaussian elimination). A transvection  $I + \lambda E_{ij}$ ,  $i \neq j$ , transports to  $1 + \lambda s_{w_i} t_{w_j}$ , a cylinder unipotent in  $H$  by Lemma A.6 (leaves of a complete leaf set are pairwise incomparable). A diagonal matrix  $\text{diag}(d_1, \dots, d_\ell)$  transports to  $\prod_i \kappa_{w_i}(d_i)$  with central scalars  $d_i \in k^\times$ ; each factor lies in  $H$  by Lemma A.8.  $\square$

**Lemma A.10** (Code changes). *Let  $(\tau_1, \dots, \tau_\ell)$  and  $(\sigma_1, \dots, \sigma_\ell)$  be complete leaf sets of the same size. Then the code change*

$$\Omega = \sum_{i=1}^{\ell} s_{\tau_i} t_{\sigma_i}$$

*is a unit of  $L$  lying in  $H$ .*

*Proof.*  $\Omega$  is a unit with inverse  $\sum_i s_{\sigma_i} t_{\tau_i}$ , by the leaf identities. Note that the  $\Omega$  are precisely the images  $U_g$  of Thompson's group  $V$  under the embedding (23), together with their compositions with index bijections; the lemma is a generation theorem for the Higman–Thompson groupoid. Induct on  $\ell$ . For  $\ell = 1$  both sets are the empty word and  $\Omega = 1$ . For  $\ell \geq 2$ , the source set, being the leaf set of a finite full binary tree, contains a sibling pair  $\sigma_a = w0$ ,  $\sigma_b = w1$ . Composing  $\Omega$  with at most two *transpositions* aligns the pairing so that the targets of  $\sigma_a, \sigma_b$  are again a sibling pair  $v0, v1$  in the same order; here a transposition is a code change that swaps two leaves  $x, y$  of one complete leaf set and fixes the rest, i.e.  $T(x, y) = s_x t_y + s_y t_x + (1 - p_x - p_y)$ , and  $T(x, y) \in H$  because it is  $\Sigma_{\mathcal{D}}(P)$  for the corresponding scalar permutation matrix  $P$  on any complete leaf set  $\mathcal{D}$  refining  $\{x, y\}$ —a code scalar (Lemma A.9). After alignment, the two sibling terms merge through the completeness relation,

KOne/CodeScalarMoves  
codeScalar\_unit\_mem

KOne/CodeChangeUnits  
codeChange\_mem\_stableUnits  
KOne/CodeChangeGlue  
codeBijection\_mem\_stableUnit  
s

$s_v t_w + s_v t_w = s_v(s_0 t_0 + s_1 t_1)t_w = s_v t_w$ , producing a code change between complete leaf sets of size  $\ell - 1$ ; the inductive hypothesis and normality of  $H$  conclude.  $\square$

**Corollary A.11** (Reshaping). *Let  $(R_i), (C_j), (P_i), (Q_j)$  be complete leaf sets, with  $|R| = |P|$  and  $|C| = |Q|$ , and let  $E_{ij} \in L$ . If*

$$u = \sum_{i,j} s_{R_i} E_{ij} t_{C_j} \quad \text{and} \quad v = \sum_{i,j} s_{P_i} E_{ij} t_{Q_j}$$

*are both units, then  $v = \Omega_1 u \Omega_2$  for code changes  $\Omega_1 = \sum_i s_{P_i} t_{R_i}$  and  $\Omega_2 = \sum_j s_{C_j} t_{Q_j}$ , and hence  $u \in H \iff v \in H$ .*

*Proof.*  $\Omega_1 u \Omega_2 = \sum s_{P_i} (t_{R_i} s_{R_a}) E_{ab} (t_{C_b} s_{C_j}) t_{Q_j} = v$  by the leaf identities; apply Lemma A.10 and normality.  $\square$

**A.3. The degree filtration and width reduction.** The defining relations of  $L$  are homogeneous for  $\deg s_i = +1$ ,  $\deg t_i = -1$ , so the free-algebra grading descends:  $L = \bigoplus_{d \in \mathbb{Z}} L_d$  is  $\mathbb{Z}$ -graded, with  $L_d$  spanned by the monomials  $s_\alpha t_\beta$ ,  $|\alpha| - |\beta| = d$ . Write

$$W[a, b] = \text{span}_k \{s_\alpha t_\beta \mid a \leq |\alpha| - |\beta| \leq b\},$$

and, for  $x \in L$ , write  $x = \sum_d x^{(d)}$  for its graded components. Elements of  $W[0, 0]$  are called *balanced*; by the leaf identities, the balanced elements supported at depth  $\leq n$  are exactly the matrices  $\sum_{\gamma, \delta} G_{\gamma\delta} s_\gamma t_\delta$  over the full code  $\{0, 1\}^n$  with scalar  $G$ , i.e. the image of  $M_{2^n}(k)$  under  $\Sigma$ -transport—so Lemma A.9 says: *balanced units are exactly the transported invertible scalar matrices, and they lie in  $H^1$ .*

**Lemma A.12** (Width reduction). *Every  $u \in L^\times$  satisfies  $[u] = [u']$  for some unit  $u'$  with value in  $W[-1, 1]$  (a narrow unit).*

*Proof.* Every element of  $L$  lies in some window  $W[-M, N]$ . We show: if  $N \geq 2$ , then  $[u] = [u'']$  for a unit  $u''$  with value in  $W[-M, \max(N - 1, 1)]$ ; the mirror statement for  $M \geq 2$  follows by the same argument applied to the anti-automorphism  $\theta$  of Lemma A.13 below, and iterating both lands in  $W[-1, 1]$ .

The engine is the *corner move*: for any  $v, w \in L$ ,

$$[u] = [s_0(u + vw)t_0 + s_0vt_1 + s_1wt_0 + s_1t_1],$$

obtained by transporting the  $2 \times 2$  identity  $\begin{pmatrix} 1 & v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ w & 1 \end{pmatrix} = \begin{pmatrix} u+vw & v \\ w & 1 \end{pmatrix}$  along  $\Theta_{C_2}$  of Proposition 5.4: the images of the two unipotent factors are the cylinder-type unipotents  $1 + s_0vt_1$  and  $1 + s_1wt_0$ , which lie in  $H$  by Lemma A.6, and the image of  $\text{diag}(u, 1)$  is  $\kappa_0(u)$ , congruent to  $u$  by Lemma A.8.

Let  $x$  be the value of  $u$ , with top component  $x^{(N)}$ ,  $N \geq 2$ . Since  $y t_0 s_0 = y$  for every  $y$ , the top component factors as  $x^{(N)} = b s_0$  with  $b = x^{(N)} t_0 \in$

<sup>1</sup>A balanced element that is a unit of  $L$  has invertible scalar matrix at its own depth—a singular matrix over a field is an annihilator of a nonzero matrix, hence a zero divisor—and therefore has balanced inverse.

KOne/PencilReshape  
reshaped\_pencil\_mem\_iff  
exists\_resaped\_pencil  
pencilVal\_window\_mem

Leavitt/LeavittWindowReduction  
exists\_window\_reduction  
Leavitt/LeavittGradingSpans  
exists\_corner\_move  
Leavitt/BinaryLeavittWindow  
exists\_narrow\_representative

$W[N-1, N-1]$ . Apply the corner move with  $v = -b$ ,  $w = s_0$ : the new value is

$$s_0(x - x^{(N)})t_0 - s_0bt_1 + s_1s_0t_0 + s_1t_1.$$

The first summand lies in  $W[-M, N-1]$  (conjugation by  $s_0(\cdot)t_0$  preserves degree), the second has degree  $N-1$ , and the last two have degrees 1 and 0. Hence the new unit has value in  $W[-M, \max(N-1, 1)]$ , as claimed.  $\square$

**A.4. Windows die.** The next stage shows that units confined to a *one-sided* window already lie in  $H$ . This is where the analytic heart of the argument sits.

**Lemma A.13** (Transpose). *The assignment  $s_i \mapsto t_i$ ,  $t_i \mapsto s_i$  extends to an anti-automorphism  $\theta$  of  $L$  with  $\theta^2 = \text{id}$ ; it maps  $W[a, b]$  onto  $W[-b, -a]$ , and  $u \in H \iff \theta(u) \in H$ .*

*Proof.* The relations are symmetric under the assignment composed with order reversal, so  $\theta$  exists; it negates degree. Applying  $\theta$  entrywise to the transpose of matrices sends  $x_{ij}(a)$  to  $x_{ji}(\theta(a))$  and  $\text{diag}(u, 1)$  to  $\text{diag}(\theta(u), 1)$ , so it preserves  $\text{EL}_2$  and hence membership in  $H$ .  $\square$

**Lemma A.14** (Pure tails are nilpotent). *Let  $1 + \eta \in L^\times$  with  $\eta \in W[1, 1]$ . Then  $\eta$  is nilpotent.*

*Proof.* Let  $y = (1 + \eta)^{-1}$ , with graded components  $y^{(d)}$  supported in a finite window. The equation  $(1 + \eta)y = 1$  reads degreewise:  $y^{(d)} + \eta y^{(d-1)} = \delta_{d,0}$ . From the bottom of the window upward,  $y^{(d)} = 0$  for  $d$  below the window; then  $y^{(d)} = -\eta y^{(d-1)}$  for  $d < 0$  forces all negative components to vanish,  $y^{(0)} = 1$ , and  $y^{(d)} = (-\eta)^d$  for  $d > 0$ . Since the window is finite,  $(-\eta)^d = 0$  for large  $d$ .  $\square$

**Lemma A.15** (Nilpotent tails die). *Let  $z \in L$  be balanced with  $s_1z$  nilpotent, and let  $u \in L^\times$  have value  $1 + s_1z$ . Then  $u \in H$ .*

*Proof.* Induct on the nilpotency index  $D$  of  $s_1z$ ; for  $D = 1$ ,  $z = t_1(s_1z) = 0$  and  $u = 1$ . For  $D \geq 2$ , put  $r = t_1(s_1z)^{D-1}$ , of pure degree  $D-2$ , so that  $(s_1z)^{D-1} = s_1r$  and  $r(s_1z) = t_1(s_1z)^D = 0$ . Factor  $r = Rs_0^{D-2}$  with  $R = rt_0^{D-2}$  balanced, take a balanced pseudo-inverse  $\Xi$  of  $R$ —balanced elements are scalar matrices along a full code, and every scalar matrix  $R$  admits  $\Xi$  with  $R\Xi R = R$ —and set

$$e = t_0^{D-2} \Xi r,$$

a balanced element with the two properties that drive the induction:  $e(s_1z) = t_0^{D-2} \Xi(r(s_1z)) = 0$ , and  $re = (rt_0^{D-2}) \Xi r = R\Xi R s_0^{D-2} = R s_0^{D-2} = r$ , whence  $(s_1z)^{D-1}e = s_1(re) = s_1r = (s_1z)^{D-1}$ . The mover  $1 - s_1(ze) = 1 - (s_1z)e$  lies in  $H$  by Lemma A.6 with  $(a, b) = (s_1z, -e)$ , since  $ba = -e(s_1z) = 0$ ; and for the same reason

$$(1 - s_1(ze))(1 + s_1z) = 1 + s_1(z - ze),$$

KOne/ThetaStable  
thetaMatUnit\_mem\_elementaryG  
roup  
thetaMatUnit\_diagUnit

KOne/PureTailNilpotency  
pure\_tail\_nilpotent

KOne/NilpotentTailKill  
nilpotent\_tail\_mem\_stableUni  
ts

Leavitt/RankNormalForm  
exists\_rank\_normal\_form

the cross term being  $-s_1(ze)(s_1z) = -s_1z(e(s_1z)) = 0$ . The new tail satisfies  $(s_1(z - ze))^{D-1} = (s_1z)^{D-1}(1 - e) = 0$ , using  $(1 - e)(s_1z) = s_1z$  to telescope and then the second property of  $e$ ; so its index is at most  $D - 1$ , and the inductive hypothesis applies to  $u$  times the mover.  $\square$

**Lemma A.16** (The keystone). *Let  $u = c + \zeta \in L^\times$  with  $c$  balanced and  $\zeta \in W[1, 1]$ . Then  $c \in L^\times$ .*

KOne/ZeroKOne  
balanced\_component\_isUnit

*Proof.* Suppose not. Balanced elements at depth  $n$  are scalar matrices  $M_{2^n}(k)$  along the full code; by the rank normal form over the field  $k$  there are balanced units  $g, h$  with  $geh = e := \sum_{\gamma \in S} p_\gamma$ , a cylinder idempotent over a subset  $S$  of the depth- $n$  code, and  $c$  is invertible exactly when  $S$  is everything. So assume  $S$  proper, and let  $f = 1 - e = \sum_{\gamma \in T} p_\gamma$  with  $T \neq \emptyset$ . Replace  $u$  by  $w = guh = e + \zeta'$  (with  $\zeta' = g\zeta h \in W[1, 1]$ ); invertibility of  $c$  is not affected. Let  $y = w^{-1}$ , with graded components  $y^{(d)}$ ,  $d \in [\text{lo}, \text{hi}]$ , and let  $M = -\text{lo}$ .

The equation  $wy = 1$  reads degree-wise  $e y^{(d)} + \zeta' y^{(d-1)} = \delta_{d,0}$ . Multiplying the  $d = 0$  equation by  $f$  on both sides ( $fe = 0$ ,  $f^2 = f$ ) gives  $f = f\zeta' y^{(-1)} f$ , and substituting the relation  $y^{(d)} = f y^{(d)} - \zeta' y^{(d-1)}$  (valid for  $d \leq -1$ , from  $e y^{(d)} = -\zeta' y^{(d-1)}$ ) repeatedly yields, after  $M$  steps (the components below the window vanish),

$$(40) \quad f = \sum_{j=0}^{M-1} \underbrace{f \zeta' (-\zeta')^j}_{\deg=j+1} \cdot \underbrace{f y^{(-1-j)} f}_{\deg=-1-j}.$$

Now count ranks at a deep uniform interface. For  $\ell$  large, every factor in (40) is represented by a scalar matrix between the full codes of appropriate depths: an element of  $W[d, d]$  supported at bounded depth is, for all large  $q$ , a  $2^{q+d} \times 2^q$  scalar matrix (pad with the completeness relation). Representing the  $j$ -th summand at interface depths  $\ell \rightarrow \ell - 1 - j \rightarrow \ell$ , the middle factor passes through the matrix of  $f$  at depth  $\ell - 1 - j$ , whose rank is exactly  $|T| 2^{\ell-1-j-n}$ . Hence the matrix of the right-hand side of (40) at depth  $\ell$  has rank at most  $\sum_{j < M} |T| 2^{\ell-1-j-n} < |T| 2^{\ell-n}$ —a geometric series short of its limit—while the matrix of  $f$  at depth  $\ell$  has rank exactly  $|T| 2^{\ell-n}$ . This contradiction proves the lemma.  $\square$

**Proposition A.17** (Windows die). *A unit of  $L$  whose value lies in  $W[0, N]$  for some  $N$ , or in  $W[-N, 0]$ , lies in  $H$ .*

KOne/WindowNonnegReduction  
window\_nonneg\_mem\_stableUnit  
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KOne/WindowNonposReduction  
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window\_nonpos\_mem\_stableUnit  
s  
KOne/WidthTwoReduction  
window\_zero\_one\_mem\_stableUnit  
its

*Proof.* By the transpose (Lemma A.13) it suffices to treat  $W[0, N]$ ; induct on  $N$ .

*Base*,  $N \leq 1$ . Let  $u = c + \zeta$ ,  $c$  balanced,  $\zeta \in W[1, 1]$ . By the keystone (Lemma A.16)  $c$  is a balanced unit, so  $c \in H$  with balanced inverse, and  $u = c(1 + \eta)$  with  $\eta = c^{-1}\zeta \in W[1, 1]$ . By Lemma A.14  $\eta$  is nilpotent. The corner insertion  $\kappa_1(1 + \eta) = 1 + s_1(\eta t_1)$  is congruent to  $1 + \eta$  (Lemma A.8), has balanced tail coefficient  $z = \eta t_1$ , and  $s_1 z = s_1 \eta t_1$  inherits nilpotency from  $\eta$  (indeed  $(s_1 \eta t_1)^j = s_1 \eta^j t_1$ ); Lemma A.15 finishes.

*Step,  $N \geq 2$ .* Write the value as  $1 + (a + \eta)$  with  $a \in W[0, N - 1]$ —subtracting 1 costs nothing since 1 is balanced—and  $\eta$  the top component, of degree  $N$ . Split  $\eta = s_0 q_0 + s_1 q_1$  with  $q_z = t_z \eta$  of degree  $N - 1$ . Perform the stable Whitehead block move along the four depth-two cylinders: with

$$X_z = s_{00} s_z t_{w_z}, \quad Y_z = s_{w_z} q_z t_{00}, \quad (w_0, w_1) = (01, 10),$$

set  $m_2 = (1 - X_0)(1 - X_1)$  and  $m_1 = (1 + Y_0)(1 + Y_1)$ , products of cylinder unipotents in  $H$  (Lemma A.6), and let  $\kappa = \kappa_{00}(u) = 1 + s_{00}(a + \eta)t_{00}$ , congruent to  $u$ . In the product  $m_2 \kappa m_1$  all cross terms vanish by incomparability of the three cylinders 00, 01, 10, except

$$X_0 Y_0 + X_1 Y_1 = s_{00} (s_0 q_0 + s_1 q_1) t_{00} = s_{00} \eta t_{00},$$

which cancels the top component inside  $\kappa$ . The value of  $m_2 \kappa m_1$  is therefore

$$1 + (s_{00} a t_{00} - X_0 - X_1 + Y_0 + Y_1),$$

whose components have degrees in  $[0, N - 1]$ : the  $a$ -part keeps its degrees,  $\deg X_z = 1$ , and  $\deg Y_z = N - 1$ . By induction  $m_2 \kappa m_1 \in H$ , hence  $u \in H$ .  $\square$

### A.5. The pencil and the elimination loop.

**Lemma A.18** (Pencil form). *Let  $x \in W[-1, 1]$ . Then there are  $m$  and scalar matrices  $A_0, A_1, C, B_0, B_1$ , indexed by the full code  $\mathcal{F} = \{0, 1\}^{m+1}$ , with*

$$x = \sum_{i,j \in \mathcal{F}} s_i E_{ij} t_j, \quad E_{ij} = A_{0,ij} t_0 + A_{1,ij} t_1 + C_{ij} \cdot 1 + B_{0,ij} s_0 + B_{1,ij} s_1.$$

*Proof.* Choose  $m$  with all monomials of  $x$  of depth  $\leq m$ , and expand  $x = \sum_{i,j} s_i (t_i x s_j) t_j$  along the full code of depth  $m + 1$  (completeness relation on both sides). For a monomial  $s_\alpha t_\beta$  of  $x$  and code words  $i, j$  of length  $m + 1$ :  $t_i s_\alpha$  vanishes unless  $\alpha$  is a prefix of  $i$ , in which case it is  $t_{i'}$  with  $|i'| = m + 1 - |\alpha| \geq 1$ , and symmetrically  $t_\beta s_j = s_{j'}$  with  $|j'| = m + 1 - |\beta| \geq 1$ ; the product  $t_{i'} s_{j'}$  collapses, by the prefix calculus, to 0, a scalar, or a single letter  $s_z$  or  $t_z$ , according to the degree  $|\alpha| - |\beta| \in \{-1, 0, 1\}$ . Hence each corner entry  $t_i x s_j$  lies in the five-dimensional scalar pencil space displayed.  $\square$

We call a unit in this form a *pencil unit* over the code pair (rows, columns); the data is the quintuple of scalar matrices, and the *stack* is the vertical block matrix  $\begin{pmatrix} B_0 \\ B_1 \end{pmatrix}$ .

**Lemma A.19** (Strict negativity). *Let  $u$  be a pencil unit over complete leaf sets  $(R_i)$ ,  $(C_j)$ , and suppose the stack admits a scalar left inverse: matrices  $G_0, G_1$  with  $G_0 B_0 + G_1 B_1 = I$ . Then there is  $N$  such that every corner entry of the inverse,*

$$y_{ji} = t_{C_j} u^{-1} s_{R_i},$$

*lies in  $W[-N, -1]$ .*

KOne/PencilForm  
exists\_pencil\_form  
KOne/RefineLoopDischarge  
pencilEntry\_mem\_window  
codePair\_expansion

KOne/StrictNegativePencil  
entry\_window\_negative\_of\_B\_f  
ull

*Proof.* All  $y_{ji}$  lie in a common finite window; let  $d^*$  be the largest degree in which any  $y_{ji}$  has a nonzero component, and suppose  $d^* \geq 0$ . From  $u u^{-1} = 1$ , sandwiching and inserting the completeness relation for  $(C_j)$ ,

$$\sum_j E_{ij} y_{ji'} = \delta_{ii'} \quad (\text{all } i, i').$$

Take the component of degree  $d^* + 1 \geq 1$  on both sides. On the left only the degree-+1 part of  $E$  can contribute against  $y^{(d^*)}$ —components of  $y$  above  $d^*$  vanish—so

$$\sum_j (B_{0,ij} s_0 + B_{1,ij} s_1) y_{ji'}^{(d^*)} = 0.$$

Multiplying on the left by  $t_0$  and by  $t_1$  separates the two shifts:

$$\sum_j B_{0,ij} y_{ji'}^{(d^*)} = 0 \quad \text{and} \quad \sum_j B_{1,ij} y_{ji'}^{(d^*)} = 0;$$

that is, the stack annihilates the matrix  $Y = (y_{ji'}^{(d^*)})$  on the left. Applying the scalar left inverse,  $Y = (G_0 B_0 + G_1 B_1) Y = 0$ , contradicting the choice of  $d^*$ . Hence all components have degree  $\leq -1$ .  $\square$

**Lemma A.20** (Refinement step). *Let  $u$  be a pencil unit over  $(R, C)$  whose column  $j_0$  is stack-free:  $B_{0,ij_0} = B_{1,ij_0} = 0$  for all  $i$ . Then  $u$  is also a pencil unit over the code pair  $(R, C')$ , where  $C'$  replaces the word  $C_{j_0}$  by its two children  $C_{j_0} 0$ ,  $C_{j_0} 1$ , with data given columnwise by*

$$(A_0, A_1, C, B_0, B_1) \cdot_{j_0 z} = (0, 0, A_{z, \cdot j_0}, [z=0] C_{\cdot j_0}, [z=1] C_{\cdot j_0}),$$

*all other columns unchanged.*

*Proof.* Insert  $1 = s_0 t_0 + s_1 t_1$  before  $t_{C_{j_0}}$ : each term  $s_{R_i} E_{ij_0} t_{C_{j_0}}$  becomes  $\sum_z s_{R_i} (E_{ij_0} s_z) t_{C_{j_0} z}$ , and for a stack-free entry the split identity

$$(a_0 t_0 + a_1 t_1 + c) s_z = a_z + c s_z$$

shows  $E_{ij_0} s_z$  is again a pencil entry, with the  $A$ -data falling to the constant slot and the constant to the  $B$ -slot—the index shift of the Wiener–Hopf analogy.  $\square$

**Lemma A.21** (Code supply). *For  $1 \leq \kappa \leq 2^r$  there is a complete leaf set of size  $\kappa$  and depth  $\leq r$ ; for  $\kappa \geq 2^M$  there is one of size  $\kappa$  and depth  $\geq M$ .*

*Proof.* Kraft counting. Shallow: start from the full code of depth  $\lceil \log_2 \kappa \rceil \leq r$  and merge sibling pairs until  $\kappa$  leaves remain (each merge reduces the count by one and cannot increase depth). Deep: start from the full code of depth  $M$ , of size  $2^M \leq \kappa$ , and split leaves (each split increases the count by one and produces words of length  $\geq M$ ).  $\square$

Figure 5 charts the elimination.

**Theorem A.22** (The elimination loop). *Every pencil unit over a row code of  $2^r$  words lies in  $H$ .*

KOne/RefineStep  
refine\_column  
pencilEntry\_mul\_s  
KOne/GLVectorNormalizatio  
n  
exists\_isUnit\_matrix\_col

KOne/CodeShapeSupply  
exists\_shallow\_code  
exists\_deep\_code

KOne/RefineLoopDischarge  
pencil\_free\_exit  
pencil\_full\_exit  
pencil\_unit\_mem\_pow

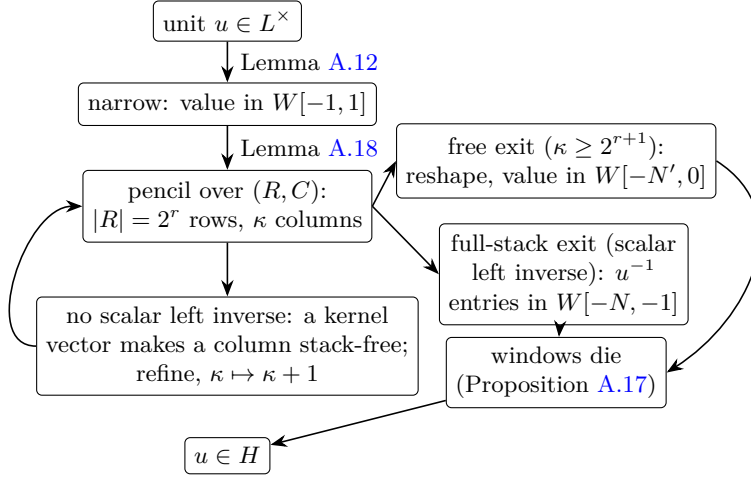


FIGURE 5. The elimination loop of Theorem A.22. Each pass either exits—shallow-versus-deep reshaping lands the unit, or its inverse, in a one-sided window—or splits one stack-free column, raising the column count; the counter  $2^{r+1} - \kappa$  bounds the number of passes.

*Proof.* Let  $u$  be a pencil unit over  $(R, C)$ ,  $|R| = 2^r$ ,  $|C| = \kappa$ . We induct on  $2^{r+1} - \kappa$  (when this is  $\leq 0$ , the first case below applies).

*Free exit* ( $\kappa \geq 2^{r+1}$ ). By Lemma A.21 choose a shallow complete code  $P$  of size  $2^r$  and depth  $\leq r$ , and a deep complete code  $Q$  of size  $\kappa$  and depth  $\geq r + 1$ . Reshape  $u$  to the pair  $(P, Q)$  (Corollary A.11). Each pencil entry lies in  $W[-1, 1]$ , so each monomial of the reshaped value has degree at most  $|P_i| + 1 - |Q_j| \leq r + 1 - (r + 1) = 0$ : the reshaped unit lies in  $W[-N', 0]$  for some  $N'$ , hence in  $H$  by Proposition A.17, hence  $u \in H$ .

*Full stack.* If the stack has a scalar left inverse, Lemma A.19 puts every corner entry  $y_{ji}$  of  $u^{-1}$  in  $W[-N, -1]$ . Expand  $u^{-1} = \sum_{j,i} s_{C_j} y_{ji} t_{R_i}$  and reshape to a shallow column code of depth  $\leq r + 1$  (possible:  $\kappa \leq 2^{r+1}$  in this branch, or the free exit applies) and a deep row code of depth  $\geq r$ . Monomial degrees are at most  $(r + 1) + (-1) - r = 0$ , so  $u^{-1} \in H$  by Proposition A.17, and  $u \in H$ .

*Refine.* Otherwise the stack, having no scalar left inverse, has a nonzero kernel vector  $v$  (over the field  $k$ , a matrix with trivial kernel has a left inverse). Choose an invertible scalar matrix  $G$  whose  $j_0$ -th column is  $v$  (extend  $v$  to a basis). The code scalar  $\Sigma_C(G)$  lies in  $H$  (Lemma A.9), and  $u \cdot \Sigma_C(G)$  is a pencil unit over  $(R, C)$  with data multiplied columnwise by  $G$ ; its column  $j_0$  is stack-free because  $v$  kills both  $B_0$  and  $B_1$ . Apply the refinement step (Lemma A.20): a pencil unit over  $(R, C')$  with  $\kappa + 1$  columns. The counter  $2^{r+1} - \kappa$  has dropped; the inductive hypothesis gives  $u \cdot \Sigma_C(G) \in H$ , hence  $u \in H$ .  $\square$

*Proof of Theorem A.2.* Let  $u \in L^\times$ . By width reduction (Lemma A.12),  $[u] = [u']$  with  $u'$  narrow; by the pencil form (Lemma A.18),  $u'$  is a pencil unit over the full code pair of depth  $m + 1$ , a row code of  $2^{m+1}$  words; by the elimination loop (Theorem A.22),  $u' \in H$ . Hence  $u \in H$ .  $\square$

### A.6. Consequences and effectivity.

**Corollary A.23.** *For every field  $k$  and  $L = L_k(1, 2)$ :  $K_1(L) = 0$ ;  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$  for every  $r \geq 2$ ; and  $L^\times$  is perfect.*

*Proof. Rank two.* For  $X \in \mathrm{GL}_2(L)$ , Lemma A.4 writes  $EXF = \mathrm{diag}(u, 1)$ , and Theorem A.2 places the diagonal factor in  $\mathrm{EL}_2(L)$ ; hence  $\mathrm{GL}_2(L) = \mathrm{EL}_2(L)$ . (This is the only use of Gaussian elimination; every higher rank now follows by transport.)

*Every rank, by flattening.* The coefficient isomorphism  $\Theta_C^{-1} : L \cong M_n(L)$  of Proposition 5.4 carries  $\mathrm{EL}_p$  onto  $\mathrm{EL}_p$  entrywise, and block flattening identifies  $\mathrm{EL}_p(M_q(L)) = \mathrm{EL}_{pq}(L)$  inside  $\mathrm{GL}_{pq}(L)$  for  $p \geq 2$  (the two inclusions proved in Proposition 5.8); both maps are restrictions of ring isomorphisms of the ambient matrix rings, so each transports the *equality*  $\mathrm{GL} = \mathrm{EL}$ . Starting from rank two:  $\mathrm{GL}_2(L) = \mathrm{EL}_2(L)$  transports through  $L \cong M_n(L)$  and flattening to  $\mathrm{GL}_{2n}(L) = \mathrm{EL}_{2n}(L)$  for every  $n$ ; and for arbitrary  $r \geq 2$ , the same two maps with the roles of  $r$  and 2 exchanged transport  $\mathrm{GL}_{2r} = \mathrm{EL}_{2r}$ , already established, back down to  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ .

*$K_1$  and perfectness.* Since  $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$  at every rank, every class of the stable quotient defining  $K_1(L)$  is trivial:  $K_1(L) = 0$ . For perfectness,  $\Theta_{C_3}$  restricts to a group isomorphism  $\mathrm{GL}_3(L) \cong L^\times$ , and  $\mathrm{GL}_3(L) = \mathrm{EL}_3(L)$  is perfect: by the Steinberg relation, every generator  $x_{ij}(a) = [x_{il}(a), x_{lj}(1)]$  with  $l \notin \{i, j\}$  is a commutator. Hence  $L^\times = [L^\times, L^\times]$ .  $\square$

The proof of Theorem A.2 is an algorithm, and every step is effective Gaussian elimination over  $k$ : the width reduction processes one graded component per corner move; the pencil form is a change of basis at depth  $m + 1$ ; each loop pass computes a kernel vector or a left inverse of the stack; and both exits are reshapes along explicitly constructed codes followed by the window inductions, each step of which is again explicit. For a unit supported on monomials of depth at most  $d$ , all codes involved have at most  $2^{d+2}$  words and the loop executes at most  $2^{d+1}$  times, giving the complexity bound of Remark 8.5: the elementary factorization of  $\mathrm{diag}(u, 1, \dots, 1)$  is computed in time polynomial in  $2^d$  and in the number of monomials of  $u$ , in the operation-counting model stated there.

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*KOne/RefineLoopDischarge  
glTwo\_eq\_elementary\_holds  
glFour\_eq\_elementary\_holds  
KOne/AllRanksElementary  
elementaryGroup\_eq\_top  
glAll\_eq\_elementary  
binaryLeavittUnits\_perfect  
KOne/WhiteheadQuotient  
BinaryLeavittWhiteheadK1  
binaryLeavittWhiteheadK1\_sub  
singleton  
The quotient  $L^\times/H$  is  
constructed as a group and  
proved trivial; the classical  
direct-limit  $K_1$  object itself is  
not formalized, and the  
equalities  $\mathrm{GL}_r = \mathrm{EL}_r$  at  
every rank are what identify  
the two.*

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